A Review of *A Posteriori* Error Estimation and Adaptive Mesh-Refinement Techniques

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Preface

In the last 15-20 years self-adaptive discretization methods have gained an enormous importance for the numerical solution of partial differential equations that arise from physical and technical applications. The aim is to obtain a numerical solution within a prescribed tolerance using a minimal amount of work. The main tool are a posteriori error estimators and indicators which are able to give global and local information on the error of the numerical solution using only the computed numerical solution and known data of the problem.

In this note we present the most frequently used error estimators which have been developed by various authors in the last two decades. We show that they are all based on the same basic principles. These principles are then used to develop an abstract framework which is able to handle general nonlinear problems. The abstract results are applied to various classes of nonlinear elliptic partial differential equations from, e.g., fluid and continuum mechanics and yield reliable and easily computable error estimators.

We restrict ourselves to stationay problems. The treatment of transient problems would be beyond the scope of this series. Moreover, for parabolic problems the theory is not as well-developed as for elliptic problems, and for hyperbolic problems it is still in its infancy.

These notes have grown out of a summer school which I held in June 1993 at the Universität Magdeburg and of a CEA-EDF-INRIA-course which I gave in September 1995 at INRIA Rocquencourt. I'm grateful to my colleagues L. Tobiska and P. Le Tallec who invited me to give these courses. My sincere thanks are due to Miss G. Schacher, Universität Zürich, and Miss M. Binert, Ruhr-Universität Bochum, for their careful and patient typing of these notes.

Bochum, October 1995

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Introduction

In the numerical solution of practical problems of physics or engineering such as, e. g., computational fluid dynamics, elasticity, or semiconductor device simulation one often encounters the difficulty that the overall accuracy of the numerical approximation is deteriorated by local singularities such as, e. g., singularities arising from re-entrant corners, interior or boundary layers, or sharp shock-like fronts. An obvious remedy is to refine the discretization near the critical regions, i. e., to place more grid-points where the solution is less regular. The question then is how to identify those regions and how to obtain a good balance between the refined and un-refined regions such that the overall accuracy is optimal.

Another closely related problem is to obtain reliable estimates of the accuracy of the computed numerical solution. A priori error estimates, as provided, e. g., by the standard error analysis for finite element or finite difference methods, are often insufficient since they only yield information on the asymptotic error behaviour and require regularity conditions of the solution which are not satisfied in the presence of singularities as described above.

These considerations clearly show the need for an error estimator which can a posteriori be extracted from the computed numerical solution and the given data of the problem. Of course, the calculation of the a posteriori error estimate should be far less expensive than the computation of the numerical solution. Moreover, the error estimator should be local and should yield reliable upper and lower bounds for the true error in a user-specified norm. In this context one should note, that global upper bounds are sufficient to obtain a numerical solution with an accuracy below a prescribed tolerance. Local lower bounds, however, are necessary to ensure that the grid is correctly refined so that one obtains a numerical solution with a prescribed tolerance using a (nearly) minimal number of grid-points.

Disposing of an a posteriori error estimator, an adaptive mesh-refinement process has the following general structure:

(1) *Construct an initial coarse mesh $\mathcal{T}_0$ representing sufficiently well the geometry of the problem. Put $k := 0$.*
(2) *Solve the discrete problem on $\mathcal{T}_k$.*
(3) *For each element T in $\mathcal{T}_k$ compute the a posteriori error estimate.*
(4) *If the estimated global error is sufficiently small then* **stop**. *Otherwise decide which elements have to be refined and construct the next mesh $\mathcal{T}_{k+1}$. Replace k by $k+1$ and return to step 2.*

The above algorithm is best suited for stationary problems. For transient calculations, some changes have to be made:

(1) *The accuracy of the computed numerical solution has to be estimated every few time-steps.*
(2) *The refinement process in space should be coupled with a time-step control.*
(3) *A partial coarsening of the mesh might be necessary.*
(4) *Occasionally, a complete re-meshing could be desirable.*

In both stationary and transient problems, the refinement / un-refinement process might also be coupled with or replaced by a moving-point technique, which keeps the number of grid-points constant but changes there relative location (cf., e.g., [5,65]).

In order to get a first impression of the capabilities of such an adaptive refinement strategy we consider a simple, but typical example. We are looking for a function u which is harmonic in the interior Ω of a circular segment centered at the origin with radius 1 and angle $\frac{3}{2}\pi$, which vanishes on the straight parts of the boundary $\partial\Omega$, and which has normal derivative $\frac{2}{3}\cos(\frac{2}{3}\varphi)$ on the curved part of $\partial\Omega$ (cf. Problem (1.1)). Using polar co-ordinates, one easily checks that $u = r^{2/3}\sin(\frac{2}{3}\varphi)$. We compute the Ritz projections u_h of u onto the spaces of continuous piecewise linear finite elements corresponding to the two triangulations shown in Figure 1 (cf. Problem (1.4)). The left triangulation is obtained by five uniform refinements of an initial triangulation $\mathcal{T}_0$ which consists of three right-angled isosceles triangles with short sides of unit length. In each refinement step every triangle is cut into four new ones by connecting the midpoints of its edges. Moreover, the midpoint of an edge having its two endpoints on $\partial\Omega$ is projected onto $\partial\Omega$. The right triangulation in Figure 1 is obtained from $\mathcal{T}_0$ by applying five steps of the adaptive refinement strategy described above using the error estimator $\eta_{R,T}$ of Equation (1.18). A triangle $T \in \mathcal{T}_k$ is divided into four new ones if $\eta_{R,T} \geq 0,5\max_{T'\in\mathcal{T}_k}\eta_{R,T'}$ (cf. Section 4.1). Midpoints of edges having their two endpoints on $\partial\Omega$ are again projected onto $\partial\Omega$. For both meshes we list in Table 1 the number NT of triangles, the number NN of unknowns, the relative error $\epsilon := \|u-u_h\|_{H^1(\Omega)}/\|u\|_{H^1(\Omega)}$, and $q := \{\sum_{T\in\mathcal{T}_k}\eta_{R,T}^2\}^{1/2}/\|u-u_h\|_{H^1(\Omega)}$ which measures the quality of the error estimator. It clearly shows the advantages of the adaptive refinement strategy and the reliability of the error estimator.

refinement	NT	NN	ε	q
uniform	3072	1552	3.8 %	0.7
adaptive	298	143	2.8 %	0.6

Table 1: comparison of uniform and adaptive refinement

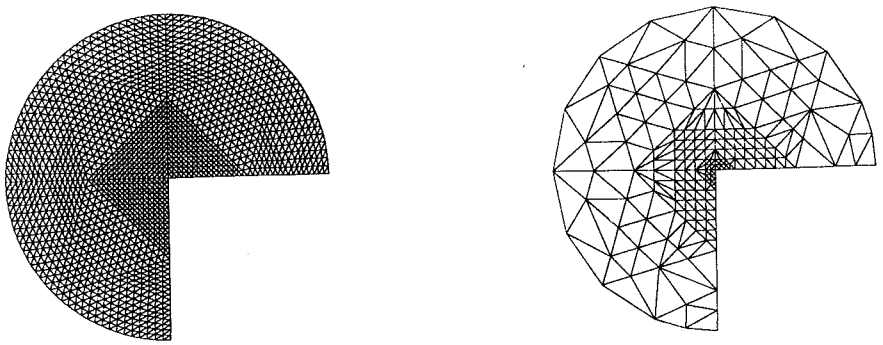


Fig. 1: Triangulations obtained by uniform (left) and adaptive refinement (right)

In what follows we will describe various possibilities for a posteriori error estimation and adaptive mesh-refinement. In Section 1 we consider a simple model problem: the two-dimensional Poisson equation discretized by continuous linear finite elements. We review several a posteriori error estimators and show that - in a certain sense - they are all equivalent and yield lower and upper bounds on the error of the finite element discretization. The estimators can roughly be classified as follows:

(1) *residual estimates:* Estimate the error of the computed numerical solution by a suitable norm of its residual with respect to the strong form of the differential equation (Section 1.2).

(2) *solution of local problems:* Solve locally discrete problems similar to, but simpler than the original problem and use appropriate norms of the local solutions for error estimation (Section 1.3).

(3) *hierarchical basis error estimates:* Evaluate the residual of the computed finite element solution with respect to another finite element space corresponding to higher order elements or to a refined grid (Section 1.4).

(4) *averaging methods:* Use some local extrapolation or averaging technique for error estimation (Section 1.5).

In Section 1.6, we shortly address the question of efficiency, i.e. whether the ratio of the estimated and the exact error remains bounded or even approaches 1 when the

meshsize converges to 0. In order to keep the notational and technical apparatus at a minimum, we consider in Section 1 only linear finite elements. The results of Section 1.2 – 1.4, however, immediately carry over to higher order finite elements (cf. Section 3.2 and [85]).

Besides presenting some of the most popular error estimation techniques, the main aim of Section 1 is to show that all these techniques rely on a common principle: the error of the computed finite element approximation may be represented as a residual in the dual of an appropriate Banach space and the norm of this residual must be calculated in a sufficiently accurate and efficient way. The main ingredients in accomplishing this task are:

(1) The stabililty of the infinite dimensional variational problem.
(2) An error representation formula.
(3) Error estimates for an interpolation operator under minimal regularity assumptions.

These observations lead to an abstract framework which is presented in Section 2.

We consider abstract nonlinear equations $F(u) = 0$ and their discretizations $F_h(u_{0h}) = 0$. Here, $F \in C^1(X, Y^*)$ and $F_h \in C(X_h, Y_h^*)$, $X_h \subset X$ and $Y_h \subset Y$ are finite dimensional subspaces of appropriate Banach spaces X and Y, and $*$ denotes the dual of a Banach space. If $u_0 \in X$ is a solution of $F(u) = 0$ such that $DF(u_0)$ is an isomorphism of X onto Y^* and DF is Lipschitz continuous at u_0, we prove in Section 2.1 that for all u in a suitable neighbourhood of u_0 the error $\|u - u_0\|_X$ is – up to multiplicative constants – bounded from below and from above by the residual $\|F(u)\|_{Y^*}$. The multiplicative constants depend on $DF(u_0)$ and $DF(u_0)^{-1}$. This equivalence is a local result. That means that it can be applied to solutions of $F_h(u_{0h}) = 0$ only if they are sufficiently close to u_0, i. e., if the discretization is "sufficiently fine". This is not surprising since we are dealing with general nonlinear problems which may have a large variety of solutions. If the continuous problem is linear, i. e., DF is constant, the afore-mentioned equivalence of course holds for all $u \in X$.

In Section 2.2 we briefly outline how the results of Section 2.1 can be extended to branches of solutions of $F(u) = 0$ including singular points such as simple limit and bifurcation points. The generalization to the case of a regular branch of solutions, i. e. situations covered by the implicit function theorem, is straightforward. The case of a simple limit or bifurcation point can be reduced as in [39] to the case of a regular branch of solutions by suitably blowing-up the spaces X and Y and modifying the function F. For practical applications it is important that the

additional spaces are low-dimensional. Thus, the cost for evaluating the residual of the modified function is essentially determined by the cost for evaluating the residual of F.

In Section 2.3, we estimate the residual $\|F(u_h)\|_{Y^*}$ where u_h is an approximate solution of $F_h(u_{0h}) = 0$. To this end we introduce a restriction operator $R_h : Y \to Y_h$, a finite dimensional subspace $\widetilde{Y}_h \subset Y$, and an approximation $\widetilde{F}_h : X_h \to Y^*$ of F at u_h which are coupled via Inequality (2.11). For practical applications, the construction of R_h and $\widetilde{F}_h$ is rather straightforward. Usually, $\widetilde{F}_h(u_h)$ is obtained by locally projecting $F(u_h)$ onto suitable finite dimensional spaces. This corresponds to the well-known technique of locally freezing the coefficients of a differential operator. The choice of $\widetilde{Y}_h$ on the other hand is less obvious. It is, however, considerably simplified by the auxiliary results of Section 3.1. We then prove that, up to multiplicative constants and additive correction terms, $\|F(u_h)\|_{Y^*}$ is bounded from below and from above by $\|\widetilde{F}_h(u_h)\|_{\widetilde{Y}_h^*}$. The latter can be evaluated quite easily since its computation is equivalent to a finite dimensional maximization problem. Moreover, sharp explicit bounds on $\|\widetilde{F}_h(u_h)\|_{\widetilde{Y}_h^*}$ are readily obtained for many practical applications. When applying the general results to finite element methods, the afore-mentioned multiplicative constants essentially depend on the element geometry and on the polynomial degree of the finite element functions. In principle, they can be estimated explicitly. The afore-mentioned correction terms consist of the following quantities:

(1) the residual $\|F_h(u_h)\|_{Y_h^*}$ of the discrete problem,
(2) the consistency error $\|F(u_h) - F_h(u_h)\|_{Y_h^*}$ of the discretization, and
(3) a term which measures the quality of the approximation of $F(u_h)$ by $\widetilde{F}_h(u_h)$.

The first quantity can be estimated from u_h and the given data. The second one can a priori be bounded. For many practical applications one can finally prove that the third quantity is a higher order perturbation when compared with $\|\widetilde{F}_h(u_h)\|_{\widetilde{Y}_h^*}$.

In this section we also give a framework which covers some of the a posteriori error estimators based on the solution of auxiliary local problems and which shows that these estimators are equivalent to the residual a posteriori error estimator considered before. When applied to partial differential equations the abstract results of Section 2 yield error estimates in a $W^{1,p}$-norm. In many applications, however, one is more interested in estimates with respect to an L^p-Norm. We also show how to modify the abstract setting such that it acomplishes this task too.

In Section 3, we apply the abstract results of Section 2 to finite element discretizations of elliptic partial differential equations. We first establish in Section 3.1 some

auxiliary results which simplify the construction of $\widetilde{Y}_h$. The main result is of the form

$$0 < \alpha \leq \inf_{u \in V_S \setminus \{0\}} \sup_{v \in V_S \setminus \{0\}} \frac{\int_S u \psi_S v}{\|u\|_{L^p(S)} \|v\|_{L^q(S)}} \leq 1.$$

Here, $1 \leq p \leq \infty$, $\frac{1}{p} + \frac{1}{q} = 1$, S is either a simplex in $\mathbb{R}^n$ or a face of such a simplex, V_S is a finite dimensional space of functions defined on S, and ψ_S is a cut-off function. It is important to note that the constant α is independent of S. Thanks to the above inequality one can show that for finite element methods $\widetilde{Y}_h$ can be chosen as the space of all linear combinations of functions $\psi_S v$ where $v \in V_S$ and S varies through all elements and their faces.

In Section 3.2 we consider once more the model problem of Section 1, now in the light of the abstract theory of Section 2. In Section 3.2–3.7, we apply the general results to finite element approximations of scalar quasilinear equations of 2nd order, the eigenvalue problem for scalar linear operators of 2nd order, the stationary, incompressible Navier-Stokes equations, the equations of linearized elasticity, and the biharmonic equation as an example for an higher order operator. In all examples we obtain upper and lower bounds for the finite element error in terms of a residual a posteriori error estimator. This error estimator consists of the elementwise error of the finite element functions with respect to the strong form of the differential equation and of jumps across inter-element boundaries of that boundary operator which naturally links the strong and weak forms of the differential equation.

Section 4 finally deals with the problem of adaptive mesh-refinement based on a posteriori error estimates. In Sections 4.1 and 4.2, we present some of the most popular mesh-refinement techniques and discuss the data-structures which are needed for their implementation. Section 4.3 gives a few numerical examples demonstrating the efficiency of the error estimation and mesh-refinement techniques.

Most chapters end with a bibliographical remark.

1. A Simple Model Problem

1.1. The Model Problem and its Discretization.

As a model problem we consider the Poisson equation with mixed Dirichlet-Neumann boundary conditions

$$\begin{aligned} -\Delta u &= f \quad \text{in} \quad \Omega \\ u &= 0 \quad \text{on} \quad \Gamma_D \\ \frac{\partial u}{\partial n} &= g \quad \text{on} \quad \Gamma_N \end{aligned} \tag{1.1}$$

in a connected, bounded, polygonal domain $\Omega \subset \mathbb{R}^2$ with boundary $\Gamma = \Gamma_D \cup \Gamma_N$, $\Gamma_D \cap \Gamma_N = \emptyset$. We assume that Γ_D is closed relative to Γ and has a positive length and that f and g are square integrable functions on Ω and Γ_N, respectively.

For any open subset ω of Ω with Lipschitz boundary γ we denote by $L^2(\omega)$, $H^1(\omega)$, and $L^2(\gamma)$ the standard Lebesgue- and Sobolev-spaces equipped with the usual norms

$$\|\varphi\|_{0,2;\omega} := \left\{ \int_\omega \varphi^2 \, dx \right\}^{1/2},$$

$$\|\varphi\|_{1,2;\omega} := \left\{ \int_\omega \varphi^2 + |\nabla \varphi|^2 \, dx \right\}^{1/2},$$

$$\|\varphi\|_{2;\gamma} := \left\{ \int_\gamma \varphi^2 \, ds \right\}^{1/2}.$$

If $\omega = \Omega$, we omit the index ω.

Set

$$X := \{ \varphi \in H^1(\Omega) : \varphi = 0 \text{ on } \Gamma_D \}. \tag{1.2}$$

The standard weak formulation of Problem (1.1) then is: Find $u \in X$ such that

$$\int_\Omega \nabla u \cdot \nabla v = \int_\Omega f v + \int_{\Gamma_N} g v \quad \forall v \in X. \tag{1.3}$$

It is well-known that Problem (1.3) admits a unique solution (cf. , e. g., [37]).

Let $\mathcal{T}_h$, $h > 0$, be a family of triangulations of Ω which satisfies the following conditions:

(1) Any two triangles in $\mathcal{T}_h$ share at most a common edge or a common vertex.

(2) The minimal angle of all triangles in the whole family $\mathcal{T}_h$ is bounded away from zero.

Denote by X_h the space of all continuous, piecewise linear finite element functions corresponding to $\mathcal{T}_h$ and vanishing on Γ_D. We then consider the following finite element discretization of Problem (1.3): Find $u_h \in X_h$ such that

$$\int_\Omega \nabla u_h \cdot \nabla v_h = \int_\Omega f\, v_h + \int_{\Gamma_N} g\, v_h \quad \forall v_h \in X_h. \tag{1.4}$$

It is well-known that Problem (1.4) admits a unique solution (cf. , e. g., [37]).

Remark 1.1. In order to keep the notational and technical apparatus at a minimum we restrict ourselves in Section 1 to linear finite elements. The results of Sections 1.2-1.4, however immediately carry over to higher oder finite elements (cf. e.g., [85] and Section 3.2). For simplicity we will assume throughout Section 1 that the integrals occuring in Equation (1.4) are evaluated exactly and that u_h is the exact solution of Problem (1.4). The abstract theory of Section 2, however, will take into account effects arising from numerical integration and inexact solution of the discrete problems. □

We end this section by introducing some additional notations which will be needed for constructing and analyzing the error estimators in the subsequent chapters.

For any $T \in \mathcal{T}_h$ we denote by $\mathcal{E}(T)$ and $\mathcal{N}(T)$ the set of its edges and vertices, respectively and set

$$\mathcal{E}_h := \bigcup_{T\in\mathcal{T}_h} \mathcal{E}(T), \quad \mathcal{N}_h := \bigcup_{T\in\mathcal{T}_h} \mathcal{N}(T).$$

We split $\mathcal{E}_h$ and $\mathcal{N}_h$ in the form

$$\mathcal{E}_h = \mathcal{E}_{h,\Omega} \cup \mathcal{E}_{h,D} \cup \mathcal{E}_{h,N}, \quad \mathcal{N}_h = \mathcal{N}_{h,\Omega} \cup \mathcal{N}_{h,D} \cup \mathcal{N}_{h,N}$$

with

$$\mathcal{E}_{h,D} := \{\, E \in \mathcal{E}_h : E \subset \Gamma_D \,\}, \quad \mathcal{E}_{h,N} := \{\, E \in \mathcal{E}_h : E \subset \Gamma_N \,\},$$
$$\mathcal{N}_{h,D} := \{\, x \in \mathcal{N}_h : x \in \Gamma_D \,\}, \quad \mathcal{N}_{h,N} := \{\, x \in \mathcal{N}_h : x \in \Gamma_N \,\}.$$

Given an $E \in \mathcal{E}_h$ we denote by $\mathcal{N}(E)$ the set of its vertices. For $T \in \mathcal{T}_h$, $E \in \mathcal{E}_h$, and $x \in \mathcal{N}_h$ we then define (see Figure 2)

$$\omega_T := \bigcup_{\mathcal{E}(T)\cap\mathcal{E}(T')\neq\emptyset} T', \quad \omega_E := \bigcup_{E\in\mathcal{E}(T')} T', \quad \omega_x := \bigcup_{x\in\mathcal{N}(T')} T',$$
$$\widetilde{\omega}_T := \bigcup_{\mathcal{N}(T)\cap\mathcal{N}(T')\neq\emptyset} T', \quad \widetilde{\omega}_E := \bigcup_{\mathcal{N}(E)\cap\mathcal{N}(T')\neq\emptyset} T'.$$

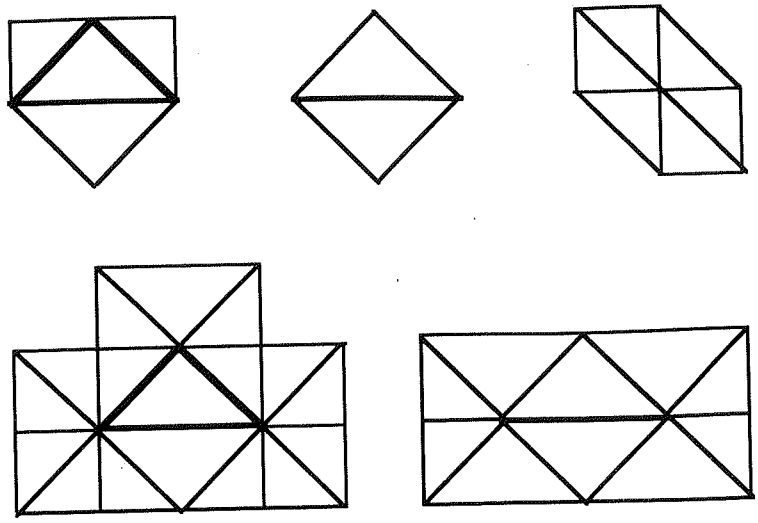


Fig. 2: domains ω_T, ω_E, ω_x, $\tilde{\omega}_T$, and $\tilde{\omega}_E$ (left to right and top to bottom)

With every edge $E \in \mathcal{E}_h$ we associate a unit vector n_E such that n_E is orthogonal to E and equals the unit exterior normal to Γ if $E \subset \Gamma$. Given any $E \in \mathcal{E}_{h,\Omega}$ and any $\varphi \in L^2(\omega_E)$ with $\varphi|_{T'} \in C(T')$, $\forall T' \subset \omega_E$, we denote by $[\varphi]_E$ the jump of φ across E in the direction n_E:

$$[\varphi]_E(x) := \lim_{t \to 0+} \varphi(x + t n_E) - \lim_{t \to 0+} \varphi(x - t n_E) \quad \forall x \in E.$$

For $T \in \mathcal{T}_h$ and $E \in \mathcal{E}_h$, let h_T and h_E be their diameter and length, respectively. We will frequently use the following simple observation.

Remark 1.2. Condition (2) above implies that the ratios h_T/h_E, $T \in \mathcal{T}_h$, $E \in \mathcal{E}(T)$, and $h_T/h_{T'}$, $T, T' \in \mathcal{T}_h$, $\mathcal{N}(T) \cap \mathcal{N}(T') \neq \emptyset$, are bounded from below and from above by constants which only depend on the lower bound in Condition (2).□

Denote by $\lambda_{T,1}, \ldots, \lambda_{T,3}$ the barycentric co-ordinates of $T \in \mathcal{T}_h$. We then define the *triangle-bubble function* b_T by

$$b_T = \begin{cases} 27\lambda_{T,1}\lambda_{T,2}\lambda_{T,3} & \text{on } T \\ 0 & \text{on } \Omega \setminus T. \end{cases} \tag{1.5}$$

Given an $E \in \mathcal{E}_{h,\Omega}$ with $\omega_E = T_1 \cup T_2$ enumerate the vertices of T_1 and T_2 such that the vertices of E are numbered first. We then define the *edge-bubble function* b_E by

$$b_E = \begin{cases} 4\lambda_{T_i,1}\lambda_{T_i,2} & \text{on } T_i,\ i = 1,2 \\ 0 & \text{on } \Omega \setminus \omega_E. \end{cases} \tag{1.6}$$

Definition (1.6) can be extended in an obvious way to boundary edges $E \subset \Gamma$. The following lemma collects some important properties of the bubble functions b_T and b_E. Its proof immediately follows from Definitions (1.5) and (1.6) and from standard scaling arguments for finite element functions.

Lemma 1.3. *Let $T \in \mathcal{T}_h$ and $E \in \mathcal{E}_h$ be arbitrary. The functions b_T and b_E have the following properties:*

$$\begin{aligned}
&\operatorname{supp} b_T \subset T, \quad 0 \le b_T \le 1, \quad \max_{x \in T} b_T(x) = 1, \\
&\operatorname{supp} b_E \subset \omega_E, \quad 0 \le b_E \le 1, \quad \max_{x \in E} b_E(x) = 1,
\end{aligned}$$

$$\begin{aligned}
c_1 h_T^2 \le \int_T b_T = \frac{9}{20}|T| \le c_2 h_T^2, \\
\int_E b_E = \frac{2}{3} h_E, \\
c_3 h_E^2 \le \int_{T'} b_E = \frac{1}{3}|T'| \le c_4 h_E^2, \quad \forall T' \subset \omega_E, \\
\|\nabla b_T\|_{0,2;T} \le c_5 h_T^{-1} \|b_T\|_{0,2;T}, \\
\|\nabla b_E\|_{0,2;T'} \le c_6 h_E^{-1} \|b_E\|_{0,2;T'} \quad \forall T' \subset \omega_E.
\end{aligned}$$

Here, $|T|$ denotes the area of T. The constants $c_1, \ldots, c_6$ only depend on the smallest angle in the triangulations. □

Finally, we denote by $I_h : X \to X_h$ the interpolation operator of Clément [38] which is defined as follows. Denote by Π_1 the space of polynomials of degree at most 1. Given $\varphi \in X$ and $x \in \mathcal{N}_h$, det $\pi_x \varphi$ be the $L^2(\omega_x)$-projection of φ onto Π_1, i.e.

$$\int_{\omega_x} \varphi\psi = \int_{\omega_x} \pi_x \varphi \psi \quad \forall \psi \in \Pi_1. \tag{1.7}$$

Then $I_h \varphi$ is uniquely defined by

$$\begin{aligned}
I_h\varphi(x) &= (\pi_x \varphi)(x) \quad \forall x \in \mathcal{N}_{h,\Omega} \cup \mathcal{N}_{h,N}, \\
I_h\varphi(x) &= 0 \qquad\quad \forall x \in \mathcal{N}_{h,D}.
\end{aligned} \tag{1.8}$$

The operator I_h satisfies the following local error estimates (cf. [38] and [37; Exercise 3.2.3]).

Lemma 1.4. *Let $T \in \mathcal{T}_h$, $E \in \mathcal{E}_h$, and $\varphi \in X$ be arbitrary. Then the following error estimates are valid*

$$\|\varphi - I_h\varphi\|_{0,2;T} \le c_7 h_T \|\varphi\|_{1,2;\widetilde{\omega}_T}$$
$$\|\varphi - I_h\varphi\|_{2;E} \le c_8 h_E^{1/2} \|\varphi\|_{1,2;\widetilde{\omega}_E}$$

The constants c_7, c_8 only depend on the smallest angle in the triangulations. □

1.2. A Residual Error Estimator.

Let $u \in X$ and $u_h \in X_h$ be the exact solutions of Problems (1.3) and (1.4), respectively. They satisfy the identity

$$\int_\Omega \nabla(u - u_h) \cdot \nabla v = \int_\Omega f\, v + \int_{\Gamma_N} g\, v - \int_\Omega \nabla u_h \cdot \nabla v \quad \forall v \in X. \tag{1.9}$$

The right-hand side of Equation (1.9) implicitly defines the *residual* of u_h as an element of the dual space of X.

Since Γ_D has a positive length, a Poincaré-Friedrichs inequality holds

$$\|v\|_{0,2} \le c_\Omega \|\nabla v\|_{0,2} \quad \forall v \in X. \tag{1.10}$$

The constant c_Ω only depends on Ω and the length of Γ_D. Inequality (1.10) and Cauchy-Schwarz's inequality imply for all $v \in X$

$$\frac{1}{1 + c_\Omega^2} \|v\|_{1,2} \le \sup_{\substack{w \in X \\ \|w\|_{1,2}=1}} \int_\Omega \nabla v \cdot \nabla w \le \|v\|_{1,2}. \tag{1.11}$$

This corresponds to the fact that the bilinear form $X \ni u, v \to \int_\Omega \nabla u \cdot \nabla v$ defines an isomorphism of X onto its dual space. The constants multiplying the first and last term in Inequality (1.11) are related to the norm of this isomorphism and of its inverse.

Equation (1.9) and Inequality (1.11) imply the estimate

$$\begin{aligned} &\sup_{\substack{w \in X \\ \|w\|_{1,2}=1}} \left\{ \int_\Omega f\, w + \int_{\Gamma_N} g\, w - \int_\Omega \nabla u_h \cdot \nabla w \right\} \\ &\le \|u - u_h\|_{1,2} \\ &\le (1 + c_\Omega^2) \sup_{\substack{w \in X \\ \|w\|_{1,2}=1}} \left\{ \int_\Omega f\, w + \int_{\Gamma_N} g\, w - \int_\Omega \nabla u_h \cdot \nabla w \right\}. \end{aligned} \tag{1.12}$$

Since the sup-term in Inequality (1.12) is equivalent to the norm of the residual in the dual space of X, we have proved: *The norm in X of the error is, up to multiplicative constants, bounded from above and from below by the norm of the residual in the dual space of X.*

Most a posteriori error estimators try to estimate this dual norm of the residual by quantities that can more easily be computed from f, g, and u_h. Here, we first observe that, since $X_h \subset X$, the error is orthogonal to X_h:

$$\int_\Omega \nabla(u - u_h) \cdot \nabla v_h = 0 \quad \forall v_h \in X_h.$$

Using Equation (1.9) this can be written as

$$\int_\Omega f\, v_h + \int_{\Gamma_N} g\, v_h - \int_\Omega \nabla u_h \cdot \nabla v_h = 0 \quad \forall v_h \in X_h. \tag{1.13}$$

Equation (1.13) reflects the fact that the discretization (1.4) is consistent and that no additional errors are introduced by numerical integration or by inexact solution of the discrete problem.

Next, elementwise integration by parts and the observation that $\Delta u_h = 0$ on all $T \in \mathcal{T}_h$ yield for all $v \in X$

$$\begin{aligned}
&\int_\Omega f\, v + \int_{\Gamma_N} g\, v - \int_\Omega \nabla u_h \cdot \nabla v \\
&= \int_\Omega f\, v + \int_{\Gamma_N} g\, v - \sum_{T \in \mathcal{T}_h} \int_T \nabla u_h \cdot \nabla v \\
&= \int_\Omega f\, v + \int_{\Gamma_N} g\, v + \sum_{T \in \mathcal{T}_h} \left\{ \int_T \Delta u_h v - \int_{\partial T} n_T \cdot \nabla u_h v \right\} \\
&= \sum_{T \in \mathcal{T}_h} \int_T f\, v + \sum_{E \in \mathcal{E}_{h,N}} \int_E (g - n_E \cdot \nabla u_h) v - \sum_{E \in \mathcal{E}_{h,\Omega}} \int_E [n_E \cdot \nabla u_h]_E\, v.
\end{aligned} \tag{1.14}$$

Here, n_T denotes the unit exterior normal to $T \in \mathcal{T}_h$.

Consider an arbitrary element $w \in X$. Equations (1.13) and (1.14) and Lemma

1.4 then imply the following estimate

$$
\begin{aligned}
&\int_\Omega f\,w + \int_{\Gamma_N} g\,w - \int_\Omega \nabla u_h \cdot \nabla w \\
&= \sum_{T\in\mathcal{T}_h} \int_T f(w - I_h w) \\
&\quad + \sum_{E\in\mathcal{E}_{h,N}} \int_E (g - n_E\cdot\nabla u_h)(w - I_h w) \\
&\quad - \sum_{E\in\mathcal{E}_{h,\Omega}} \int_E [n_E\cdot\nabla u_h]_E (w - I_h w) \\
&\le \sum_{T\in\mathcal{T}_h} c_7 h_T \|f\|_{0,2;T} \|w\|_{1,2;\widetilde{\omega}_T} \\
&\quad + \sum_{E\in\mathcal{E}_{h,N}} c_8 h_E^{1/2} \|g - n_E\cdot\nabla u_h\|_{2;E} \|w\|_{1,2;\widetilde{\omega}_E} \\
&\quad + \sum_{E\in\mathcal{E}_{h,\Omega}} c_8 h_E^{1/2} \|[n_E\cdot\nabla u_h]_E\|_{2;E} \|w\|_{1,2;\widetilde{\omega}_E} \\
&\le \max\{c_7,c_8\}\Big\{ \sum_{T\in\mathcal{T}_h} h_T^2 \|f\|_{0,2;T}^2 \\
&\qquad\qquad + \sum_{E\in\mathcal{E}_{h,N}} h_E \|g - n_E\cdot\nabla u_h\|_{2;E}^2 \\
&\qquad\qquad + \sum_{E\in\mathcal{E}_{h,\Omega}} h_E \|[n_E\cdot\nabla u_h]_E\|_{2;E}^2 \Big\}^{1/2} \\
&\qquad\qquad \Big\{ \sum_{T\in\mathcal{T}_h} \|w\|_{1,2;\widetilde{\omega}_T}^2 + \sum_{E\in\mathcal{E}_{h,N}\cup\mathcal{E}_{h,\Omega}} \|w\|_{1,2;\widetilde{\omega}_E}^2 \Big\}^{1/2} \\
&\le c_{\mathcal{T}} \max\{c_7,c_8\} \|w\|_{1,2} \Big\{ \sum_{T\in\mathcal{T}_h} h_T^2 \|f\|_{0,2;T}^2 \\
&\qquad\qquad + \sum_{E\in\mathcal{E}_{h,N}} h_E \|g - n_E\cdot\nabla u_h\|_{2;E}^2 \\
&\qquad\qquad + \sum_{E\in\mathcal{E}_{h,\Omega}} h_E \|[n_E\cdot\nabla u_h]_E\|_{2;E}^2 \Big\}^{1/2}, \qquad (1.15)
\end{aligned}
$$

where $c_{\mathcal{T}}$ depends on the smallest angle in the triangulation. Combining Estimates

(1.12) and (1.15), we get

$$\|u-u_h\|_{1,2} \le c\Big\{ \sum_{T\in\mathcal{T}_h} h_T^2\|f\|_{0,2;T}^2 + \sum_{E\in\mathcal{E}_{h,N}} h_E\|g-n_E\cdot\nabla u_h\|_{2;E}^2 + \sum_{E\in\mathcal{E}_{h,\Omega}} h_E\|[n_E\cdot\nabla u_h]_E\|_{2;E}^2\Big\}^{1/2}, \tag{1.16}$$

where $c := c_{\mathcal{T}}\max\{c_7,c_8\}(1+c_\Omega^2)$. The right-hand side of Equation (1.16) can be used as an a posteriori error estimator since it only involves the known data f and g, the solution u_h of the discrete problem, and the geometrical data of the triangulation. For general functions f and g, however, the exact evaluation of the integrals occurring on the right-hand side of Equation (1.16) might be prohibitively expensive or even impossible. They must then be approximated by suitable quadrature formulae. The arguments given below and the general results of Section 3.1 on the other hand show that it is desirable to approximate f and g by functions in suitable finite dimensional spaces. Both approaches – numerical quadrature and approximation by simpler functions combined with exact integration of the latter – are often equivalent and yield similar a posteriori error estimators. Here, we restrict ourselves to one of the simplest approximations and replace f and g by their L^2-projections onto the spaces of piecewise constant functions with respect to $\mathcal{T}_h$ and $\mathcal{E}_{h,N}$, respectively. To this end we define for $T\in\mathcal{T}_h$ and $E\in\mathcal{E}_{h,N}$

$$f_T := \frac{1}{|T|}\int_T f, \quad g_E = h_E^{-1}\int_E g. \tag{1.17}$$

A residual a posteriori error estimator is then given by (cf., e. g., [12, 13, 78, 85])

$$\eta_{R,T} := \Big\{ h_T^2\|f_T\|_{0,2;T}^2 + \frac{1}{2}\sum_{E\in\mathcal{E}(T)\cap\mathcal{E}_{h,\Omega}} h_E\|[n_E\cdot\nabla u_h]_E\|_{2;E}^2 + \sum_{E\in\mathcal{E}(T)\cap\mathcal{E}_{h,N}} h_E\|g_E-n_E\cdot\nabla u_h\|_{2,E}^2\Big\}^{1/2}. \tag{1.18}$$

Observing that interior edges are counted twice, Equations (1.16), (1.18), and the

triangle inequality imply

$$\|u-u_h\|_{1,2} \le c\Big\{ \sum_{T\in\mathcal{T}_h} \eta_{R,T}^2 + \sum_{T\in\mathcal{T}_h} h_T^2 \|f-f_T\|_{0,2;T}^2 + \sum_{E\in\mathcal{E}_{h,N}} h_E \|g-g_E\|_{2;E}^2 \Big\}^{1/2}. \tag{1.19}$$

We want to show that, in a certain sense, the converse of Inequality (1.19) is also true. Therefore, we first consider an arbitrary $T \in \mathcal{T}_h$ and define

$$w_T := f_T b_T. \tag{1.20}$$

Lemma 1.3 implies

$$\int_T f_T w_T = \frac{9}{20}|T|\,|f_T|^2 = \frac{9}{20}\|f_T\|_{0,2;T}^2. \tag{1.21}$$

Since $\operatorname{supp} w_T \subset T$, Equations (1.9), (1.14), and Lemma 1.3 on the other hand yield

$$\begin{aligned}
\int_T f_T w_T &= \int_T f\, w_T + \int_T (f_T - f) w_T \\
&= \int_\Omega f\, w_T + \int_{\Gamma_N} g\, w_T - \int_\Omega \nabla u_h \cdot \nabla w_T + \int_T (f_T - f) w_T \\
&= \int_T \nabla(u-u_h)\cdot\nabla w_T + \int_T (f_T - f) w_T \\
&\le \|u-u_h\|_{1,2;T} \|\nabla w_T\|_{0,2;T} + \|f-f_T\|_{0,2;T} \|w_T\|_{0,2;T} \\
&\le \|u-u_h\|_{1,2;T} c_5 h_T^{-1} |f_T|\, \|b_T\|_{0,2;T} + \|f-f_T\|_{0,2;T} |f_T|\, \|b_T\|_{0,2;T} \\
&\le \|u-u_h\|_{1,2;T} c_5 h_T^{-1} |f_T| \Big\{ \int_T b_T \Big\}^{1/2} + \|f-f_T\|_{0,2;T} |f_T| \Big\{ \int_T b_T \Big\}^{1/2} \\
&\le \sqrt{\frac{9}{20}} \|f_T\|_{0,2;T} \Big\{ c_5 h_T^{-1} \|u-u_h\|_{1,2;T} + \|f-f_T\|_{0,2;T} \Big\}.
\end{aligned} \tag{1.22}$$

Combining Inequalities (1.21) and (1.22), we obtain

$$\|f_T\|_{0,2;T} \le \sqrt{\frac{20}{9}} c_5 h_T^{-1} \|u-u_h\|_{1,2;T} + \sqrt{\frac{20}{9}} \|f-f_T\|_{0,2;T}. \tag{1.23}$$

Next, we consider an arbitrary $E \in \mathcal{E}_{h,\Omega}$ and define

$$w_E := [n_E \cdot \nabla u_h]_E\, b_E. \tag{1.24}$$

Lemma 1.3 implies

$$\int_E [n_E \cdot \nabla u_h]_E w_E = \frac{2}{3} h_E |[n_E \cdot \nabla u_h]_E|^2 = \frac{2}{3} \|[n_E \cdot \nabla u_h]_E\|_{2;E}^2. \tag{1.25}$$

Since $\operatorname{supp} w_E \subset \omega_E$, Equations (1.9), (1.14), Lemma 1.3, and Inequality (1.23) on the other hand yield

$$\begin{aligned}
&\int_E [n_E \cdot \nabla u_h]_E \, w_E \\
&= \sum_{T' \subset \omega_E} \int_{T'} f \, w_E - \int_\Omega f \, w_E - \int_{\Gamma_N} g \, w_E + \int_\Omega \nabla u_h \cdot \nabla w_E \\
&= \sum_{T' \subset \omega_E} \int_{T'} f \, w_E - \int_{\omega_E} \nabla(u - u_h) \cdot \nabla w_E \\
&\le \|f\|_{0,2;\omega_E} \|w_E\|_{0,2;\omega_E} + \|u - u_h\|_{1,2;\omega_E} \|\nabla w_E\|_{0,2;\omega_E} \\
&\le \|f\|_{0,2;\omega_E} |[n_E \cdot \nabla u_h]_E| \, \|b_E\|_{0,2;\omega_E} \\
&\qquad + \|u - u_h\|_{1,2;\omega_E} |[n_E \cdot \nabla u_h]_E| c_6 h_E^{-1} \|b_E\|_{0,2;\omega_E} \\
&\le \|f\|_{0,2;\omega_E} |[n_E \cdot \nabla u_h]_E| \left\{ \int_{\omega_E} b_E \right\}^{1/2} \\
&\qquad + \|u - u_h\|_{1,2;\omega_E} |[n_E \cdot \nabla u_h]_E| c_6 h_E^{-1} \left\{ \int_{\omega_E} b_E \right\}^{1/2} \\
&\le \sqrt{2c_4} \|[n_E \cdot \nabla u_h]_E\|_{2;E} \{ \, h_E^{1/2} \|f\|_{0,2;\omega_E} + c_6 h_E^{-1/2} \|u - u_h\|_{1,2;\omega_E} \, \} \\
&\le \sqrt{2c_4} \|[n_E \cdot \nabla u_h]_E\|_{2;E} \Big\{ h_E^{1/2} \sum_{T' \subset \omega_E} \|f - f_{T'}\|_{0,2;T'} \\
&\qquad\qquad + h_E^{1/2} \sum_{T' \subset \omega_E} \|f_{T'}\|_{0,2;T'} \\
&\qquad\qquad + c_6 h_E^{-1/2} \|u - u_h\|_{1,2;\omega_E} \Big\} \\
&\le c \|[n_E \cdot \nabla u_h]_E\|_{2;E} \Big\{ h_E^{1/2} \sum_{T' \subset \omega_E} \|f - f_{T'}\|_{0,2;T'} \\
&\qquad\qquad + h_E^{-1/2} \|u - u_h\|_{1,2;\omega_E} \Big\},
\end{aligned} \tag{1.26}$$

where

$$c = \sqrt{2c_4} \max\left\{ 1 + \sqrt{\frac{20}{9}}, c_6 + \sqrt{\frac{20}{9}} c_5 \max_{T' \subset \omega_E} h_E / h_{T'} \right\}.$$

Combining Inequalities (1.25) and (1.26) we obtain

$$\|[n_E \cdot \nabla u_h]_E\|_{2;E} \leq \frac{3}{2} c \left\{ h_E^{1/2} \sum_{T' \subset \omega_E} \|f - f_{T'}\|_{0,2;T'} + h_E^{-1/2} \|u - u_h\|_{1,2;\omega_E} \right\}. \quad (1.27)$$

Finally, we consider an arbitrary $E \in \mathcal{E}_{h,N}$ and define

$$w_E := (g_E - n_E \cdot \nabla u_h) b_E. \quad (1.28)$$

With the same arguments as above, we now obtain

$$\int_E (g_E - n_E \cdot \nabla u_h) w_E = \frac{2}{3} \|g_E - n_E \cdot \nabla u_h\|_{2;E}^2 \quad (1.29)$$

and

$$\begin{aligned}
&\int_E (g_E - n_E \cdot \nabla u_h) w_E \\
&= \int_E (g - n_E \cdot \nabla u_h) w_E + \int_E (g_E - g) w_E \\
&= \int_\Omega f\, w_E + \int_{\Gamma_N} g\, w_E - \int_\Omega \nabla u_h \cdot \nabla w_E - \int_{\omega_E} f\, w_E + \int_E (g_E - g) w_E \\
&= \int_{\omega_E} \nabla (u - u_h) \cdot \nabla w_E - \int_{\omega_E} f\, w_E + \int_E (g_E - g) w_E \\
&\leq \|g_E - n_E \cdot \nabla u_h\|_{2;E} \Big\{ \sqrt{c_4} c_6 h_E^{-1/2} \|u - u_h\|_{1,2;\omega_E} \\
&\qquad + \sqrt{c_4} h_E^{1/2} \|f\|_{0,2;\omega_E} + \sqrt{\frac{2}{3}} h_E^{1/2} \|g - g_E\|_{2;E} \Big\} \\
&\leq c' \|g_E - n_E \cdot \nabla u_h\|_{2;E} \{ h_E^{-1/2} \|u - u_h\|_{1,2;\omega_E} + h_E^{1/2} \|f - f_{T'}\|_{0,2;T'} \\
&\qquad + h_E^{1/2} \|g - g_E\|_{2;E} \}, \qquad (1.30)
\end{aligned}$$

where $T' = \omega_E$ and

$$c' = \max \left\{ \sqrt{\frac{2}{3}}, \sqrt{c_4} \left(1 + \sqrt{\frac{20}{9}} \right), \sqrt{c_4} \left(c_6 + \sqrt{\frac{20}{9}} c_5 h_E / h_{T'} \right) \right\}.$$

Combining Inequalities (1.29) and (1.30) we obtain

$$\begin{aligned}
\|g_E - n_E \cdot \nabla u_h\|_{2;E} \leq \frac{3}{2} c' \{ & h_E^{-1/2} \|u - u_h\|_{1,2;\omega_E} \\
& + h_E^{1/2} \|f - f_{T'}\|_{0,2;T'} + h_E^{1/2} \|g - g_E\|_{2;E} \}. \quad (1.31)
\end{aligned}$$

Collecting the estimates (1.19), (1.23), (1.27), and (1.31) proves:

Proposition 1.5. *Let u and u_h be the exact solutions of Problems (1.3) and (1.4). There are two constants $\underline{c}$, $\bar{c}$, which only depend on the smallest angle in the triangulation, such that the estimates*

$$\|u-u_h\|_{1,2} \leq \bar{c}\Big\{\sum_{T\in\mathcal{T}_h}\eta_{R,T}^2+\sum_{T\in\mathcal{T}_h}h_T^2\|f-f_T\|_{0,2;T}^2+\sum_{E\in\mathcal{E}_{h,N}}h_E\|g-g_E\|_{2;E}^2\Big\}^{1/2}$$

and

$$\eta_{R,T} \leq \underline{c}\Big\{\|u-u_h\|_{1,2;\omega_T}^2 + \sum_{T'\subset\omega_T}h_{T'}^2\|f-f_{T'}\|_{0,2;T'}^2+\sum_{E\in\mathcal{E}(T)\cap\mathcal{E}_{h,N}}h_E\|g-g_E\|_{2;E}^2\Big\}^{1/2}$$

hold for all $T\in\mathcal{T}_h$. Here, f_T, g_E, and $\eta_{R,T}$ are given by Equations (1.17) and (1.18). □

Remark 1.6. The first term in $\eta_{R,T}$ is related to the residual of u_h with respect to the strong form of the differential equation. The second and third term in $\eta_{R,T}$ are related to that boundary operator which is canonically associated with the strong and weak form of the differential equation. These boundary terms are crucial when considering low order finite element discretizations as done here. Consider e. g. Problem (1.1) in the unit square $(0,1)^2$ with Dirichlet boundary conditions on the left and bottom part and exact solution $u(x)=x_1x_2$. When using a triangulation consisting of right angled isosceles triangles and evaluating the line integrals by the trapezoidal rule, the solution of Problem (1.4) satisfies $u_h(x)=u(x)\ \forall x\in\mathcal{N}_h$ but $u_h\neq u$. The second and third term in $\eta_{R,T}$ reflect the fact that $u_h\notin H^2(\Omega)$ and that u_h does not exactly satisfy the Neumann boundary condition. □

Remark 1.7. The correction terms $h_T\|f-f_T\|_{0,2;T}$ and $h_E^{1/2}\|g-g_E\|_{2;E}$ in the estimates of Proposition 1.5 are in general higher order perturbations of the other terms. In special situations, however, they can be dominant. To see this, choose a triangle $T_0\in\mathcal{T}_h$ and a non-zero function $\varrho_0\in C_0^\infty(\dot{T}_0)$ and consider Problem (1.1) with $f:=-\Delta\varrho_0$ and $\Gamma_D=\Gamma$. Since

$$\int_\Omega f\,v_h=-\int_T\Delta\varrho_0 v_h=-\int_T\varrho_0\Delta v_h=0\quad\forall v_h\in X_h,$$

we have $f_T=0\ \forall T\in\mathcal{T}_h$ and the exact solution of Problem (1.4) is $u_h=0$. Hence, we have $\eta_{R,T}=0\ \forall T\in\mathcal{T}_h$, but $\|u-u_h\|_{1,2}\neq 0$. This effect is not restricted to the

particular approximation of f considered here. Since $\varrho_0 \in C_0^\infty(\dot{T}_0)$ is completely arbitrary, we will always encounter similar difficulties as long as we do not evaluate $\|f\|_{0,2;T}$ exactly – which in general is impossible. Obviously, this problem is cured when further refining the mesh. □

Remark 1.8. The error estimator $\eta_{R,T}$ was first proposed and analyzed for Problem (1.1) in one dimension in [12, 13]. It was generalized to mixed finite element approximations of the Stokes and Navier-Stokes equations in [78, 80]. The idea of using suitable local test-functions for establishing local lower bounds was introduced in [78]. The present analysis is a slight modification of the one given in [85] and takes into account the abstract framework of Section 2 and of [86]. A quite different analysis is given in [9]. □

1.3. Error Estimators Based on the Solution of Local Problems.

The results of the previous section show that we must reliably estimate the norm of the residual as an element of the dual space of X. This could be achieved by lifting the residual to a suitable subspace of X by solving auxiliary problems similar to, but simpler than the original discrete Problem (1.4). Practical considerations and the results of the previous section suggest that the auxiliary problems should satisfy the following conditions:

(1) In order to get an information on the local behaviour of the error, they should involve only small subdomains of Ω.
(2) In order to yield an accurate information on the error, they should be based on finite element spaces which are more accurate than the original one.
(3) In order to keep the computational work at a minimum, they should involve as few degrees of freedom as possible.
(4) To each edge and to each triangle there should correspond at least one degree of freedom in at least one of the auxiliary problems.
(5) The solution of all auxiliary problems should not cost more than the assembly of the stiffness matrix of Problem (1.4).

There are many possible ways to satisfy these conditions. Here, we present three of them.

First, we decide to impose Dirichlet boundary conditions on the auxiliary problems. Condition (4) then implies that the corresponding subdomains must consist of more than one triangle. A reasonable choice is to consider all nodes $x \in \mathcal{N}_{h,\Omega} \cup \mathcal{N}_{h,N}$ and the corresponding domains ω_x. The above conditions then lead to the following

definition: Let for all $x \in \mathcal{N}_{h,\Omega} \cup \mathcal{N}_{h,N}$

$$V_x := \text{span}\{\, b_T, b_E, b_e : x \in \mathcal{N}(T); x \in \mathcal{N}(E), E \in \mathcal{E}_{h,\Omega}; e \subset \partial\omega_x \cap \Gamma_N \,\} \tag{1.32}$$

and

$$\eta_{D,x} := \|\nabla v_x\|_{0,2;\omega_x} \tag{1.33}$$

where $v_x \in V_x$ is the unique solution of

$$\int_{\omega_x} \nabla v_x \cdot \nabla w = \sum_{x \in \mathcal{N}(T)} \int_T f_T w + \sum_{e \subset \partial\omega_x \cap \Gamma_N} \int_e g_e w - \int_{\omega_x} \nabla u_h \cdot \nabla w \quad \forall w \in V_x. \tag{1.34}$$

In order to get a different interpretation of Problem (1.34), set

$$u_x := u_h + v_x.$$

Then

$$\eta_{D,x} = \|\nabla(u_x - u_h)\|_{0,2;\omega_x}$$

and $u_x \in u_h + V_x$ is the unique solution of

$$\int_{\omega_x} \nabla u_x \cdot \nabla w = \sum_{x \in \mathcal{N}(T)} \int_T f_T w + \sum_{e \subset \partial\omega_x \cap \Gamma_N} \int_e g_e w \quad \forall w \in V_x.$$

This is a discrete analogue of the following Dirichlet problem

$$\begin{aligned} -\Delta\varphi &= f \quad \text{in} \quad \omega_x \\ \varphi &= u_h \quad \text{on} \quad \partial\omega_x \setminus \Gamma_N \\ \frac{\partial\varphi}{\partial n} &= g \quad \text{on} \quad \partial\omega_x \cap \Gamma_N. \end{aligned}$$

Hence, we can interpret the error estimator $\eta_{D,x}$ in two ways:

(1) We solve a local analogue of the residual equation (1.9) using a higher order finite element approximation and use a suitable norm of the solution as error estimator.

(2) We solve a local, discrete analogue of the original problem (1.1) using a higher order finite element space and compare the solution of this problem to the one of Problem (1.4).

Thus, in a certain sense, $\eta_{D,x}$ is based on an extrapolation technique. The following proposition shows that it yields upper and lower bounds on the error $u - u_h$ and that it is comparable to the estimator $\eta_{R,T}$.

Proposition 1.9. *Let u and u_h be the unique solutions of Problems (1.3) and (1.4). There are constants $c_{\mathcal{N},1}, \ldots, c_{\mathcal{N},4}$, which only depend on the smallest angle in the triangulation, such that the estimates*

$$\eta_{D,x} \le c_{\mathcal{N},1} \left\{ \sum_{x \in \mathcal{N}(T)} \eta_{R,T}^2 \right\}^{1/2},$$

$$\eta_{R,T} \le c_{\mathcal{N},2} \left\{ \sum_{x \in \mathcal{N}(T) \setminus \mathcal{N}_{h,D}} \eta_{D,x}^2 \right\}^{1/2},$$

$$\eta_{D,x} \le c_{\mathcal{N},3} \left\{ \|u - u_h\|_{1,2;\omega_x}^2 + \sum_{x \in \mathcal{N}(T)} h_T^2 \|f - f_T\|_{0,2;T}^2 + \sum_{e \subset \partial\omega_x \cap \Gamma_N} h_e \|g - g_e\|_{2;e}^2 \right\}^{1/2},$$

$$\|u - u_h\|_{1,2} \le c_{\mathcal{N},4} \left\{ \sum_{x \in \mathcal{N}_{h,\Omega} \cup \mathcal{N}_{h,N}} \eta_{D,x}^2 + \sum_{T \in \mathcal{T}_h} h_T^2 \|f - f_T\|_{0,2;T}^2 + \sum_{E \in \mathcal{E}_{h,N}} h_E \|g - g_E\|_{2;E}^2 \right\}^{1/2}$$

hold for all $x \in \mathcal{N}_{h,\Omega} \cup \mathcal{N}_{h,N}$ and all $T \in \mathcal{T}_h$. Here, f_T, g_E, $\eta_{R,T}$, and $\eta_{D,x}$ are given by Equations (1.17), (1.18), and (1.33).

PROOF: Since the functions in V_x vanish at all nodes in ω_x, standard scaling arguments for finite elements imply

$$\|\varphi\|_{0,2;T} \le c_{I1} h_T \|\nabla\varphi\|_{0,2;T}, \tag{1.35}$$

$$\|\varphi\|_{2;E} \le c_{I2} h_E^{1/2} \|\nabla\varphi\|_{0,2;T} \tag{1.36}$$

for all $\varphi \in V_x$, $x \in \mathcal{N}_{h,\Omega} \cup \mathcal{N}_{h,N}$, $T \subset \omega_x$, $E \in \mathcal{E}(T) \setminus \mathcal{E}_{h,D}$, where c_{I1} and c_{I2} only depend on the smallest angle in the triangulation.
Consider an arbitrary $x \in \mathcal{N}_{h,\Omega} \cup \mathcal{N}_{h,N}$. Equations (1.33)–(1.36) and (1.18) and

elementwise integration by parts then yield

$$\begin{aligned}
\eta_{D,x}^2 &= \sum_{x\in\mathcal{N}(T)} \int_T f_T v_x + \sum_{e\subset\partial\omega_x\cap\Gamma_N} \int_e g_e v_x - \sum_{x\in\mathcal{N}(T)} \int_T \nabla u_h \cdot \nabla v_x \\
&= \sum_{x\in\mathcal{N}(T)} \int_T f_T v_x - \sum_{x\in\mathcal{N}(E),E\in\mathcal{E}_{h,\Omega}} \int_E [n_E \cdot \nabla u_h]_E \, v_x \\
&\quad + \sum_{e\subset\partial\omega_x\cap\Gamma_N} \int_e (g_e - n_e \cdot \nabla u_h) v_x \\
&\le c_{I1} \sum_{x\in\mathcal{N}(T)} h_T \|f_T\|_{0,2;T} \|\nabla v_x\|_{0,2;T} \\
&\quad + c_{I2} \sum_{x\in\mathcal{N}(E),E\in\mathcal{E}_{h,\Omega}} h_E^{1/2} \|[n_E \cdot \nabla u_h]_E\|_{2;E} \|\nabla v_x\|_{0,2;\omega_E} \\
&\quad + c_{I2} \sum_{e\subset\partial\omega_x\cap\Gamma_N} h_e^{1/2} \|g_e - n_e \cdot \nabla u_h\|_{2;e} \|\nabla v_x\|_{0,2;\omega_e} \\
&\le 2\max\{c_{I1}, c_{I2}\} \Big\{ \sum_{x\in\mathcal{N}(T)} \eta_{R,T}^2 \Big\}^{1/2} \eta_{D,x}.
\end{aligned}$$

This proves the first inequality.
Next, we observe that the functions w_T and w_E defined in Equations (1.20), (1.24), and (1.28) are contained in V_x whenever $x \in \mathcal{N}(T)$ or $x \in \mathcal{N}(E)$. Lemma 1.3 and Equations (1.21), (1.33), and (1.34) therefore imply for all $T \in \mathcal{T}_h$ and all $x \in \mathcal{N}(T)$

$$\begin{aligned}
\frac{9}{20} \|f_T\|_{0,2;T}^2 &= \int_T f_T w_T \\
&= \int_T \nabla v_x \cdot \nabla w_T + \int_T \nabla u_h \cdot \nabla w_T \\
&= \int_{\omega_x} \nabla v_x \cdot \nabla w_T \\
&\le \eta_{D,x} |f_T| \, \|\nabla b_T\|_{0,2;T} \\
&\le \eta_{D,x} |f_T| c_5 h_T^{-1} \|b_T\|_{0,2;T} \\
&\le \sqrt{\frac{9}{20}} c_5 h_T^{-1} \|f_T\|_{0,2;T} \eta_{D,x}
\end{aligned}$$

and thus

$$h_T \|f_T\|_{0,2;T} \le \sqrt{\frac{20}{9}} c_5 \eta_{D,x}. \tag{1.37}$$

Similarly, Lemma 1.3 and Equations (1.25), (1.33), (1.34), and (1.37) yield for all $E \in \mathcal{E}_{h,\Omega}$ and all $x \in \mathcal{N}(E)$

$$\begin{aligned}
\frac{2}{3}\|[n_E \cdot \nabla u_h]_E\|_{2;E}^2 &= \int_E [n_E \cdot \nabla u_h]_E \, w_E \\
&= \int_{\omega_E} \nabla u_h \cdot \nabla w_E \\
&= \sum_{T \subset \omega_E} \int_T f_T w_E - \int_{\omega_T} \nabla v_x \cdot \nabla w_E \\
&\leq \sum_{T \subset \omega_E} \|f_T\|_{0,2;T} \|w_E\|_{0,2;T} + \eta_{D,x} \|\nabla w_E\|_{0,2;\omega_E} \\
&\leq \sum_{T \in \omega_E} \{ \|f_T\|_{0,2;T} |[n_E \cdot \nabla u_h]_E| c_4^{1/2} h_E \\
&\qquad + \eta_{D,x} |[n_E \cdot \nabla u_h]_E| c_6 c_4^{1/2} \} \\
&\leq \|[n_E \cdot \nabla u_h]_E\|_{2;E} \Big\{ 2 c_6 c_4^{1/2} \eta_{D,x} h_E^{-1/2} \\
&\qquad + \sum_{T \subset \omega_E} c_4^{1/2} h_E^{1/2} \|f_T\|_{0,2;T} \Big\} \\
&\leq c \, h_E^{-1/2} \eta_{D,x} \|[n_E \cdot \nabla u_h]_E\|_{2;E}
\end{aligned}$$

where

$$c = 2\sqrt{c_4} \Big\{ c_6 + c_5 \sqrt{\frac{20}{9}} \max_{T \subset \omega_E} h_E / h_T \Big\}.$$

Hence, we have

$$h_E^{1/2} \|[n_E \cdot \nabla u_h]_E\|_{2;E} \leq \frac{3}{2} c \, \eta_{D,x}. \tag{1.38}$$

With the same arguments, we conclude that

$$h_E^{1/2} \|g_E - n_E \cdot \nabla u_h\|_{2;E} \leq \frac{3}{2} c \, \eta_{D,x} \tag{1.39}$$

holds with a different constant c for all $E \in \mathcal{E}_{h,N}$ and all $x \in \mathcal{N}(E)$. Combining Inequalities (1.37)–(1.39) and using the definition (1.18) of $\eta_{R,T}$, we immediately obtain the second estimate of the proposition.

Consider once more an arbitrary $x \in \mathcal{N}_{h,\Omega} \cup \mathcal{N}_{h,N}$. Equations (1.33)–(1.36) and

(1.9) then imply

$$\begin{aligned}\eta_{D,x}^2 &= \int_{\omega_x} \nabla(u-u_h)\cdot\nabla v_x + \sum_{x\in\mathcal{N}(T)} \int_T (f_T - f)v_x + \sum_{e\subset\partial\omega_x\cap\Gamma_N} \int_e (g_e - g)v_x \\ &\le \|u-u_h\|_{1,2;\omega_x}\,\eta_{D,x} \\ &\quad + \sum_{x\in\mathcal{N}(T)} c_{I1} h_T \|f-f_T\|_{0,2;T}\,\eta_{D,x} + \sum_{e\subset\partial\omega_x\cap\Gamma_N} c_{I2} h_e^{1/2}\|g-g_e\|_{2;e}\,\eta_{D,x}.\end{aligned}$$

This proves the third estimate of the proposition.
The fourth estimate of Proposition 1.9 finally follows from its second estimate and the first estimate of Proposition 1.5. □

We now consider an estimator which is a slight variation of the preceding one. Instead of all $x \in \mathcal{N}_{h,\Omega} \cup \mathcal{N}_{h,N}$ and the corresponding domains ω_x we consider all $T \in \mathcal{T}_h$ and the corresponding sets ω_T. Conditions (1)–(5) from above then lead to the following definition: Let for all $T \in \mathcal{T}_h$

$$\widetilde{V}_T := \operatorname{span}\{\, b_{T'}, b_E, b_e : T' \subset \omega_T; E \in \mathcal{E}(T)\cap\mathcal{E}_{h,\Omega}; e \subset \partial\omega_T \cap \Gamma_N \,\} \tag{1.40}$$

and

$$\eta_{D,T} := \|\nabla\widetilde{v}_T\|_{0,2;\omega_T} \tag{1.41}$$

where $\widetilde{v}_T \in \widetilde{V}_T$ is the unique solution of

$$\begin{aligned}\int_{\omega_T} \nabla\widetilde{v}_T\cdot\nabla w &= \sum_{T'\subset\omega_T} \int_{T'} f_{T'} w + \sum_{e\subset\partial\omega_T\cap\Gamma_N} \int_e g_e w \\ &\quad - \int_{\omega_T} \nabla u_h\cdot\nabla w \quad \forall w \in \widetilde{V}_T.\end{aligned} \tag{1.42}$$

As before we can interprete $u_h + \widetilde{v}_T$ as an approximate solution of the following Dirichlet problem

$$\begin{aligned} -\Delta\varphi &= f \quad \text{in} \quad \omega_T \\ \varphi &= u_h \quad \text{on} \quad \partial\omega_T \setminus \Gamma_N \\ \frac{\partial\varphi}{\partial n} &= g \quad \text{on} \quad \partial\omega_T \cap \Gamma_N. \end{aligned}$$

The following proposition shows that $\eta_{D,T}$ yields upper and lower bounds on the error $u - u_h$ and that it is comparable to $\eta_{D,x}$ and $\eta_{R,T}$. Its proof is completely analogous to the one of Proposition 1.9 and is therefore left to the reader.

Proposition 1.10. *Let u and u_h be the unique solutions of Problem (1.3) and (1.4). There are constants $c_{T,1}, \dots, c_{T,4}$, which only depend on the smallest angle in the triangulation, such that the estimates*

$$\eta_{D,T} \le c_{T,1} \Big\{ \sum_{T' \subset \omega_T} \eta_{R,T'}^2 \Big\}^{1/2},$$

$$\eta_{R,T} \le c_{T,2} \Big\{ \sum_{T' \subset \omega_T} \eta_{D,T'}^2 \Big\}^{1/2},$$

$$\eta_{D,T} \le c_{T,3} \Big\{ \|u - u_h\|_{1,2;\omega_T}^2 + \sum_{T' \subset \omega_T} h_{T'}^2 \|f - f_{T'}\|_{0,2;T'}^2 + \sum_{e \subset \partial\omega_T \cap \Gamma_N} h_e \|g - g_e\|_{2;e}^2 \Big\}^{1/2},$$

$$\|u - u_h\|_{1,2} \le c_{T,4} \Big\{ \sum_{T \in \mathcal{T}_h} \eta_{D,T}^2 + \sum_{T \in \mathcal{T}_h} h_T^2 \|f - f_T\|_{0,2;T}^2 + \sum_{E \in \mathcal{E}_{h,N}} h_E \|g - g_E\|_{2;E}^2 \Big\}^{1/2}$$

hold for all $T \in \mathcal{T}_h$. Here, f_T, g_E, $\eta_{R,T}$, and $\eta_{D,T}$ are defined by Equations (1.17), (1.18), and (1.41). □

For the third estimator we decide to impose Neumann boundary conditions on the auxiliary problems. Now it is possible to choose the triangles in $\mathcal{T}_h$ as the corresponding subdomains. Conditions (1)–(5) from above then lead to the definition: Let for alle $T \in \mathcal{T}_h$

$$V_T := \operatorname{span}\{ b_T, b_E : E \in \mathcal{E}(T) \setminus \mathcal{E}_{h,D} \} \tag{1.43}$$

and

$$\eta_{N,T} := \|\nabla v_T\|_{0,2;T} \tag{1.44}$$

where v_T is the unique solution of

$$\int_T \nabla v_T \cdot \nabla w = \int_T f_T w - \frac{1}{2} \sum_{E \in \mathcal{E}(T) \cap \mathcal{E}_{h,\Omega}} \int_E [n_E \cdot \nabla u_h]_E \, w + \sum_{E \in \mathcal{E}(T) \cap \mathcal{E}_{h,N}} \int_E (g_E - n_E \cdot \nabla u_h) w \quad \forall w \in V_T. \tag{1.45}$$

Problem (1.45) can be interpreted as a discrete analogue of the following Neumann problem

$$\begin{aligned} -\Delta\varphi &= f && \text{in} \quad T \\ \varphi &= 0 && \text{on} \quad \partial T \cap \Gamma_D \\ \frac{\partial\varphi}{\partial n} &= R(u_h) && \text{on} \quad \partial T \setminus \Gamma_D \end{aligned}$$

where

$$R(u_h)|_E := \begin{cases} -\frac{1}{2}[n_E \cdot \nabla u_h]_E & \text{if } E \in \mathcal{E}_{h,\Omega} \\ g - n_E \cdot \nabla u_h & \text{if } E \in \mathcal{E}_{h,N}. \end{cases}$$

The following proposition shows that $\eta_{N,T}$ yields upper and lower bounds on the error and that it is comparable to $\eta_{R,T}$.

Proposition 1.11. *Let u and u_h be the unique solutions of Problems (1.3) and (1.4). There are constants $c_{T,5}, \ldots, c_{T,8}$, which only depend on the smallest angle in the triangulation, such that the estimates*

$$\begin{aligned} \eta_{N,T} &\le c_{T,5}\, \eta_{R,T}, \\ \eta_{R,T} &\le c_{T,6} \Big\{ \sum_{T' \subset \omega_T} \eta_{N,T'}^2 \Big\}^{1/2}, \\ \eta_{N,T} &\le c_{T,7} \Big\{ \|u - u_h\|_{1,2;\omega_T}^2 + \sum_{T' \subset \omega_T} h_{T'}^2 \|f - f_{T'}\|_{0,2;T'}^2 \\ &\qquad + \sum_{E \in \mathcal{E}(T) \cap \mathcal{E}_{h,N}} h_E \|g - g_E\|_{2;E}^2 \Big\}^{1/2}, \\ \|u - u_h\|_{1,2} &\le c_{T,8} \Big\{ \sum_{T \in \mathcal{T}_h} \eta_{N,T}^2 + \sum_{T \in \mathcal{T}_h} h_T^2 \|f - f_T\|_{0,2;T}^2 \\ &\qquad + \sum_{E \in \mathcal{E}_{h,N}} h_E \|g - g_E\|_{2;E}^2 \Big\}^{1/2} \end{aligned}$$

hold for all $T \in \mathcal{T}_h$. Here, f_T, g_E, $\eta_{R,T}$, and $\eta_{N,T}$ are given by Equations (1.17), (1.18), and (1.44).

PROOF: Since the functions in V_T vanish at the vertices of T and have the same polynomial degree as the functions in V_x, Inequalities (1.35) and (1.36) remain

valid. Together with Equations (1.18) and (1.44) they imply

$$\begin{aligned}
\eta_{N,T}^2 &= \int_T f_T v_T - \frac{1}{2} \sum_{E \in \mathcal{E}(T) \cap \mathcal{E}_{h,\Omega}} \int_E [n_E \cdot \nabla u_h]_E \, v_T \\
&\quad + \sum_{E \in \mathcal{E}(T) \cap \mathcal{E}_{h,N}} \int_E (g_E - n_E \cdot \nabla u_h) v_T \\
&\leq \eta_{N,T} \Bigg\{ c_{I1} h_T \|f_T\|_{0,2;T} + c_{I2} \frac{1}{2} \sum_{E \in \mathcal{E}(T) \cap \mathcal{E}_{h,\Omega}} h_E^{1/2} \|[n_E \cdot \nabla u_h]_E\|_{2;E} \\
&\qquad + c_{I2} \frac{1}{2} \sum_{E \in \mathcal{E}(T) \cap \mathcal{E}_{h,N}} h_E^{1/2} \|g_E - n_E \cdot \nabla u_h\|_{2;E} \Bigg\} \\
&\leq \sqrt{6} \max\{c_{I1}, c_{I2}\} \eta_{N,T} \, \eta_{R,T}.
\end{aligned}$$

This proves the first inequality.

In order to prove the second inequality, we observe that the functions w_T and w_E defined in Equations (1.20), (1.24), and (1.28) are contained in V_T. The same arguments as used for the proof of the second estimate of Proposition 1.9 therefore imply

$$\frac{9}{20} \|f_T\|_{0,2;T}^2 = \int_T f_T w_T = \int_T \nabla v_T \cdot \nabla w_T \leq \sqrt{\frac{9}{20}} c_5 h_T^{-1} \|f_T\|_{0,2;T} \, \eta_{N,T}$$

hence

$$h_T \|f_T\|_{0,2;T} \leq \sqrt{\frac{20}{9}} c_5 \eta_{N,T} \tag{1.46}$$

and – using Inequality (1.46) –

$$\begin{aligned}
\frac{1}{3} \|[n_E \cdot \nabla u_h]_E\|_{2;E}^2 &= \frac{1}{2} \int_E [n_E \cdot \nabla u_h]_E \, w_E \\
&= \sum_{T \subset \omega_E} \left\{ \int_T f_T w_E - \int_T \nabla v_T \cdot \nabla w_E \right\} \\
&\leq c \sum_{T \subset \omega_E} \{ \, h_E^{1/2} \|[n_E \cdot \nabla u_h]_E\|_{2;E} \|f_T\|_{0,2;T} \\
&\qquad + h_E^{-1/2} \|[n_E \cdot \nabla u_h]_E\|_{2;E} \, \eta_{N,T} \, \} \\
&\leq c' \|[n_E \cdot \nabla u_h]_E\|_{2;E} h_E^{-1/2} \sum_{T \subset \omega_E} \eta_{N,T}
\end{aligned}$$

thus

$$h_E^{1/2}\|[n_E \cdot \nabla u_h]_E\|_{2;E} \leq 3c' \sum_{T \in \omega_E} \eta_{N,T}. \tag{1.47}$$

With the same arguments we get for all $E \in \mathcal{E}_{h,N}$

$$h_E^{1/2}\|g_E - n_E \cdot \nabla u_h\|_{2;E} \leq \frac{3}{2}c'' \sum_{T \subset \omega_E} \eta_{N,T}. \tag{1.48}$$

Combining inequalities (1.46)–(1.48), we immediately obtain the second estimate of the proposition.

The third and fourth estimate of Proposition 1.11 finally follow from its first and second estimate and Proposition 1.5. □

Remark 1.12. For the present model problem it is easy to see that the auxiliary problems (1.34), (1.42), and (1.45) admit unique solutions. For more complex problems such as, e. g., the Stokes or Navier-Stokes equations, this is far less obvious (cf., e. g., [78, 80, 84] or Section 3.5). Figure 2 of Section 1.1 shows the generic form of ω_T and ω_x. From this it is obvious that, in the generic case, Problems (1.34), (1.42), and (1.45) have the dimensions 12, 7, and 4, respectively. In any case, the computation of $\eta_{D,x}$, $\eta_{D,T}$, and $\eta_{N,T}$ is more expensive than the one of $\eta_{R,T}$. □

Remark 1.13. The estimator $\eta_{D,x}$ was first introduced in [12]. Apparently, the estimator $\eta_{D,T}$ was first introduced in [27]. The estimator $\eta_{N,T}$, finally, is a slight modification of an estimator which was first introduced and analyzed in [22] and then extended to the Stokes and Navier-Stokes equations in [23, 24, 78, 80]. In its original form presented in [22], the estimator $\eta_{N,T}$ is based on the solution of the auxiliary problem (1.45) incorporating only the three degrees of freedom corresponding to the edge-bubble functions. In [22], the proof that $\eta_{N,T}$ yields upper and lower bounds on the eror is based on a saturation assumption. Recently it was shown that this extra condition is superflous (cf. [60]). The present analysis is a slight modification of the one given in [85] and takes into account the abstract framework of Section 2 and of [86]. A quite different analysis is given in [43, 44]. For a comparison of these error estimators see, e. g., [14, 59]. □

1.4. Hierarchical Basis Error Estimators.

The hierarchical basis error estimators try to bound the error $u - u_h$ by evaluating the residual of u_h with respect to certain basis functions of another finite element space W_h which satisfies $X_h \subset W_h \subset X$ and which either consists of higher order finite elements or corresponds to a refinement of $\mathcal{T}_h$. When suitably scaling the

corresponding basis functions, Equation (1.9) and the Cauchy-Schwarz inequality immediately imply that one thus obtains lower bounds on the error. The two following propositions show that a judicious choice of W_h allows to obtain upper bounds, too. Their proofs are based on the results of Section 1.2.

Proposition 1.14. *Denote by u and u_h the unique solutions of Problems (1.3) and (1.4), respectively and by W_h a finite element space corresponding to $\mathcal{T}_h$ with $X_h \subset W_h \subset X$. Suppose that to each edge $E \in \mathcal{E}_{h,\Omega} \cup \mathcal{E}_{h,N}$ there corresponds a function $\varphi_E \in W_h \cap C_0(\omega_E, I\!R)$ with the following properties*

$$\begin{aligned} 0 \le \varphi_E &\le 1, \\ c_1 h_E &\le \int_E \varphi_E, \\ \|\varphi_E\|_{1,2;\omega_E} &\le c_2 h_E^{-1} \|\varphi_E\|_{0,2;\omega_E}. \end{aligned}$$

Let

$$r_E := \int_\Omega f\,\varphi_E + \int_{\Gamma_N} g\,\varphi_E - \int_\Omega \nabla u_h \cdot \nabla \varphi_E \tag{1.49}$$

be the residual of u_h with respect to φ_E, $E \in \mathcal{E}_{h,\Omega} \cup \mathcal{E}_{h,N}$. Then the following a posteriori error estimates hold

$$|r_E| \le c_3 \|u - u_h\|_{1,2;\omega_E} \quad \forall E \in \mathcal{E}_{h,\Omega} \cup \mathcal{E}_{h,N} \tag{1.50}$$

$$\begin{aligned} \|u - u_h\|_{1,2} \le c_4 \Bigg\{ & \sum_{E \in \mathcal{E}_{h,\Omega} \cup \mathcal{E}_{h,N}} r_E^2 \\ & + \sum_{T \in \mathcal{T}_h} h_T^2 \|f\|_{0,2;T}^2 + \sum_{E \in \mathcal{E}_{h,N}} h_E \|g - g_E\|_{2,E}^2 \Bigg\}^{1/2}. \end{aligned} \tag{1.51}$$

PROOF: The conditions on φ_E in particular imply that

$$\|\varphi_E\|_{1,2;\omega_E} \le c_2 h_E^{-1} \|\varphi_E\|_{0,2;\omega_E} \le c_2 h_E^{-1} \left\{ \int_{\omega_E} 1 \right\}^{1/2} \le c_2'.$$

This together with Equation (1.9) and the Cauchy-Schwarz inequality yields Inequality (1.50).

Consider an arbitrary $E \in \mathcal{E}_{h,\Omega}$. The conditions on φ_E and Equations (1.14) and

(1.49) imply

$$\begin{aligned} c_1\|[n_E\cdot\nabla u_h]_E\|_{2;E}^2 &\le \int_E [n_E\cdot\nabla u_h]_E^2\,\varphi_E \\ &= -[n_E\cdot\nabla u_h]_E\, r_E + \sum_{T\subset\omega_E}\int_T f[n_E\cdot\nabla u_h]_E\,\varphi_E \\ &\le c\,h_E^{-1/2}\|[n_E\cdot\nabla u_h]_E\|_{2;E}\left\{r_E^2+\sum_{T\subset\omega_E} h_T^2\|f\|_{0,2;T}^2\right\}^{1/2} \end{aligned}$$

and thus

$$h_E^{1/2}\|[n_E\cdot\nabla u_h]_E\|_{2;E} \le c\left\{r_E^2+\sum_{T\subset\omega_E} h_T^2\|f\|_{0,2;T}^2\right\}^{1/2}. \tag{1.52}$$

Similarly, we obtain for all $E\in\mathcal{E}_{h,N}$

$$\begin{aligned} &c_1\|g_E - n_E\cdot\nabla u_h\|_{2;E}^2 \\ &\le \int_E (g_E - n_E\cdot\nabla u_h)^2\varphi_E \\ &= (g_E - n_E\cdot\nabla u_h)r_E - \sum_{T\subset\omega_E}\int_T f(g_E - n_E\cdot\nabla u_h)\varphi_E \\ &\quad + \int_E (g_E - g)(g_E - n_E\cdot\nabla u_h)\varphi_E \\ &\le c\,h_E^{-1/2}\|g_E - n_E\cdot\nabla u_h\|_{2;E}\left\{r_E^2+\sum_{T\subset\omega_E} h_T^2\|f\|_{0,2;T}^2 + h_E\|g-g_E\|_{2;E}^2\right\}^{1/2} \end{aligned}$$

and hence

$$h_E^{1/2}\|g_E - n_E\cdot\nabla u_h\|_{2;E} \le c\left\{r_E^2+\sum_{T\subset\omega_E} h_T^2\|f\|_{0,2;T}^2 + h_E\|g-g_E\|_{2;E}^2\right\}^{1/2}. \tag{1.53}$$

Estimate (1.51) now follows from Inequalities (1.12), (1.15), (1.52), and (1.53). □

Proposition 1.15. *Let the notations and assumptions be as in Proposition 1.14. Suppose in addition that to each $T\in\mathcal{T}_h$ there corresponds a function $\varphi_T\in W_h\cap C_0(T,\Omega)$ with the following properties*

$$\begin{aligned} 0\le\varphi_T&\le 1,\\ c_5 h_T^2 &\le \int_T \varphi_T,\\ \|\varphi_T\|_{1,2;T} &\le c_6 h_T^{-1}\|\varphi_T\|_{0,2;T}. \end{aligned}$$

Let

$$r_T := \int_\Omega f\,\varphi_T + \int_{\Gamma_N} g\,\varphi_T - \int_\Omega \nabla u_h \cdot \nabla \varphi_T \tag{1.54}$$

be the residual of u_h with respect to φ_T, $T \in \mathcal{T}_h$. Then the following a posteriori error estimates

$$|r_T| \le c_7 \|u - u_h\|_{1,2;T} \quad \forall T \in \mathcal{T}_h \tag{1.55}$$

$$\|u - u_h\|_{1,2} \le c_8 \Big\{ \sum_{E \in \mathcal{E}_{h,\Omega} \cup \mathcal{E}_{h,N}} r_E^2 + \sum_{T \in \mathcal{T}_h} r_T^2 + \sum_{T \in \mathcal{T}_h} h_T^2 \|f - f_T\|_{0,2;T}^2 + \sum_{E \in \mathcal{E}_{h,N}} h_E \|g - g_E\|_{2;E}^2 \Big\} \tag{1.56}$$

hold in addition to those of Proposition 1.14.

PROOF: The conditions on φ_T imply

$$\|\varphi\|_{1,2;T} \le c_6 h_T^{-1} \|\varphi_T\|_{0,2;T} \le c_6 h_T^{-1} \Big\{ \int_T 1 \Big\}^{1/2} \le c_6'.$$

Together with Equation (1.9) and the Cauchy-Schwarz inequality this proves Inequality (1.55).
Consider an arbitrary $T \in \mathcal{T}_h$. The conditions on φ_T and Equations (1.9) and (1.54) yield

$$\begin{aligned} c_6 \|f_T\|_{0,2;T}^2 &\le \int_T f_T^2 \varphi_T = f_T\, r_T + \int_T (f_T - f) f_T\, \varphi_T \\ &\le c\, h_T^{-1} \|f_T\|_{0,2;T} \{ r_T^2 + h_T^2 \|f - f_T\|_{0,2;T}^2 \}^{1/2} \end{aligned}$$

and thus

$$h_T \|f_T\|_{0,2;T} \le c \{ r_T^2 + h_T^2 \|f - f_T\|_{0,2;T}^2 \}^{1/2}. \tag{1.57}$$

Estimates (1.51) and (1.57) immediately prove Inequality (1.56). □

In the following example we show a few possibilities to verify the conditions of Propositions 1.14 and 1.15.

Example 1.16. (1) Let W_h be the space of all continuous, piecewise quadratic functions vanishing on Γ_D. Lemma 1.3 shows that we may choose $\varphi_E := b_E$, $E \in \mathcal{E}_{h,\Omega} \cup \mathcal{E}_{h,N}$.

(2) Let W_h be the space of all continuous piecewise cubic functions vanishing on Γ_D. Lemma 1.3 shows that we may choose $\varphi_E := b_E$, $E \in \mathcal{E}_{h,\Omega} \cup \mathcal{E}_{h,N}$, and $\varphi_T := b_T$, $T \in \mathcal{T}_h$. This gives a different interpretation of the error estimator $\eta_{R,T}$.

(3) Denote by $\mathcal{T}_{h/2}$ the triangulation which is obtained by a uniform refinement of $\mathcal{T}_h$, i. e., every $T \in \mathcal{T}_h$ is cut into four new triangles by joining the midpoints of its edges. Let W_h be the space of all continuous, piecewise linear functions corresponding to $\mathcal{T}_{h/2}$ and vanishing on Γ_D. Standard scaling arguments for finite elements then show that we may choose for φ_E, $E \in \mathcal{E}_{h,\Omega} \cup \mathcal{E}_{h,N}$, the nodal basis function of W_h corresponding to the midpoint of E (see Figure 3).

(4) Let $\mathcal{T}_{h/4}$ be the triangulation, which is obtained by a uniform refinement of $\mathcal{T}_{h/2}$, and let W_h be the space of all continuous, piecewise linear functions corresponding to $\mathcal{T}_{h/4}$ and vanishing on Γ_D. The functions φ_E, $E \in \mathcal{E}_{h,\Omega} \cup \mathcal{E}_{h,N}$, may be choosen as in part (3). Consider an arbitrary $T \in \mathcal{T}_h$ and denote by $T' \in \mathcal{T}_{h/4}$ that triangle that has all its vertices in the interior of T. Then we may choose φ_T as that function in W_h which is identically one on T' and which identically vanishes on ∂T (see Figure 3).

(5) Cut every triangle $T \in \mathcal{T}_h$ into six new triangles by joining every vertex of T to the midpoint of the edge opposite to it. Let W_h be the space of all continuous, piecewise linear functions corresponding to the resulting triangulation and vanishing on Γ_D. Then we may choose φ_T, $T \in \mathcal{T}_h$, and φ_E, $E \in \mathcal{E}_{h,\Omega} \cup \mathcal{E}_{h,N}$, as the nodal basis functions of W_h corresponding to the barycentres of triangles in $\mathcal{T}_h$ and to the midpoints of edges (see Figure 3). □

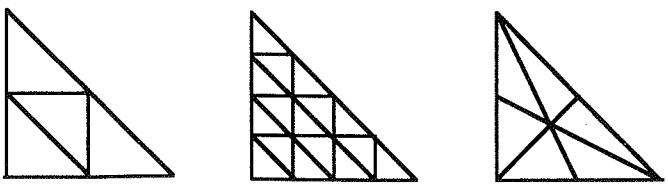


Fig. 3: partition of a triangle into 4, 16, and 6 new triangles

Usually, hierarchical basis error estimators are presented in the following slightly different form (cf., e.g., [21,31,40]). Split W_h in the form

$$W_h = X_h \oplus Z_h.$$

We assume that X_h and Z_h satisfy a strenghtened Cauchy-Schwarz inequality, i.e., there is a constant $0 \leq \gamma < 1$ such

$$\int_\Omega \nabla v_h \nabla z_h \leq \gamma \|\nabla v_h\|_{0,2} \|\nabla z_h\|_{0,2} \quad \forall v_h \in X_h, z_h \in Z_h. \tag{1.58}$$

On Z_h we introduce a symmetric bilinear form which is equivalent to the standard Dirichlet form, i.e.

$$\|\nabla z_h\|_{0,2}^2 \leq b(z_h, z_h) \leq \frac{1}{\alpha}\|\nabla z_h\|_{0,2}^2 \quad \forall z_h \in Z_h \tag{1.59}$$

with a constant $0 < \alpha \leq 1$. Denote by $z_h^* \in Z_h$ the solution of

$$b(z_h^*, \varphi_h) = \int_\Omega f\varphi_h + \int_{\Gamma_N} g\varphi_h - \int_\Omega \nabla u_h \nabla \varphi_h \quad \forall \varphi_h \in Z_h \tag{1.60}$$

and set

$$\eta_H := b(z_h^*, z_h^*)^{1/2}. \tag{1.61}$$

Here, u_h is the solution of Problem (1.4). The main point in introducing the bilinear form b is that Problem (1.60) should be much easier to solve than the corresponding problem with b replaced by a (cf. Example 1.18 below).

Proposition 1.17. *Assume that the space W_h satisfies a saturation assumption with respect to X_h, i.e., there is a constant $0 \leq \beta < 1$ such that*

$$\min_{\psi_h \in W_h} \|\nabla(v - \psi_h)\|_{0,2} \leq \beta \min_{\varphi_h \in X_h} \|\nabla(v - \varphi_h)\|_{0,2} \quad \forall v \in X. \tag{1.62}$$

Then the following a posteriori error estimate holds for the solutions u and u_h of Problems (1.3) and (1.4)

$$(1-\beta)\sqrt{\frac{\alpha(1-\gamma)}{1+c_\Omega^2}}\|u - u_h\|_{1,2} \leq \eta_H \leq \frac{1}{\sqrt{\alpha}}\|u - u_h\|_{1,2}.$$

Here, η_H is given by Equation (1.61) and c_Ω denotes the constant in the Poincaré-Friedrichs inequality (1.10).

PROOF: First, we observe that due to inequality (1.10)

$$\frac{1}{\sqrt{1+c_\Omega^2}}\|u - u_h\|_{1,2} \leq \|\nabla(u - u_h)\|_{0,2} \leq \|u - u_h\|_{1,2}. \tag{1.63}$$

Next, denote by $w_h \in W_h$ the unique solution of

$$\int_\Omega \nabla w_h \nabla \varphi_h = \int_\Omega f\varphi_h + \int_{\Gamma_N} g\varphi_h \quad \forall \varphi_h \in W_h.$$

Since

$$\int_\Omega \nabla(w_h - u_h)\nabla\varphi_h = \int_\Omega \nabla(u - u_h)\nabla\varphi_h \quad \forall \varphi_h \in W_h, \tag{1.64}$$

we have

$$\|\nabla(w_h - u_h)\|_{0,2} \leq \|\nabla(u - u_h)\|_{0,2}. \tag{1.65}$$

Since

$$w_h = \operatorname*{argmin}_{\psi_h \in W_h} \|\nabla(u - \psi_h)\|_{0,2},$$

the saturation assumption (1.62) on the other hand yields

$$\begin{aligned}\|\nabla(u - u_h)\|_{0,2} \leq & \|\nabla(u - w_h)\|_{0,2} + \|\nabla(w_h - u_h)\|_{0,2} \\ \leq & \beta\|\nabla(u - u_h)\|_{0,2} + \|\nabla(w_h - u_h)\|_{0,2}\end{aligned}$$

and therefore

$$(1 - \beta)\|\nabla(u - u_h)\|_{0,2} \leq \|\nabla(w_h - u_h)\|_{0,2}. \tag{1.66}$$

Denote by $\widetilde{z}_h \in Z_h$ the unique solution of

$$\int_\Omega \nabla\widetilde{z}_h \nabla\varphi_h = \int_\Omega f\varphi_h + \int_{\Gamma_N} g\varphi_h - \int_\Omega \nabla u_h \nabla\varphi_h \quad \forall\varphi_h \in Z_h. \tag{1.67}$$

Obviously, we have

$$\int_\Omega \nabla\widetilde{z}_h \nabla\varphi_h = \int_\Omega \nabla(w_h - u_h)\nabla\varphi_h \quad \forall\varphi_h \in Z_h \tag{1.68}$$

and therefore

$$\|\nabla\widetilde{z}_h\|_{0,2} \leq \|\nabla(w_h - u_h)\|_{0,2}. \tag{1.69}$$

Now, we split $w_h - u_h$ in the form

$$w_h - u_h = \overline{v}_h + \overline{z}_h$$

with $\overline{v}_h \in X_h$ and $\overline{z}_h \in Z_h$. The strengthened Cauchy-Schwarz inequality (1.58) implies that

$$\begin{aligned}\|\nabla(\overline{v}_h + \overline{z}_h)\|_{0,2}^2 \geq & \|\nabla\overline{v}_h\|_{0,2}^2 - 2\gamma\|\nabla\overline{v}_h\|_{0,2}\|\nabla\overline{z}_h\|_{0,2} + \|\nabla\overline{z}_h\|_{0,2}^2 \\ \geq & (1 - \gamma)[\|\nabla\overline{v}_h\|_{0,2}^2 + \|\nabla\overline{z}_h\|_{0,2}^2].\end{aligned}$$

This, in particular, yields

$$\|\nabla\overline{z}_h\|_{0,2} \leq \frac{1}{\sqrt{1 - \gamma}}\|\nabla(w_h - u_h)\|_{0,2}. \tag{1.70}$$

Equation (1.64) implies the orthogonality relation

$$\int_\Omega \nabla(w_h - u_h)\nabla v_h = 0 \quad \forall v_h \in X_h.$$

Together with Equations (1.68) and Inequality (1.70) this gives

$$\begin{aligned}
\|\nabla(w_h - u_h)\|_{0,2}^2 &= \int_\Omega \nabla(w_h - u_h)\nabla(\overline{v}_h + \overline{z}_h) \\
&= \int_\Omega \nabla(w_h - u_h)\nabla\overline{z}_h \\
&= \int_\Omega \nabla\widetilde{z}_h\nabla\overline{z}_h \\
&\leq \|\nabla\widetilde{z}_h\|_{0,2}\|\nabla\overline{z}_h\|_{0,2} \\
&\leq \frac{1}{\sqrt{1-\gamma}}\|\nabla(w_h - u_h)\|_{0,2}\|\nabla\widetilde{z}_h\|_{0,2}.
\end{aligned}$$

Hence, we have

$$\sqrt{1-\gamma}\|\nabla(w_h - u_h)\|_{0,2} \leq \|\nabla\widetilde{z}_h\|_{0,2}. \tag{1.71}$$

Finally, a comparison of Equation (1.60) and (1.67) shows that

$$b(z_h^*, \varphi_h) = \int_\Omega \nabla\widetilde{z}_h\nabla\varphi_h \quad \forall\varphi_h \in Z_h.$$

Together with Inequality (1.59) this proves that

$$\sqrt{\alpha}\|\nabla\widetilde{z}_h\|_{0,2} \leq \eta_H \leq \frac{1}{\sqrt{\alpha}}\|\nabla\widetilde{z}_h\|_{0,2}. \tag{1.72}$$

Combining Estimates (1.63), (1.65), (1.66), (1.69), (1.71), and (1.72), we obtain the desired estimates. □

Example 1.18. We choose W_h as in Example 1.16 and set

$$Z_h := \{\varphi_h \in W_h : \varphi_h(x) = 0 \quad \forall x \in \mathcal{N}_h\}.$$

The strengthened Cauchy-Schwarz inequality can be proven by transforming to the reference element, solving a small eigenvalue problem there, and transforming back (cf., e.g., [45, 57]). The saturation assumption holds exept in trivial cases where

$$\lim_{h\to 0} h^{-1}\|u - u_h\|_{1,2} = 0.$$

In order to define $\mathfrak{b}$, denote by $\overline{\mathcal{N}}_h$ the nodes corresponding to W_h, e.g. $\mathcal{N}_{h/2}$ in examples (1) and (3) and $\mathcal{N}_{h/4}$ in example (4). Set

$$\mathfrak{b}(\varphi_h, \psi_h) := \sum_{x \in \overline{\mathcal{N}}_h \setminus \mathcal{N}_h} \varphi_h(x)\psi_h(x).$$

Transforming to the reference element, using the equivalence of norms on finite dimensional spaces, and transforming back, one easily checks that $\mathfrak{b}(\varphi_h, \varphi_h)^{1/2}$ and $h^{-1}\|\varphi_h\|_{0,2}$ are equivalent norms on Z_h with constants being independent of h. Since the functions in Z_h vanish at the nodes corresponding to X_h one proves with the same arguments that $\|\nabla\varphi_h\|_{0,2}$ and $h^{-1}\|\varphi_h\|_{0,2}$ are equivalent norms on Z_h. This proves Estimate (1.59). With this choice of $\mathfrak{b}$ the matrix of Problem (1.60) becomes diagonal with diagonal entries of order 1. Hence, up to multiplicative constants of order 1, η_H is equivalent to

$$\{\sum_{T \in \mathcal{T}_h} r_T^2 + \sum_{E \in \mathcal{E}_{h,\Omega} \cup \mathcal{E}_{h,N}} r_E^2\}^{1/2}.$$

Thus we recover the estimators of Proposition 1.14 and 1.15. □

Remark 1.19. The saturation assumption (1.62) is used to prove Estimate (1.66). One can prove that (1.66) in turn implies (1.62) (cf. [31; Theorem 2.1]). In establishing Propositions 1.5, 1.14, and 1.15 we did not need to resort to the saturation assumption. We had to pay this by admitting the perturbation term $\{\sum_{T \in \mathcal{T}_h} h_T^2 \|f - f_T\|_{0,T}^2\}^{1/2}$. Indeed, Proposition 1.5 may be used to show that the saturation assumption holds if the right-hand side f of Problem (1.1) is piecewise constant on $\mathcal{T}_h$ (cf. [31; §4]. □

Remark 1.20. Proposition 1.17 together with some extensions may be found in, e.g., [21, 31, 40]. Proposition 1.14 and 1.15 seem to be new. □

1.5. An Error Estimator Based on an Averaging of the Gradient.

The error estimator of this chapter is based on the following ideas. Consider Problem (1.1) with $\Gamma_N = \emptyset$ and denote by u and u_h the unique solutions of Problems (1.3) and (1.4). Suppose that we dispose of an easily computable approximation Gu_h of ∇u_h such that

$$\|\nabla u - Gu_h\|_{0,2} \leq \beta \|\nabla u - \nabla u_h\|_{0,2} \tag{1.73}$$

holds with a constant $0 \leq \beta < 1$. We then have

$$\frac{1}{1+\beta}\|Gu_h - \nabla u_h\|_{0,2} \leq \|\nabla u - \nabla u_h\|_{0,2} \leq \frac{1}{1-\beta}\|Gu_h - \nabla u_h\|_{0,2} \tag{1.74}$$

and we may therefore choose $\|Gu_h - \nabla u_h\|_{0,2}$ as error estimator. Since ∇u_h is a piecewise constant vector-field we may hope that its L^2-projection onto the continuous, piecewise linear vector-fields satisfies Inequality (1.73). The computation of this projection, however, is as expensive as the solution of Problem (1.4). We therefore replace the L^2-scalar product by an approximation which leads to a more tractable auxiliary problem.

In order to make these ideas more precise, we denote by W_h the space of all piecewise linear vector-fields and set $V_h := W_h \cap C(\Omega, I\!R^2)$. Note that $\nabla X_h \subset W_h$. We define a mesh-dependent scalar product $(\cdot,\cdot)_h$ on W_h by

$$(\underline{v}, \underline{w})_h := \sum_{T \in \mathcal{T}_h} \frac{|T|}{3} \left\{ \sum_{x \in \mathcal{N}(T)} \underline{v}|_T(x) \cdot \underline{w}|_T(x) \right\}. \tag{1.75}$$

Here, $|T|$ denotes the area of T and

$$\underline{\varphi}|_T(x) := \lim_{\substack{y \to x \\ y \in T}} \underline{\varphi}(y) \quad \forall \underline{\varphi} \in W_h, T \in \mathcal{T}_h, x \in \mathcal{N}(T).$$

Since the quadrature formula

$$\int_T \varphi \sim \frac{|T|}{3} \sum_{x \in \mathcal{N}(T)} \varphi(x)$$

is exact for all linear functions, we have

$$(\underline{v}, \underline{w})_h = \int_\Omega \underline{v} \cdot \underline{w} \tag{1.76}$$

if both arguments are elements of W_h and at least one of them is piecewise constant. Moreover, one easily checks that

$$\frac{1}{4}\|\underline{v}\|_{0,2}^2 \leq (\underline{v}, \underline{v})_h \leq \|\underline{v}\|_{0,2}^2 \quad \forall \underline{v} \in W_h \tag{1.77}$$

and

$$(\underline{v}, \underline{w})_h = \frac{1}{3} \sum_{x \in \mathcal{N}_h} |\omega_x| \underline{v}(x) \cdot \underline{w}(x) \quad \forall \underline{v}, \underline{w} \in V_h. \tag{1.78}$$

Let $Gu_h \in V_h$ be the $(\cdot,\cdot)_h$-projection of ∇u_h onto V_h, i.e.

$$(Gu_h, \underline{v}_h)_h = (\nabla u_h, \underline{v}_h)_h \quad \forall \underline{v}_h \in V_h. \tag{1.79}$$

Equations (1.76) and (1.78) imply that

$$Gu_h(x) = \sum_{T \subset \omega_x} \frac{|T|}{|\omega_x|} \nabla u_{h|T} \quad \forall x \in \mathcal{N}_h. \tag{1.80}$$

Thus, Gu_h may be computed by a local averaging of ∇u_h. We finally set

$$\eta_{Z,T} := \|Gu_h - \nabla u_h\|_{0,2;T} \tag{1.81}$$

and

$$\eta_Z := \left\{ \sum_{T \in \mathcal{T}_h} \eta_{Z,T}^2 \right\}^{1/2}. \tag{1.82}$$

In order to prove that η_Z is an error estimator, we will not establish Inequality (1.73). Instead, we will compare $\eta_{Z,T}$ with a modification $\widetilde{\eta}_{R,T}$ of the residual error estimator of Section 1.2 which is obtained by omitting the element-residuals:

$$\widetilde{\eta}_{R,T} := \left\{ \frac{1}{2} \sum_{E \in \mathcal{E}(T) \cap \mathcal{E}_{h,\Omega}} h_E \|[n_E \cdot \nabla u_h]_E\|_{0;E}^2 \right\}^{1/2}. \tag{1.83}$$

Set

$$\widetilde{\eta}_R := \left\{ \sum_{T \in \mathcal{T}_h} \widetilde{\eta}_{R,T}^2 \right\}^{1/2}. \tag{1.84}$$

We then have:

Proposition 1.21. *There are constants $c_1, \ldots, c_4$, which only depend on the smallest angle in the triangulation, such that the following estimates hold*

$$\|u - u_h\|_{1,2} \le c_1 \left\{ \widetilde{\eta}_R^2 + \sum_{T \in \mathcal{T}_h} h_T^2 \|f\|_{0,2;T}^2 \right\}^{1/2},$$

$$\widetilde{\eta}_{R,T} \le c_2 \left\{ \|u - u_h\|_{1,2;\omega_T}^2 + \sum_{T' \subset \omega_T} h_{T'}^2 \|f\|_{0,2;T'}^2 \right\}^{1/2},$$

$$c_3 \widetilde{\eta}_R \le \eta_Z \le c_4 \widetilde{\eta}_R.$$

Here, η_Z, $\widetilde{\eta}_{R,T}$, and $\widetilde{\eta}_R$ are given by Equations (1.82), (1.83), and (1.84).

PROOF: Recall that $\Gamma_N = \emptyset$ in this section. The first estimate then is Inequality (1.16).

The second one follows from the proof of Inequality (1.27) simply by disregarding

the local L^2-projection f_T.
In order to prove the third estimate, we first rewrite $\widetilde{\eta}_R$:

$$\widetilde{\eta}_R^2 = \sum_{T\in\mathcal{T}_h} \frac{1}{2} \sum_{E\in\mathcal{E}_{h,\Omega}\cap\mathcal{E}(T)} h_E \|[n_E\cdot\nabla u_h]_E\|_{0;E}^2 = \sum_{E\in\mathcal{E}_{h,\Omega}} h_E^2 [n_E\cdot\nabla u_h]_E^2.$$

Next we observe that, due to the minimal angle condition, the right hand side of the previous equation is bounded from above and from below by multiples of

$$\sum_{x\in\mathcal{N}_h} |\omega_x| \widetilde{\eta}_{R,x}^2$$

where

$$\widetilde{\eta}_{R,x} := \Big\{ \sum_{\substack{E\in\mathcal{E}_{h,\Omega}\\ x\in\mathcal{N}(E)}} [n_E\cdot\nabla u_h]_E^2 \Big\}^{1/2}. \tag{1.85}$$

Thanks to Inequality (1.77) on the other hand, η_Z is bounded from above and from below by multiples of $(Gu_h - \nabla u_h, Gu_h - \nabla u_h)_h$. But Equations (1.76), (1.78), (1.79), and (1.80) imply that

$$\begin{aligned}
&(Gu_h - \nabla u_h, Gu_h - \nabla u_h)_h \\
&= (\nabla u_h, \nabla u_h)_h - (Gu_h, Gu_h)_h \\
&= \sum_{T\in\mathcal{T}_h} |T|\,|\nabla u_{h|T}|^2 - \sum_{x\in\mathcal{N}_h} \frac{1}{3}|\omega_x|\,|Gu_h(x)|^2 \\
&= \sum_{x\in\mathcal{N}_h} \Big\{ \sum_{T\subset\omega_x} \frac{1}{3}|T|\,|\nabla u_{h|T}|^2 - \frac{1}{3}|\omega_x| \Big| \sum_{T\subset\omega_x} \frac{|T|}{|\omega_x|} \nabla u_{h|T} \Big|^2 \Big\} \\
&= \frac{1}{3} \sum_{x\in\mathcal{N}_h} |\omega_x| \eta_{Z,x}^2
\end{aligned}$$

where

$$\eta_{Z,x} := \Big\{ \sum_{T\subset\omega_x} \frac{|T|}{|\omega_x|} |\nabla u_{h|T}|^2 - \Big| \sum_{T\subset\omega_x} \frac{|T|}{|\omega_x|} \nabla u_{h|T} \Big|^2 \Big\}^{1/2}. \tag{1.86}$$

Therefore, it is sufficient to prove that

$$c_5 \widetilde{\eta}_{R,x}^2 \le \eta_{Z,x}^2 \le c_6 \widetilde{\eta}_{R,x}^2 \tag{1.87}$$

holds for all $x \in \mathcal{N}_h$ with constants c_5, c_6 that only depend on the smallest angle in the triangulation.

In order to establish Inequality (1.87) we consider an arbitrary $x \in \mathcal{N}_{h,\Omega}$. The case of a boundary node is completely similar. Let N be the number of triangles in ω_x and enumerate the triangles in counter-clockwise ordering starting with 0 as depicted in Figure 4.

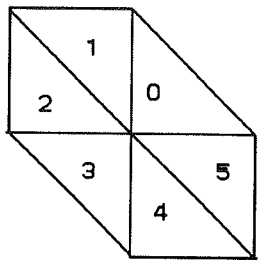


Fig. 4: enumeration of triangles in ω_x

Set

$$\begin{aligned} T_N &:= T_0, \\ v_i &:= \partial_x u|_{T_i},\ w_i := \partial_y u|_{T_i},\ \mu_i := \frac{|T_i|}{|\omega_x|}, \quad 0 \le i \le N, \\ v &:= (v_1, \ldots, v_N)^t,\ w := (w_1, \ldots, w_N)^t. \end{aligned}$$

Since the tangential components of ∇u_h are continuous across edges, we have

$$\widetilde{\eta}^2_{R,x} = \sum_{i=1}^{N} \{ (v_i - v_{i-1})^2 + (w_i - w_{i-1})^2 \} = v^t B v + w^t B w$$

and

$$\begin{aligned} \eta^2_{Z,x} &= \sum_{i=1}^{N} \mu_i v_i^2 - \left(\sum_{i=1}^{N} \mu_i v_i \right)^2 + \sum_{i=1}^{N} \mu_i w_i^2 - \left(\sum_{i=1}^{N} \mu_i w_i \right)^2 \\ &= v^t A v + w^t A w \end{aligned}$$

with symmetric, positive semi-definite $N \times N$ matrices A and B. Since $\sum_{i=1}^{N} \mu_i = 1$, we have

$$e := (1, \ldots, 1)^t \in \ker(A) \cap \ker(B).$$

For any $z \in \mathbb{R}^N$ set

$$\widetilde{z} := (z_1 - z_N, \ldots, z_{N-1} - z_N)^t \in \mathbb{R}^{N-1}$$

and denote by $\widetilde{A}$ and $\widetilde{B}$ the $(N-1)\times(N-1)$ matrices which are obtained by deleting the last column and the last row of A and B. We then have for all $z \in I\!R^N$

$$z^t A z = (z - z_N e)^t A (z - z_N e) = \widetilde{z}^t \widetilde{A} \widetilde{z}$$

and

$$z^t B z = (z - z_N e)^t B (z - z_N e) = \widetilde{z}^t \widetilde{B} \widetilde{z}.$$

Since

$$\widetilde{B} = \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & -1 \\ & & \ddots & -1 & -2 \end{pmatrix},$$

its eigenvalues are given by

$$\lambda_k(\widetilde{B}) = 4\sin^2 \frac{k\pi}{2N}, \quad 1 \le k \le N-1.$$

Gerschgorin's theorem on the other hand implies that the eigenvalues $\lambda_k(\widetilde{A})$ of $\widetilde{A}$ are bounded by

$$\min_{1\le i\le N} \mu_i^2 \le \lambda_k(\widetilde{A}) \le 2 \quad \forall 1 \le k \le N-1.$$

Hence, we have for all $z \in I\!R^N$

$$\frac{1}{4} \min_{1\le i\le N} \mu_i^2 z^t B z \le z^t A z, \quad 2\sin^2 \frac{\pi}{2N} z^t A z \le z^t B z.$$

This proves Inequality (1.87). □

Remark 1.22. The estimator η_Z was first proposed in [92]. The present analysis follows the lines of [68]. Similar estimators based on a local averaging or extrapolation of the gradient are analyzed in [43]. □

1.6. Efficiency of the Error Estimators.

The quality of an a posteriori error estimator is often measured by its *efficiency index,* i. e. the ratio of the estimated error and of the true error. An error estimator is called *efficient* if its efficiency index together with its inverse remain bounded for all meshsizes. It is called *asymptotically exact* if its efficiency index tends to one when the meshsize converges to zero.

In the generic case we have

$$\{\sum_{T\in\mathcal{T}_h} h_T^2 \|f - f_T\|_{0,2;T}^2\}^{1/2} = o(h)$$

and

$$\{ \sum_{E \in \mathcal{E}_{h,N}} h_E \|g - g_E\|_{2;E}^2 \}^{1/2} = o(h),$$

where $h := \max_{T \in \mathcal{T}_h} h_T$ denotes the maximal meshsize. On the other hand, the solutions of Problems (1.3) und (1.4) satisfy $\|u - u_h\|_{1,2} \geq ch$ always but in trivial cases. Hence, Propositions 1.5, 1.9, 1.10, 1.11, and 1.15 imply that the corresponding error estimators are efficient. Their efficiency indices can in principle be estimated explicitly since the constants in the above propositions only depend on local approximation and inverse estimates. The same holds for the estimator η_Z of Section 1.5 (cf. [14] and [69]).

Using superconvergence results one can also prove that on special meshes the error estimators of Sections 1.3–1.5 are asymptotically exact. As an example, we will show this for the estimator $\eta_{N,T}$ defined by Equation (1.44). To do this we need the following notations. Ω_0 and Ω_1 are two open subsets of Ω such that

$$\overline{\Omega}_0 \subset \Omega_1 \quad \text{and} \quad \overline{\Omega}_0 = \bigcup_{\substack{T \in \mathcal{T}_h \\ T \subset \Omega_0}} T.$$

The triangulation $\mathcal{T}_h$ is called *parallel* in Ω_1 if ω_E is a parallelogram for all edges $E \subset \Omega_1$. Set

$$e := u - u_h$$

where u and u_h are the solutions of Problems (1.3) and (1.4). Denote by v_T, $T \in \mathcal{T}_h$, the unique solution of Problem (1.45) and define $\varepsilon \in L^2(\Omega)$ and $\nabla_h \varepsilon \in L^2(\Omega)^2$ by

$$\varepsilon|_T := v_T, \quad (\nabla_h \varepsilon)|_T := \nabla v_T \quad \forall T \in \mathcal{T}_h.$$

The following superconvergence result is proven in [90].

Lemma 1.23. *Assume that $\mathcal{T}_h$ is parallel in Ω_1 and that $u|_{\Omega_1} \in H^3(\Omega_1)$. Then the following superconvergence result holds*

$$\|\nabla(i_h u - u_h)\|_{0,2;\Omega_0} \leq c\{ h^2 \|u\|_{3,2;\Omega_1} + \|e\|_{0,2;\Omega_1} \}.$$

Here, $i_h u$ denotes the piecewise linear interpolation of u in the nodes $\mathcal{N}_h$, i. e.

$$i_h u(x) = u(x) \quad \forall x \in \mathcal{N}_h.$$

□

With the help of Lemma 1.23 we can prove the following auxiliary result.

Lemma 1.24. *Assume that the conditions of Lemma 1.23 are satisfied. Then the estimate*

$$\|\nabla e - \nabla_h \varepsilon\|_{0,2;\Omega_0} \leq c\{\, h^2 \|u\|_{3,2;\Omega_1} + \|e\|_{0,2;\Omega_1} \,\}$$

holds for sufficiently small h.

PROOF: Assume that h is small enough, such that $\omega_T \subset \Omega_1$ for all $T \in \mathcal{T}_h$ with $T \subset \Omega_0$. Given an arbitrary $T \in \mathcal{T}_h$ with $T \subset \Omega_0$ and an arbitrary $\varphi \in H^2(\omega_T)$ we then denote by $v_\varphi \in V_T$ the unique solution of

$$\int_T \nabla v_\varphi \cdot \nabla w = -\int_T (\Delta\varphi)_T\, w - \frac{1}{2} \sum_{E \in \mathcal{E}(T)} \int_E [n_E \cdot \nabla i_h \varphi]_E\, w \quad \forall w \in V_T. \tag{1.88}$$

Here, V_T is as in Equation (1.43). With this notation, we obtain for all $T \in \mathcal{T}_h$ with $T \subset \Omega_0$

$$\begin{aligned} &\|\nabla e - \nabla_h \varepsilon\|_{0,2;T} \\ &= \|\nabla u - \nabla u_h - \nabla v_T\|_{0,2;T} \\ &\leq \|\nabla u - \nabla i_h u - \nabla v_u\|_{0,2;T} + \|\nabla i_h u - \nabla u_h\|_{0,2;T} + \|\nabla v_u - \nabla v_T\|_{0,2;T} \\ &:= S_{1,T} + S_{2,T} + S_{3,T}. \end{aligned} \tag{1.89}$$

Lemma 1.23 implies

$$\left\{ \sum_{\substack{T \in \mathcal{T}_h \\ T \subset \Omega_0}} S_{2,T}^2 \right\}^{1/2} = c\{\, h^2 \|u\|_{3,2;\Omega_1} + \|e\|_{0,2;\Omega_1} \,\}. \tag{1.90}$$

Consider an arbitrary $T \in \mathcal{T}_h$ with $T \subset \Omega_0$ and an arbitrary quadratic polynomial φ on ω_T. We claim that

$$v_\varphi = \varphi - i_h \varphi \quad \text{on } T. \tag{1.91}$$

To see this, we first observe that $(\varphi - i_h\varphi)|_T \in V_T$. Next, we recall from [43] that

$$n_T \cdot \nabla(\varphi - i_h\varphi)(x_E) = -\frac{1}{2}[n_E \cdot \nabla i_h \varphi]_E \tag{1.92}$$

holds for all $E \in \mathcal{E}(T)$, where x_E denotes the midpoint of E. The fact that $\Delta\varphi$ is constant on T, integration by parts, application of Simpson's rule, and equation

(1.92) imply for all $w \in V_T$

$$\begin{aligned}
&\int_T \nabla(\varphi - i_h\varphi)\cdot\nabla w \\
&= -\int_T \Delta(\varphi - i_h\varphi)w + \sum_{E\in\mathcal{E}(T)}\int_E n_T\cdot\nabla(\varphi - i_h\varphi)w \\
&= -\int_T (\Delta\varphi)_T\, w + \sum_{E\in\mathcal{E}(T)}\frac{2}{3}h_E n_T\cdot\nabla(\varphi - i_h\varphi)(x_E)w(x_E) \\
&= -\int_T (\Delta\varphi)_T\, w - \frac{1}{2}\sum_{E\in\mathcal{E}(T)}\int_E [n_E\cdot\nabla i_h\varphi]_E\, w.
\end{aligned}$$

Since the solution of Problem (1.88) is unique, this proves Equation (1.91). Inequalities (1.35) and (1.36) and a standard local inverse estimate imply that

$$\|\nabla v_\varphi\|_{0,2;T} \le c\, h_T \|\varphi\|_{2,2;\omega_T}$$

and thus

$$\|\nabla\varphi - \nabla i_h\varphi - \nabla v_\varphi\|_{0,2;T} \le c\, h_T\|\varphi\|_{2,2;\omega_T} \tag{1.93}$$

hold for all $\varphi \in H^2(\omega_T)$. Equation (1.91), Inequality (1.93), and the Bramble-Hilbert lemma yield

$$\|\nabla\varphi - \nabla i_h\varphi - \nabla v_\varphi\|_{0,2;T} \le c\, h_T^2\|\varphi\|_{3,2;\omega_T} \quad \forall\varphi \in H^3(\omega_T). \tag{1.94}$$

Applying Estimate (1.94) to u gives

$$\left\{\sum_{\substack{T\in\mathcal{T}_h\\ T\subset\Omega_0}} S_{1,T}^2\right\}^{1/2} \le c\, h^2\|u\|_{3,2;\Omega_1}. \tag{1.95}$$

Equations (1.45) and (1.88) finally imply

$$\int_T \nabla(v_u - v_T)\cdot\nabla w = \frac{1}{2}\sum_{E\in\mathcal{E}(T)}\int_E [n_E\cdot\nabla(u_h - i_h u)]_E\, w \quad \forall w \in V_T.$$

Together with Inequalities (1.35) and (1.36) and a standard local inverse estimate this shows that

$$\|\nabla v_u - \nabla v_T\|_{0,2;T} \le c\|\nabla(u_h - i_h u)\|_{0,2;\omega_T}. \tag{1.96}$$

Estimate (1.96) and Lemma 1.23 yield

$$\left\{\sum_{\substack{T\in\mathcal{T}_h\\ T\subset\Omega_0}} S_{3,T}^2\right\}^{1/2} \le c\{\, h^2\|u\|_{3,2;\Omega_1} + \|e\|_{0,2;\Omega_1}\,\}. \tag{1.97}$$

Inequalities (1.89), (1.90), (1.95), and (1.97) establish the desired result. □

Under suitable assumptions, Lemma 1.24 implies the asymptotic exactness of $\eta_{N,T}$.

Proposition 1.25. *Assume that*

(1) $\mathcal{T}_h$ *is parallel on* Ω_0,
(2) $u|_{\Omega_1} \in H^3(\Omega_1)$,
(3) $\|\nabla e\|_{0,2;\Omega_0} \geq c\,h$,
(4) $\|e\|_{0,2;\Omega_1} \leq c\,h^{1+\varepsilon}$ *for some* $\varepsilon > 0$.

Then the error estimator $\eta_{N,T}$ *of Equation (1.44) is asymptotically exact, i. e.*

$$\left\{ \sum_{\substack{T\in\mathcal{T}_h \\ T\subset\Omega_0}} \eta_{N,T}^2 \right\}^{1/2} / \|\nabla e\|_{0,2;\Omega_0} \underset{h\to 0}{\longrightarrow} 1.$$

PROOF: Note, that

$$\sum_{\substack{T\in\mathcal{T}_h \\ T\subset\Omega_0}} \eta_{N,T}^2 = \|\nabla_h \varepsilon\|_{0,2;\Omega_0}^2.$$

Moreover, assumptions (1)–(4) and Lemma 1.24 give

$$\|\nabla e - \nabla_h \varepsilon\|_{0,2;\Omega_0} / \|\nabla e\|_{0,2;\Omega_0} \leq c\{\, h\|u\|_{3,2;\Omega_1} + h^{\varepsilon}\,\} \underset{h\to 0}{\longrightarrow} 0.$$

□

Remark 1.26. Assumption (3) of Proposition 1.25 is satisfied in all but trivial cases. Assumption (4) of Proposition 1.25 follows, e. g. from standard regularity theory if $f \in L^2(\Omega)$ and g is the restriction to Γ_N of an H^1-function. □

The following example shows that asymptotic exactness may not hold on general meshes even if they are strongly structured.

Example 1.27. Consider Problem (1.1) on the unit square $\Omega = (0,1)^2$ with $\Gamma_N = (0,1)\times\{0\} \cup (0,1)\times\{1\}$, $g = 0$ and $f = 1$. The exact solution is $u(x,y) = \frac{1}{2}x(1-x)$. The triangulation $\mathcal{T}_h$ is obtained as follows: Ω is divided into n^2 squares with sides of length $h := \frac{1}{n}$, $n \in I\!N^*$; each square is cut into four triangles by drawing the two diagonals. This triangulation, which is often called a *criss-cross grid*, is obviously not parallel. Since the solution u of Problem (1.1) is quadratic and the Neumann boundary conditions are homogeneous, one easily checks that the solution u_h of Problem (1.4) is given by

$$u_h(P) = \begin{cases} u(P) & \text{if } P \text{ is a vertex of a square} \\ u(P) - \frac{h^2}{24} & \text{if } P \text{ is a midpoint of a square.} \end{cases}$$

Using this expression for u_h one can explicitly calculate the error and the error estimator. After some computations one obtains for any square Q, which is disjoint from Γ_N^*,

$$\left\{ \sum_{\substack{T \in \mathcal{T}_h \\ T \subset Q}} \eta_{N,T}^2 \right\}^{1/2} / \|\nabla e\|_{0,2;Q} = \sqrt{\frac{17}{6}} \approx 1.68.$$

Hence, the error estimator cannot be asymptotically exact. □

Remark 1.28. The present analysis follows the lines of [44]. The asymptotic exactness of other error estimators is proved, e. g., in [43, 68] using similar superconvergence results. An analogue of Example 1.27 is also established there. The efficiency of the residual error estimator $\eta_{R,T}$ is established in [9] by comparing the linear finite element solution to solutions obtained by the so-called p-version of the finite element method. This analysis also gives realistic bounds on the efficiency index of $\eta_{R,T}$. Different error estimators are compared, e. g., in [14, 59]. □

2. Abstract Nonlinear Equations

2.1. Error Estimates for Isolated Solutions.

Let X, Y be two Banach spaces with norms $\|\cdot\|_X$ and $\|\cdot\|_Y$. For any element $u \in X$ and any real number $R > 0$ set $B(u,R) := \{ v \in X : \|u - v\|_X < R \}$. $\mathcal{L}(X,Y)$ and $\mathrm{Isom}(X,Y) \subset \mathcal{L}(X,Y)$ denote the Banach space of continuous linear maps of X in Y equipped with the operator-norm $\|\cdot\|_{\mathcal{L}(X,Y)}$ and the open subset of linear homeomorphisms of X onto Y. $Y^* := \mathcal{L}(Y, I\!R)$ and $\langle \cdot, \cdot \rangle$ are the dual space of Y and the corresponding duality pairing. Finally, $A^* \in \mathcal{L}(Y^*, X^*)$ denotes the adjoint of a given linear operator $A \in \mathcal{L}(X,Y)$.

Let $F \in C^1(X, Y^*)$ be a given continuously differentiable function. The following proposition yields a posteriori error estimates for elements in a neighbourhood of a solution of

$$F(u) = 0. \tag{2.1}$$

Proposition 2.1. *Let $u_0 \in X$ be a regular solution of Equation (2.1), i. e. $DF(u_0) \in \mathrm{Isom}(X, Y^*)$. Assume that DF is Lipschitz continuous at u_0, i. e., there is an $R_0 > 0$ such that*

$$\gamma := \sup_{u \in B(u_0, R_0)} \frac{\|DF(u) - DF(u_0)\|_{\mathcal{L}(X,Y^*)}}{\|u - u_0\|_X} < \infty.$$

Set

$$R := \min\{ R_0, \gamma^{-1} \|DF(u_0)^{-1}\|^{-1}_{\mathcal{L}(Y^*,X)}, 2\gamma^{-1} \|DF(u_0)\|_{\mathcal{L}(X,Y^*)} \}.$$

Then the following error estimates hold for all $u \in B(u_0, R)$:

$$\begin{aligned} &\frac{1}{2} \|DF(u_0)\|^{-1}_{\mathcal{L}(X,Y^*)} \|F(u)\|_{Y^*} \\ &\leq \|u - u_0\|_X \\ &\leq 2 \|DF(u_0)^{-1}\|_{\mathcal{L}(Y^*,X)} \|F(u)\|_{Y^*}. \end{aligned} \tag{2.2}$$

PROOF: Let $u \in B(u_0, R)$. We then have

$$u - u_0 = DF(u_0)^{-1} \left\{ F(u) + \int_0^1 [DF(u_0) - DF(u_0 + t(u - u_0))](u - u_0) dt \right\}$$

and thus

$$\begin{aligned}\|u-u_0\|_X &\le \|DF(u_0)^{-1}\|_{\mathcal{L}(Y^*,X)}\Big\{\|F(u)\|_{Y^*}\\ &\quad+\int_0^1 \|DF(u_0)-DF(u_0+t(u-u_0))\|_{\mathcal{L}(X,Y^*)}\|u-u_0\|_X\,dt\Big\}\\ &\le \|DF(u_0)^{-1}\|_{\mathcal{L}(Y^*,X)}\Big\{\|F(u)\|_{Y^*}+\frac{1}{2}\gamma\|u-u_0\|_X^2\Big\}\\ &\le \|DF(u_0)^{-1}\|_{\mathcal{L}(Y^*,X)}\|F(u)\|_{Y^*}+\frac{1}{2}\|u-u_0\|_X.\end{aligned}$$

This yields the second inequality in (2.2).
On the other hand, we have for all $\varphi \in Y$ with $\|\varphi\|_Y = 1$

$$\begin{aligned}&\langle F(u),\varphi\rangle\\ &=\langle DF(u_0)(u-u_0),\varphi\rangle\\ &\quad+\Big\langle\int_0^1 [DF(u_0+t(u-u_0))-DF(u_0)](u-u_0)dt,\varphi\Big\rangle\end{aligned}\tag{2.3}$$

and thus

$$\begin{aligned}\|F(u)\|_{Y^*} &\le \|DF(u_0)\|_{\mathcal{L}(X,Y^*)}\|u-u_0\|_X\\ &\quad+\int_0^1 \|DF(u_0+t(u-u_0))-DF(u_0)\|_{\mathcal{L}(X,Y^*)}\|u-u_0\|_X\,dt\\ &\le \|DF(u_0)\|_{\mathcal{L}(X,Y^*)}\|u-u_0\|_X+\frac{1}{2}\gamma\|u-u_0\|_X^2\\ &\le 2\|DF(u_0)\|_{\mathcal{L}(X,Y^*)}\|u-u_0\|_X.\end{aligned}$$

This proves the first inequality in (2.2). □

Remark 2.2. In the examples of Section 3, X and Y are closed subspaces of suitable Sobolev spaces of functions defined on an open set $\Omega \subset \mathbb{R}^n$. When considering in Equation (2.3) only functions φ with support in a given open subset $\omega \subset \Omega$, one then often obtains lower bounds for $u - u_0$ restricted to ω. □

Remark 2.3. The conditions about F can be weakened. Assume, e. g., that $F \in C(X, Y^*)$, $F(u_0) = 0$ and that there are an $R > 0$ and two monotonically increasing homeomorphisms ϱ, σ of $[0,\infty)$ onto itself such that

$$\varrho(\|u-u_0\|_X) \le \|F(u)\|_{Y^*} \le \sigma(\|u-u_0\|_X) \quad \forall u \in B(u_0,R). \tag{2.4}$$

Then we trivially have

$$\sigma^{-1}(\|F(u)\|_{Y^*}) \leq \|u - u_0\|_X \leq \varrho^{-1}(\|F(u)\|_{Y^*}) \quad \forall u \in B(u_0, R).$$

The first inequality in (2.4) is satisfied if, e. g., F is strongly monotone in a neighbourhood of u_0. The second inequality in (2.4) holds if, e. g., F is Hölder continuous at u_0. □

Remark 2.4. When applying Proposition 2.1 to elliptic partial differential equations, one obtains bounds on the error measured in a $W^{1,p}$-norm. For problems in continuum mechanics, e.g., this corresponds to estimating the strain energy. Sometimes, however, one is more interested in an error control with respect to an L^p-norm. For problems in fluid mechanics, e.g., this corresponds to estimating the kinetic energy. In order to cover this case by the present abstract framework, we must enlarge the space X and restrict the space Y. We therefore consider three additional Banach spaces X_-, X_+, and Y_+ such that $X_+ \subset X \subset X_-$ and $Y_+ \subset Y$ with continuous and dense injections. Here, the $+/-$ sign indicates a space with a stronger/weaker norm. We assume that X_- is reflexive. In addition to the assumptions of Proposition 2.1 we now suppose that $u_0 \in X_+$, that $DF(u_0)^* \in \mathrm{Isom}(Y_+, X_-^*)$, and that there are two numbers $R_0 > 0$ and $\beta > 0$ such that

$$\begin{aligned}\|[DF(u_0) - DF(u_0 + tw)]w\|_{Y_+^*} \leq & \beta t \|w\|_{X_+} \|w\|_{X_-} \\ & \forall t \in [0,1], w \in X_+, \|w\|_{X_+} < R_0.\end{aligned}$$

Set

$$R := \min\{R_0, \beta^{-1}\|DF(u_0)^{*-1}\|^{-1}_{\mathcal{L}(X_-^*, Y_+)}, 2\beta^{-1}\|DF(u_0)^*\|_{\mathcal{L}(Y_+, X_-^*)}\}.$$

Then a straightforward modification of the proof of Proposition 2.1 (cf. [88; Proposition 2.3]) shows that the error estimate

$$\begin{aligned}&\frac{1}{2}\|DF(u_0)^*\|^{-1}_{\mathcal{L}(Y_+, X_-^*)}\|F(u_0)\|_{Y_+^*} \\ \leq &\|u - u_0\|_{X_-} \\ \leq &2\|DF(u_0)^{*-1}\|_{\mathcal{L}(X_-^*, Y_+)}\|F(u)\|_{Y_+^*}\end{aligned} \tag{2.5}$$

holds for all $u \in X_+$ satisfying $\|u - u_0\|_{X_+} < R$.
The conditions of Proposition 2.1 imply that $DF(u_0)^* \in \mathrm{Isom}(Y^{**}, X^*)$. The

condition $DF(u_0)^* \in \mathrm{Isom}(Y_+, X_-^*)$ is more restrictive. For partial differential equations, it is equivalent to an additional regularity assumption. For linear problems, i.e. when DF is constant, one may extend F by continuity to a continuously differentiable map of X_- to Y_+^*. Then the space X_+ is not needed. For nonlinear problems, however, this extension often is impossible or the derivative of the extension is no longer Lipschitz continuous. This is the place where the space X_+ comes into play. □

2.2. Error Estimates for Branches of Solutions.

In this chapter we shortly outline how the results of the previous chapter may be extended to branches of solutions of Equation (2.1) including simple limit and bifurcation points. To this end we assume that $X = \mathbb{R}^m \times V$, $m \geq 1$, and that $u_0 = (\lambda_0, v_0)$ is a solution of Equation (2.1).

We first consider the case that u_0 is a *regular point,* i. e. $D_V F(u_0) \in \mathrm{Isom}(V, Y^*)$. The implicit function theorem then implies that there are neighbourhoods I of λ_0 in $\mathbb{R}^m$ and U of v_0 in V and a continuous map $\lambda \to v_\lambda$ from I into U such that $v_{\lambda_0} = v_0$ and every $u_\lambda := (\lambda, v_\lambda)$ is a solution of Equation (2.1) with $D_V F(u_\lambda) \in \mathrm{Isom}(V, Y^*)$. Assume that there is an $R_0^* > 0$ such that

$$\gamma^* := \sup_{\lambda \in I} \sup_{v \in B(v_\lambda, R_0^*)} \frac{\|D_V F(\lambda, v) - D_V F(\lambda, v_\lambda)\|_{\mathcal{L}(V,Y^*)}}{\|v - v_\lambda\|_V} < \infty$$

and set

$$R^* := \min\Big\{R_0^*, \gamma^{*^{-1}} \sup_{\lambda \in I} \|D_V F(u_\lambda)^{-1}\|^{-1}_{\mathcal{L}(Y^*,V)}, 2\gamma^{*^{-1}} \sup_{\lambda \in I} \|D_V F(u_\lambda)\|_{\mathcal{L}(V,Y^*)}\Big\}.$$

With the same arguments as in the proof of Proposition 2.1 we then obtain for all $\lambda \in I$ and all $v \in B(v_\lambda, R^*)$ the estimates

$$\begin{aligned} &\frac{1}{2}\|D_V F(u_\lambda)\|^{-1}_{\mathcal{L}(V,Y^*)}\|F(\lambda, v)\|_{Y^*} \\ &\leq \|v - v_\lambda\|_V \\ &\leq 2\|D_V F(u_\lambda)^{-1}\|_{\mathcal{L}(Y^*,V)}\|F(\lambda, v)\|_{Y^*}. \end{aligned} \tag{2.6}$$

As described in [39], the case, that u_0 is not a regular point, but a simple limit or bifurcation point, may be reduced to the case of a regular point by suitably blowing-up the spaces X and Y and modifying the function F. For completeness, we shortly describe this procedure.

Consider first the case that u_0 is a *simple limit point,* i. e., $DF(u_0)$ is a Fredholm operator of X onto Y^* with index m and $\mathrm{Range}(DF(u_0)) = Y^*$ but $D_V F(u_0) \notin$

$\mathrm{Isom}(V, Y^*)$. Choose a linear operator $B \in \mathcal{L}(X, \mathbb{R}^m)$ with $\ker(B) \cap \ker(DF(u_0)) = \{0\}$ and define $\Phi \in C^1(\mathbb{R}^m \times X,\ \mathbb{R}^m \times Y^*)$ by

$$\Phi(t,u) := (B(u-u_0) - t, F(u)).$$

Then, $(0, u_0)$ is a regular point of Φ (with respect to the parameter t) and we are back to the situation described in the first part of this section. Since B is linear, conditions about the Lipschitz continuity of $D\Phi$ reduce to those on DF. Equation (2.6) yields in this case estimates of the form

$$\begin{aligned} &\underline{c}\{\, \|B(u-u_0) - t\|_{\mathbb{R}^m} + \|F(u)\|_{Y^*} \,\} \\ &\le \|\lambda - \lambda_t\|_{\mathbb{R}^m} + \|v - v_t\|_V \\ &\le \overline{c}\{\, \|B(u-u_0) - t\|_{\mathbb{R}^m} + \|F(u)\|_{Y^*} \,\} \end{aligned} \tag{2.7}$$

for all t in a suitable neighbourhood of 0 and all $u = (\lambda, v)$ in a suitable neighbourhood of $u_t = (\lambda_t, v_t)$. Here, $t \to u_t$ is a regular branch of solutions of $\Phi(t,u) = 0$. Note, that Bu_0 is often known explicitly and that the estimation of $\|B(u-u_0) - t\|_{\mathbb{R}^m}$ is straightforward since it is a low dimensional maximization problem. The term $\|F(u)\|_{Y^*}$, on the other hand, may be estimated by the methods of the next chapter as in the case of regular solutions.

Next, we consider the case of a *simple bifurcation from the trivial branch.* I. e., we assume that $u_0 = (\lambda_0, 0)$ and that $D_V F(u_0)$ is a Fredholm operator with index 0 and $\dim\ker(D_V F(u_0)) = 1$. Choose a $w_0 \in \ker(D_V F(u_0)) \setminus \{0\}$ and a linear functional $l \in \mathcal{L}(V, \mathbb{R})$ with $l(w_0) = 1$. Define the function $\Phi \in C(\mathbb{R}\times X,\ \mathbb{R}\times Y^*)$ by

$$\Phi(t,u) := \begin{cases} \left(l(v) - 1,\ \frac{1}{t}F(\lambda, tv)\right) & t \neq 0, \quad u = (\lambda, v) \in X \\ \left(l(v) - 1,\ D_V F(\lambda, 0)v\right) & t = 0, \quad u = (\lambda, v) \in X. \end{cases}$$

Conditions about the Lipschitz continuity of $D\Phi$ now reduce to those on D^2F. Obviously, we have $\Phi(0, \widetilde{u}_0) = 0$ where $\widetilde{u}_0 := (\lambda_0, w_0)$. If F is of class C^2 in a neighbourhood of u_0 and $D^2_{\lambda v} F(u_0)w_0 \notin \mathrm{Range}\, D_V F(u_0)$, we conclude that $\widetilde{u}_0$ is a regular point and we are once more back to the situation described in the first part of this section. Equation (2.6) now yields estimates of the form

$$\begin{aligned} &\underline{c}\{\, |l(w) - 1| + \|D_V F(\lambda, 0)w\|_{Y^*} \,\} \\ &\le \|\lambda - \lambda_0\|_{\mathbb{R}^m} + \|w - w_0\|_V \\ &\le \overline{c}\{\, |l(w) - 1| + \|D_V F(\lambda, 0)w\|_{Y^*} \,\} \end{aligned} \tag{2.8}$$

for all (λ, w) in a suitable neighbourhood of $\widetilde{u}_0$ and

$$\begin{aligned} &\underline{c}\Big\{|l(w)-1| + \Big\|\frac{1}{t}F(\lambda, tw)\Big\|_{Y^*}\Big\} \\ &\leq \|\lambda - \lambda_t\|_{\mathbb{R}^m} + \|w - w_t\|_V \\ &\leq \overline{c}\Big\{|l(w)-1| + \Big\|\frac{1}{t}F(\lambda, tw)\Big\|_{Y^*}\Big\} \end{aligned} \tag{2.9}$$

for all $t \neq 0$ in a neighbourhood of 0 and all (λ, w) in a suitable neighbourhood of $\widetilde{u}_t = (\lambda_t, w_t)$. Here, $t \to \widetilde{u}_t$ is a regular branch of solutions of $\Phi(t, u) = 0$. Note, that the constants in Equations (2.8), (2.9) now depend on second derivatives of F.

Finally, we consider the case of a *simple bifurcation point*, i.e. $D_V F(u_0)$ is a Fredholm operator of index 0 and $q := \dim(\ker DF(u_0)) - m \geq 1$. Choose a basis $\varphi_1^*, \ldots, \varphi_q^*$ of $Y^* \setminus \text{Range}(DF(u_0))$, set $\widehat{X} := \mathbb{R}^q \times X$, $\widehat{u}_0 := (0, u_0)$, and define the function $\widehat{F} \in C^1(\widehat{X}, Y^*)$ by

$$\widehat{F}(\widehat{u}) := F(u) - \sum_{i=1}^{q} f_i \varphi_i^* \quad \forall \widehat{u} = (f, u) \in \widehat{X}.$$

Obviously, we have $\widehat{F}(\widehat{u}_0) = 0$. Moreover, $D\widehat{F}(\widehat{u}_0)$ is a Fredholm operator with index $m + q$ and $\text{Range}(D\widehat{F}(\widehat{u}_0)) = Y^*$. Replacing X, u_0, and F by $\widehat{X}$, $\widehat{u}_0$, and $\widehat{F}$, respectively, we are thus back to the situation considered in the second part of this section.

2.3. Estimation of the Residual.

Let $X_h \subset X$ and $Y_h \subset Y$ be finite dimensional subspaces and $F_h \in C(X_h, Y_h^*)$ be an approximation of F. We want to estimate $\|F(u_h)\|_{Y^*}$ where $u_h \in X_h$ is an approximate solution of

$$F_h(u_{0h}) = 0. \tag{2.10}$$

In what follows, $c, c_0, c_1, \ldots$ denote various constants which are independent of h.

Proposition 2.5. *Let $u_h \in X_h$ be an approximate solution of Equation (2.10), i.e., $\|F_h(u_h)\|_{Y_h^*}$ is "small". Assume that there are a restriction operator $R_h \in \mathcal{L}(Y, Y_h)$, a finite dimensional subspace $\widetilde{Y}_h \subset Y$, and an approximation $\widetilde{F}_h : X_h \to Y^*$ of F at u_h such that*

$$\|(\text{Id}_Y - R_h)^* \widetilde{F}_h(u_h)\|_{Y^*} \leq c_0 \|\widetilde{F}_h(u_h)\|_{\widetilde{Y}_h^*}. \tag{2.11}$$

Then the following estimates hold

$$\begin{aligned}\|F(u_h)\|_{Y^*} \le c_0\|\widetilde{F}_h(u_h)\|_{\widetilde{Y}_h^*} &+ \|(\mathrm{Id}_Y - R_h)^*[F(u_h) - \widetilde{F}_h(u_h)]\|_{Y^*} \\ &+ \|R_h\|_{\mathcal{L}(Y,Y_h)}\|F(u_h) - F_h(u_h)\|_{Y_h^*} \\ &+ \|R_h\|_{\mathcal{L}(Y,Y_h)}\|F_h(u_h)\|_{Y_h^*} \end{aligned} \tag{2.12}$$

and

$$\|\widetilde{F}_h(u_h)\|_{\widetilde{Y}_h^*} \le \|F(u_h)\|_{\widetilde{Y}_h^*} + \|F(u_h) - \widetilde{F}_h(u_h)\|_{\widetilde{Y}_h^*}. \tag{2.13}$$

□

Remark 2.6. In the examples of Section 3, X_h and Y_h are suitable finite element spaces. The choice of R_h then is quite natural. $\widetilde{F}_h(u_h)$ is obtained by projecting $F(u_h)$ elementwise onto suitable finite dimensional spaces. This construction is rather canonical too. The main difficulty is to find a space $\widetilde{Y}_h$ such that Inequality (2.11) is satisfied. This task is simplified by the auxiliary results of Section 3.1. The second terms on the right-hand sides of Equations (2.12) and (2.13) measure the quality of the approximation $\widetilde{F}_h(u_h)$ to $F(u_h)$. Usually they are higher order terms when compared with $\|\widetilde{F}_h(u_h)\|_{\widetilde{Y}_h^*}$. The term $\|F(u_h) - F_h(u_h)\|_{Y_h^*}$ is the consistency error of the discretization. The term $\|F_h(u_h)\|_{Y_h^*}$ measures the residual of the algebraic equation (2.10) and must be evaluated seperately. □

PROOF OF PROPOSITION 2.5: Consider an arbitrary element $\varphi \in Y$ with $\|\varphi\|_Y = 1$. We then have

$$\begin{aligned}\langle F(u_h), \varphi\rangle &= \langle \widetilde{F}_h(u_h), \varphi - R_h\varphi\rangle + \langle F(u_h) - \widetilde{F}_h(u_h), \varphi - R_h\varphi\rangle \\ &\quad + \langle F(u_h) - F_h(u_h), R_h\varphi\rangle + \langle F_h(u_h), R_h\varphi\rangle \\ &\le \|(\mathrm{Id}_Y - R_h)^*\widetilde{F}_h(u_h)\|_{Y^*} + \|(\mathrm{Id}_Y - R_h)^*[F(u_h) - \widetilde{F}_h(u_h)]\|_{Y^*} \\ &\quad + \|R_h\|_{\mathcal{L}(Y,Y_h)}\|F(u_h) - F_h(u_h)\|_{Y_h^*} + \|R_h\|_{\mathcal{L}(Y,Y_h)}\|F_h(u_h)\|_{Y_h^*}.\end{aligned}$$

Together with Inequality (2.11), this proves Estimate (2.12).
Estimate (2.13) follows from the triangle inequality. □

When combining Propositions 2.1 and 2.5 we obtain a residual a posteriori error estimator. The following proposition together with Proposition 2.1 yields a framework for some of the a posteriori error estimators of Section 1.3, which are based on the solution of auxiliary local problems.

Proposition 2.7. *Let $u_h \in X_h$ be an approximate solution of Equation (2.10). Assume that there are finite dimensional subspaces $\widehat{X}_h \subset X$ and $\widehat{Y}_h \subset Y$ and a linear operator $B \in \mathrm{Isom}(\widehat{X}_h, \widehat{Y}_h^*)$ such that $\widetilde{Y}_h \subset \widehat{Y}_h$ and*

$$\|\widetilde{F}_h(u_h)\|_{\widehat{Y}_h^*} \le c_1\|\widetilde{F}_h(u_h)\|_{\widetilde{Y}_h^*}. \tag{2.14}$$

Let $\widehat{u}_h \in \widehat{X}_h$ be the unique solution of

$$\langle B\widehat{u}_h, \varphi\rangle = \langle \widetilde{F}_h(u_h), \varphi\rangle \quad \forall \varphi \in \widehat{Y}_h. \tag{2.15}$$

Then the following estimates hold

$$\|B\|^{-1}_{\mathcal{L}(\widehat{X}_h, \widehat{Y}_h^*)} \|\widetilde{F}_h(u_h)\|_{\widetilde{Y}_h^*} \leq \|u_h\|_{\widehat{X}_h} \leq c_1 \|B^{-1}\|_{\mathcal{L}(\widehat{Y}_h^*, \widehat{X}_h)} \|\widetilde{F}_h(u_h)\|_{\widetilde{Y}_h^*}. \tag{2.16}$$

PROOF: Since $B \in \mathrm{Isom}(\widehat{X}_h, \widehat{Y}_h^*)$, we immediately obtain from Equation (2.15) the estimate

$$\|B\|^{-1}_{\mathcal{L}(\widehat{X}_h, \widehat{Y}_h^*)} \|\widetilde{F}_h(u_h)\|_{\widehat{Y}_h^*} \leq \|u_h\|_{\widehat{X}_h} \leq \|B^{-1}\|_{\mathcal{L}(\widehat{Y}_h^*, \widehat{X}_h)} \|\widetilde{F}_h(u_h\|_{\widehat{Y}_h^*}. \tag{2.17}$$

Together with Inequality (2.14) this proves the upper bound of Inequality (2.16). Since $\widetilde{Y}_h \subset \widehat{Y}_h$, we have

$$\|\widetilde{F}_h(u_h)\|_{\widetilde{Y}_h^*} \leq \|\widetilde{F}_h(u_h)\|_{\widehat{Y}_h^*}.$$

Together with Inequality (2.17) this proves the lower bound of Inequality (2.16).□

Remark 2.8. Usually B is some approximation of $DF(u_h)$. The construction of $\widehat{Y}_h$ and the proof of Inequality (2.14) are similar to the construction of $\widetilde{Y}_h$ and the proof of Inequality (2.11) and are simplified by the auxiliary results of Section 3.1. Once B and $\widehat{Y}_h$ are chosen, the construction of $\widehat{X}_h$ is quite obvious from the condition that $B \in \mathrm{Isom}(\widehat{X}_h, \widehat{Y}_h^*)$. □

Remark 2.9. Propositions 2.5 und 2.7 can easily be modified such that they cover the situation described in Remark 2.4 und yield upper and lower bounds on $\|F(u_h)\|_{Y_+^*}$ (cf. [88; Proposition 2.4]). To this end one has to proceed as follows:

(1) Replace Y by Y_+.
(2) Equip the spaces $\widetilde{Y}_h$ and $\widehat{Y}_h$ with the norm of Y_+ and equip $\widehat{X}_h$ with the norm of X_-.
(3) Replace the third and fourth term in Inequality (2.12) by

$$\|R_h^*[F(u_h) - F_h(u_h)]\|_{Y_+^*} + \|R_h^* F_h(u_h)\|_{Y_+^*}.$$

A non-optimal estimate of the quantities in item (3) above is given by

$$\|Id_{Y_+}\|_{\mathcal{L}(Y_+, Y)} \|R_h\|_{\mathcal{L}(Y, Y_h)} \{\|F(u_h) - F_h(u_h)\|_{Y_h^*} + \|F_h(u_h)\|_{Y_h^*}\}.$$

Here – as in Propositions 2.5 and 2.7 – Y_h is equipped with the norm of Y. The condition $\widetilde{Y}_h \subset \widehat{Y}_h \subset Y_+$ is stronger than the assumption $\widetilde{Y}_h \subset \widehat{Y}_h \subset Y$ needed in Propositions 2.5 and 2.7. When applied, e.g., to partial differential equations of 2nd oder it implies that the functions in $\widetilde{Y}_h$ and $\widehat{Y}_h$ must at least be continuously differentiable across inter-element boundaries. The space Y_h, however, must only be contained in Y. Furthermore, Remark 2.4 requires that $X_h \subset X_+$. For practical applications, however, this is not a restriction since typically $X_+ \subset W^{1,p}(\Omega)$ with $p \leq \infty$ and $X_h \subset W^{1,\infty}(\Omega)$. □

3. Finite Element Discretizations of Elliptic PDEs

3.1. Auxiliary Results.

In this section we extend the notations and auxiliary results of Section 1.1 to domains in $\mathbb{R}^n$, $n \geq 2$, and to L^p-spaces.

Let Ω be a bounded, connected, open domain in $\mathbb{R}^n$, $n \geq 2$, with polyhedral boundary Γ. For any open subset $\omega \subset \Omega$ with Lipschitz boundary γ, we denote by $W^{k,s}(\omega)$, $k \in \mathbb{N}$, $1 \leq s \leq \infty$, $L^s(\Omega) := W^{0,s}(\omega)$, and $L^s(\gamma)$ the usual Sobolev- and Lebesgue-spaces equipped with the standard norms $\|\cdot\|_{k,s;\omega} := \|\cdot\|_{W^{k,s}(\omega)}$ and $\|\cdot\|_{s;\gamma} := \|\cdot\|_{L^s(\gamma)}$ (cf. [1]). Note, that $H^1(\Omega) = W^{1,2}(\Omega)$. If $\omega = \Omega$, we omit the index ω. We use the same notation for the corresponding norms of vector-valued functions.

Let $\mathcal{T}_h$, $h > 0$, be a family of partitions of Ω into n-simplices which satisfies the following conditions:

(1) Any two simplices in $\mathcal{T}_h$ are either disjoint or share a complete smooth submanifold of their boundaries.
(2) The ratio h_T/ϱ_T is bounded from above independently of $T \in \mathcal{T}_h$ and $h > 0$.

Here, h_T, ϱ_T, and h_E denote the diameter of $T \in \mathcal{T}_h$, the diameter of the largest ball inscribed into T, and the diameter of a face E of T. Condition (2) is the n-dimensional analogue of Condition (2) of Section 1.1. It allows the use of locally refined meshes and it implies that the ratio h_T/h_E is for all $T \in \mathcal{T}_h$ and all faces E of T bounded from above and from below by constants which are independent of h, T, and E.

In analogy to Section 1.1 we denote for any $T \in \mathcal{T}_h$ by $\mathcal{E}(T)$ and $\mathcal{N}(T)$ the set of its faces and vertices, respectively and set

$$\mathcal{E}_h := \bigcup_{T \in \mathcal{T}_h} \mathcal{E}(T), \quad \mathcal{N}_h := \bigcup_{T \in \mathcal{T}_h} \mathcal{N}(T).$$

$\mathcal{E}_h$ and $\mathcal{N}_h$ may be decomposed as $\mathcal{E}_h = E_{h,\Omega} \cup \mathcal{E}_{h,\Gamma}$ and $\mathcal{N}_h = \mathcal{N}_{h,\Omega} \cup \mathcal{N}_{h,\Gamma}$ where $\mathcal{E}_{h,\Gamma}$ and $\mathcal{N}_{h,\Gamma}$ denote the set of all faces respective vertices lying on Γ. Note that the faces $E \in \mathcal{E}_h$ are $(n-1)$-simplices. $\mathcal{N}(E)$, $E \in \mathcal{E}_h$, denotes the set of the vertices of E. Given $T \in \mathcal{T}_h$, $E \in \mathcal{E}_h$, and $x \in \mathcal{N}_h$ we define (see Figure 2 in

Section 1.1 for the two-dimensional situation)

$$\omega_T := \bigcup_{\mathcal{E}(T)\cap\mathcal{E}(T')\neq\emptyset} T', \quad \omega_E := \bigcup_{E\in\mathcal{E}(T')} T', \qquad \omega_x := \bigcup_{x\in\mathcal{N}(T')} T',$$
$$\widetilde{\omega}_T := \bigcup_{\mathcal{N}(T)\cap\mathcal{N}(T')\neq\emptyset} T', \quad \widetilde{\omega}_E := \bigcup_{\mathcal{N}(E)\cap\mathcal{N}(T')\neq\emptyset} T'.$$

For any $E \in \mathcal{E}_h$ and any piecewise continuous function φ, we denote by $[\varphi]_E$ the jump of φ across E in the direction n_E:

$$[\varphi]_E(x) := \lim_{t\to 0+} \varphi(x + tn_e) - \lim_{t\to 0+} \varphi(x - tn_E)\ \forall x \in E.$$

Here, φ is continued by 0 outside Ω and n_E is a unit vector orthogonal to E which coincides with the exterior normal of Γ if $E \in \mathcal{E}_{h,\Gamma}$.

For $k \in I\!N$, we define

$$S_h^{k,-1} := \{\, \varphi : \Omega \to I\!R : \varphi|_T \in \Pi_k\ \forall T \in \mathcal{T}_h \,\},$$
$$S_h^{k,0} := S_h^{k,-1} \cap C(\bar{\Omega}).$$

Here, Π_k, $k \geq 0$, is the space of polynomials of degree at most k. Moreover, we denote by $\pi_{k,S}$, $S \in \mathcal{T}_h \cup \mathcal{E}_h$, the L^2-projection of $L^1(S)$ onto $\Pi_{k|S}$.

As in Section 1.1 we denote by $I_h : L^1(\Omega) \to S_h^{1,0}$ the interpolation operator of Clément [38] given by Equations (1.7) and (1.8). Adapting the usual convention that $1/\infty := 0$, the following analogue of Lemma 1.4 holds:

Lemma 3.1. *For any $T \in \mathcal{T}_h$, $E \in \mathcal{E}_h$, and $1 \leq q \leq \infty$ the following error estimates are valid*

$$\|\varphi - I_h\varphi\|_{k,q;T} \leq c_1 h_T^{l-k} \|\varphi\|_{l,q;\widetilde{\omega}_T} \quad \forall 0 \leq k \leq l \leq 2,\ \varphi \in W^{l,q}(\widetilde{\omega}_T)$$
$$\|\varphi - I_h\varphi\|_{q;E} \leq c_2 h_E^{l-1/q} \|\varphi\|_{l,q;\widetilde{\omega}_E} \quad \forall 1 \leq l \leq 2,\ \varphi \in W^{l,q}(\widetilde{\omega}_E).$$

The constants c_1 and c_2 only depend on the upper bound in Condition (2) above. □

Denote by $\widehat{T} := \{\, \widehat{x} \in I\!R^n : \sum_{i=1}^n \widehat{x}_i \leq 1, \widehat{x}_j \geq 0,\ 1 \leq j \leq n \,\}$ the reference n-simplex. Set $\widehat{E} := \widehat{T} \cap \{\, \widehat{x} \in I\!R^n : \widehat{x}_n = 0 \,\}$. Let $\psi_{\widehat{T}}, \psi_{\widehat{E}} \in C^\infty(\widehat{T}, I\!R)$ be two functions with the following properties

$$0 \leq \psi_{\widehat{T}} \leq 1,\ \max_{\widehat{x}\in\widehat{T}} \psi_{\widehat{T}}(\widehat{x}) = 1, \quad \psi_{\widehat{T}} = 0 \text{ on } \partial\widehat{T},$$
$$0 \leq \psi_{\widehat{E}} \leq 1,\ \max_{\widehat{x}\in\widehat{E}} \psi_{\widehat{E}}(\widehat{x}) = 1, \quad \psi_{\widehat{E}} = 0 \text{ on } \partial\widehat{T} \setminus \widehat{E}.$$

We define a continuation operator $\widehat{P}\colon L^\infty(\widehat{E}) \to L^\infty(\widehat{T})$ by

$$\widehat{P}\widehat{u}(\widehat{x}_1,\dots,\widehat{x}_n) := \widehat{u}(\widehat{x}_1,\dots,\widehat{x}_{n-1}) \quad \forall \widehat{x} \in \widehat{T},\ \widehat{u} \in L^\infty(\widehat{E}).$$

Finally, $V_{\widehat{T}} \subset L^\infty(\widehat{T})$ and $V_{\widehat{E}} \subset L^\infty(\widehat{E})$ are two arbitrary finite dimensional spaces which are kept fixed throughout this section.

Let $T \in \mathcal{T}_h$ be an arbitrary n-simplex and $E \subset \partial T$ be a face of T. There is an invertible affine mapping $F_T\colon \widehat{T} \to T$, $\widehat{x} \to x := F_T(\widehat{x}) = b_T + B_T\widehat{x}$ such that $\widehat{T}$ is mapped onto T and $\widehat{E}$ is mapped onto E. Denote by B'_T the matrix, which is obtained from B_T by discarding its last column, and set $\beta_T := \det({B'}^t_T B'_T)^{1/2}$. β_T is the Gram determinant of the transformation $\widehat{E} \to E$. Set

$$\psi_T := \psi_{\widehat{T}} \circ F_T^{-1}, \qquad \psi_E := \psi_{\widehat{E}} \circ F_T^{-1},$$
$$V_T := \{\widehat{u} \circ F_T^{-1} : \widehat{u} \in V_{\widehat{T}}\}, \qquad V_E := \{\widehat{\sigma} \circ F_T^{-1} : \widehat{\sigma} \in V_{\widehat{E}}\}.$$

Finally, we define the continuation operator $P\colon L^\infty(E) \to L^\infty(T)$ by

$$P\sigma := [\widehat{P}(\sigma \circ F_T)] \circ F_T^{-1}.$$

Figure 5 shows the level lines of $P\sigma$ in two dimensions.

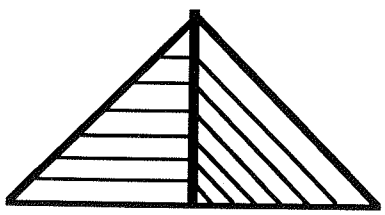


Fig. 5: level lines of $P\sigma$

Remark 3.2. Enumerate the vertices of $\widehat{T}$ such that the vertices of $\widehat{E}$ are numbered first and denote by $\widehat{\lambda}_1,\dots,\widehat{\lambda}_{n+1}$ the corresponding barycentric co-ordinates. The functions

$$\psi_{\widehat{T}} := (n+1)^{n+1} \prod_{i=1}^{n+1} \widehat{\lambda}_i, \quad \psi_{\widehat{E}} := n^n \prod_{i=1}^{n} \widehat{\lambda}_i$$

then fulfill the above conditions. One easily checks that in the case $n = 2$ one recovers the function $\mathit{6}_T$ and $\mathit{6}_E$ of Section 1.1, i. e.

$$\psi_T = \mathit{6}_T, \quad \psi_E = \mathit{6}_E \quad \forall T \in \mathcal{T}_h,\ E \in \mathcal{E}_h.$$

The prolongation operator P was not needed in Section 1. Since there we only considered piecewise constant functions on the edges. In most applications, $V_{\widehat{T}}$ and $V_{\widehat{E}}$ will be suitable spaces of polynomials. □

In what follows, p, q are two fixed real numbers with $1 \le p \le \infty$ and $\frac{1}{p} + \frac{1}{q} = 1$ and $|||\cdot|||$ denotes the spectral norm on $\mathbb{R}^{n\times n}$.

Lemma 3.3. *There are constants $c_1, \ldots, c_7$, which only depend on the spaces $V_{\widehat{T}}$ and $V_{\widehat{E}}$, the functions $\psi_{\widehat{T}}$ and $\psi_{\widehat{E}}$, the number p, and the ratio h_T/ϱ_T, such that the following inequalities hold for all $T \in \mathcal{T}_h$, $E \in \mathcal{E}(T)$, $u \in V_T$, and $\sigma \in V_E$:*

$$c_1\|u\|_{0,p;T} \le \sup_{v\in V_T} \frac{\int_T u\,\psi_T\, v}{\|v\|_{0,q;T}} \le \|u\|_{0,p;T}, \tag{3.1}$$

$$c_2\|\sigma\|_{p;E} \le \sup_{\tau\in V_E} \frac{\int_E \sigma\,\psi_E\,\tau}{\|\tau\|_{q;E}} \le \|\sigma\|_{p;E}, \tag{3.2}$$

$$c_3 h_T^{-1}\|\psi_T u\|_{0,q;T} \le \|\nabla(\psi_T u)\|_{0,q;T} \le c_4 h_T^{-1}\|\psi_T u\|_{0,q;T}, \tag{3.3}$$

$$c_5 h_T^{-1}\|\psi_E P\sigma\|_{0,q;T} \le \|\nabla(\psi_E P\sigma)\|_{0,q;T} \le c_6 h_T^{-1}\|\psi_E P\sigma\|_{0,q;T}, \tag{3.4}$$

$$\|\psi_E P\sigma\|_{0,q;T} \le c_7 h_T^{1/q}\|\sigma\|_{q;E}. \tag{3.5}$$

PROOF: The upper bounds of Equations (3.1), (3.2) immediately follow from Hölder's inequality and $0 \le \psi_T \le 1$, $0 \le \psi_E \le 1$.
In order to prove the lower bound of Equation (3.1), one easily checks that the mapping

$$\widehat{u} \to \sup_{\widehat{v}\in V_{\widehat{T}}} \frac{\int_{\widehat{T}} \widehat{u}\,\psi_{\widehat{T}}\,\widehat{v}}{\|\widehat{v}\|_{0,q;\widehat{T}}}$$

defines a norm on $V_{\widehat{T}}$. Since $\dim V_{\widehat{T}} < \infty$, there is a constant $\widehat{c} > 0$ such that

$$\widehat{c}\,\|\widehat{u}\|_{0,p;\widehat{T}} \le \sup_{\widehat{v}\in V_{\widehat{T}}} \frac{\int_{\widehat{T}} \widehat{u}\,\psi_{\widehat{T}}\,\widehat{v}}{\|\widehat{v}\|_{0,q;\widehat{T}}} \quad \forall \widehat{u} \in V_{\widehat{T}}.$$

Now, take an arbitrary $u \in V_T$. Set $\widehat{u} := u \circ F_T \in V_{\widehat{T}}$ and choose a $\widehat{w} \in V_{\widehat{T}}$ such that

$$\|\widehat{w}\|_{0,q;\widehat{T}} = 1 \quad \text{and} \quad \int_{\widehat{T}} \widehat{u}\,\psi_{\widehat{T}}\,\widehat{w} \ge \widehat{c}\,\|\widehat{u}\|_{0,p;\widehat{T}}.$$

With $w := \widehat{w} \circ F_T^{-1}$ we then obtain

$$\begin{aligned}\sup_{v\in V_T} \frac{\int_T u\,\psi_T\, v}{\|v\|_{0,q;T}} &\ge \frac{\int_T u\,\psi_T w}{\|w\|_{0,q;T}} = |\det B_T|^{1-1/q} \int_{\widehat{T}} \widehat{u}\,\psi_{\widehat{T}}\,\widehat{w} \\ &\ge \widehat{c}\,|\det B_T|^{1/p}\|\widehat{u}\|_{0,p;\widehat{T}} = \widehat{c}\,\|u\|_{0,p;T}.\end{aligned}$$

The proof of the lower bound of Equation (3.2) is completely analogous. One only has to replace $|\det B_T|$ by β_T.
Since $\psi_{\widehat{T}}$ and $\psi_{\widehat{E}}$ vanish at the vertices of $\widehat{T}$, the mappings

$$\widehat{u} \to \|\nabla(\psi_{\widehat{T}}\,\widehat{u})\|_{0,q;\widehat{T}} \quad \text{and} \quad \widehat{\sigma} \to \|\nabla(\psi_{\widehat{E}}\widehat{P}\widehat{\sigma})\|_{0,q;\widehat{T}}$$

define norms on $V_{\widehat{T}}$ and $V_{\widehat{E}}$. Since $\dim V_{\widehat{T}} < \infty$ and $\dim V_{\widehat{E}} < \infty$, these norms are equivalent to $\|\psi_{\widehat{T}}\,\widehat{u}\|_{0,q;\widehat{T}}$ and $\|\psi_{\widehat{E}}\widehat{P}\widehat{\sigma}\|_{0,q;\widehat{T}}$, respectively. Estimates (3.3), (3.4) now follow in the usual way by transforming to $\widehat{T}$, using the equivalence of norms there, and transforming back to T.

With the same arguments as above we finally conclude that there is a constant $\widetilde{c} > 0$ such that

$$\|\psi_{\widehat{E}}\widehat{P}\widehat{\sigma}\|_{0,q;\widehat{T}} \le \widetilde{c}\|\widehat{\sigma}\|_{q;\widehat{E}} \quad \forall \widehat{\sigma} \in V_{\widehat{E}}.$$

Since

$$|\det B_T| \le |||B_T|||^n \le c\,h_T^n, \quad \beta_T^{-1} \le |||B_T^{-1}|||^{n-1} \le c\,h_T^{1-n},$$

this implies for all $\sigma \in V_E$

$$\begin{aligned}\|\psi_E P\,\sigma\|_{0,q;T} &= |\det B_T|^{1/q}\|\psi_{\widehat{E}}\widehat{P}\,\widehat{\sigma}\|_{0,q;\widehat{T}} \le \widetilde{c}\,|\det B_T|^{1/q}\|\widehat{\sigma}\|_{q;\widehat{E}} \\ &= \widetilde{c}\,|\det B_T|^{1/q}\beta_T^{-1/q}\|\sigma\|_{q;E} \le c_7\,h_T^{1/q}\|\sigma\|_{q;E}.\end{aligned}$$

□

Remark 3.4. The estimates of Lemma 3.3 also hold for "slightly curved" simplices. More precisely, assume that the transformation F_T is no longer affine, but that it still is a diffeomorphism. Let $A_T\colon \widehat{T} \to I\!R^n$ be the invertible affine mapping which is uniquely determined by the condition that $A_T^{-1} \circ F_T$ leaves the vertices of $\widehat{T}$ invariant. Denote by α_T the Gram determinant of the transformation of $\widehat{E}$ induced by A_T. A simple perturbation argument then shows that the estimates of Lemma 3.3 remain valid, provided

$$\begin{aligned}&\| \,|||I - DF_T^{-1}DA_T|||\,\|_{0,\infty;\widehat{T}}, && \| \,|||I - DA_T^{-1}DF_T|||\,\|_{0,\infty;\widehat{T}}, \\ &\|1 - |\det DF_T|^{-1}|\det DA_T|\,\|_{0,\infty;\widehat{T}}, && \|1 - \beta_T^{-1}\alpha_T\|_{0,\infty;\widehat{T}}\end{aligned}$$

are smaller than a positive threshold which only depends on the constants in the corresponding estimates on $\widehat{T}$. Note that the faces and vertices of a curved element T are defined as the images under F_T of the faces respective vertices of $\widehat{T}$. □

Remark 3.5. Partitions into generalized n-cubes may be treated in the same way as those into curved simplices. More precisely, denote by $\widehat{Q} := [0,1]^n$ the reference n-cube and by $\widehat{S} := \widehat{Q} \cap \{\widehat{x} \in I\!R^n : \widehat{x}_n = 0\}$ its reference face. Assume that for every element $Q \in \mathcal{T}_h$ and every complete, smooth $(n-1)$-dimensional submanifold S of ∂Q there exists a diffeomorphism $F_Q : \widehat{Q} \to Q$ which maps $\widehat{S}$ onto S. Denote by β_Q the Gram determinant of the transformation $\widehat{S} \to S$ induced by F_Q. In most applications, the components of F_Q are n-linear functions. The edges of Q then are

straight lines. Isoparametric elements of higher order, however, also fall into the category considered here. Let $A_Q : \widehat{T} \to I\!R^n$ be the invertible affine transformation which is uniquely determined by the condition that $A_Q^{-1} \circ F_Q$ leaves invariant the vertices of the reference simplex $\widehat{T}$. Denote by α_Q the Gram determinant of the transformation of $\widehat{E}$ induced by A_Q. Replacing $\widehat{T}, \widehat{E}, T, E$, and F_T by $\widehat{Q}, \widehat{S}, Q, S$, and F_Q respectively, the results of Lemma 3.3 remain valid provided

$$\| \, |||I - DF_Q^{-1} DA_Q||| \, \|_{0,\infty;\widehat{Q}}, \qquad \| \, |||I - DA_Q^{-1} DF_Q||| \, \|_{0,\infty;\widehat{Q}},$$
$$\|1 - |\det DF_Q|^{-1} |\det DA_Q| \, \|_{0,\infty;\widehat{Q}}, \quad \|1 - \beta_Q^{-1} \alpha_Q\|_{0,\infty;\widehat{Q}}$$

is smaller than a positive threshold which only depends on the corresponding estimates on $\widehat{Q}$. Geometrically this means that each $Q \in \mathcal{T}_h$ is close to a parallelogram if $n = 2$ or to a parallelepiped if $n = 3$. □

Thanks to Lemma 3.3 we may construct in the next sections spaces $\widetilde{Y}_h$ and $\widehat{Y}_h$ satisfying the conditions of Propositions 2.5 and 2.7 by considering all linear combinations of functions $\psi_T v$ and $\psi_E P\sigma$ where v and σ vary in suitable spaces V_T and V_E, respectively, and T and E run through all simplices and faces of the finite element partition.

Remark 3.6. In order to apply Remark 2.9 we must construct spaces $\widetilde{Y}_h$ and $\widehat{Y}_h$ consisting of functions which are globally k-times continuously differentiable. Here $k \geq 1$ is a fixed integer; for 2nd order problems $k = 1$ is sufficient. To achieve this we must modify the previous constructions (cf. [88, §3] for a more detailed presentation). First, we must in addition require that

$$\nabla^l \psi_{\widehat{T}} = 0 \quad \text{on} \quad \partial\widehat{T}, 1 \leq l \leq k.$$

Using the notations of Remark 3.2 this can be satisfied, e.g., by setting

$$\psi_{\widehat{T}} := [(n+1)^{n+1} \prod_{i=1}^{n+1} \widehat{\lambda}_i]^{k+1}.$$

Second, the functions $\psi_E, E \in \mathcal{E}_{h,\Omega}$, must globally be of Class C^k, too. To achieve this consider an element $E = T_1 \cap T_2 \in \mathcal{E}_{h,\Omega}$. Enumerate the vertices of T_1 and T_2 such that the vertices of E are numbered first. Then we may set

$$\psi_E = \begin{cases} [n^{2n} \prod_{i=1}^{n} (\lambda_{T_1 i} \lambda_{T_2 i})]^{k+1} & \text{on } \omega_E, \\ 0 & \text{on } \Omega \backslash \omega_E. \end{cases} \tag{3.6}$$

Here – as usual – $\lambda_{T1}, ..., \lambda_{T(n+1)}$ denote the barycentric co-ordinates of $T \in \mathcal{T}_h$. Third, the continuation operator P must be replaced by the operator $\widetilde{P}$ which we will now construct. Consider once more an element E of $\mathcal{E}_{h,\Omega}$. Denote by $x_E = (x_{E1}, ..., x_{En})$ an Euclidean co-ordinate system such that E is contained in the hyperplane $\{x_{En} = 0\}$. Set $x'_E := (x_{E1}, ..., x_{E(n-1)})$ and define

$$\widetilde{P}\sigma(x_E) := \begin{cases} \psi_E(x'_E, 0)\sigma(x'_E, 0) & \text{if } (x'_E, 0) \in E \\ 0 & \text{if } (x'_E, 0) \notin E. \end{cases}$$

Here, ψ_E is the cut-off function defined above. One easily checks that $\widetilde{P}\sigma \in C^k(\Omega)$ if $\sigma \in C^k(E)$. With these modifications Lemma 3.3 still holds (cf. [88; Lemma 3.8]). Moreover, estimates (3.3) and (3.4) can be extended to

$$\widetilde{c}_3 h_T^{-l} \|\psi_T u\|_{0,q;T} \leq \|\nabla^l(\psi_T u)\|_{0,q;T} \leq \widetilde{c}_4 h_T^{-l} \|\psi_T u\|_{0,q;T}, \tag{3.7}$$

$$\widetilde{c}_5 h_T^{-l} \|\psi_E \widetilde{P}\sigma\|_{0,q;T} \leq \|\nabla^l(\psi_E \widetilde{P}\sigma)\|_{0,q;T} \leq \widetilde{c}_6 h_T^{-l} \|\psi_E \widehat{P}\sigma\|_{0,q;T} \tag{3.8}$$

for all $1 \leq l \leq k+1$. □

Remark 3.7. Lemma 3.3 is a generalization to L^p-spaces of Lemma 4.1 in [85] where the L^2-setting is considered. □

3.2. The Model Problem Revisited.

In this section we want to show how the results of Sections 1.2 and 1.3 fit into the abstract framework of Section 2. To this end, we set

$$X = Y := \{ u \in W^{1,2}(\Omega) : u = 0 \text{ on } \Gamma_D \},$$
$$\|\cdot\|_X = \|\cdot\|_Y := \|\cdot\|_{1,2}$$
$$\langle F(u), \varphi \rangle := \int_\Omega \nabla u \cdot \nabla \varphi - \int_\Omega f\,\varphi - \int_{\Gamma_N} g\,\varphi.$$

Then $u \in X$ is a solution of Problem (2.1), i.e. $F(u) = 0$, if and only if it is a solution of Problem (1.3). Since the bilinear form $u, v \to \int_\Omega \nabla u \cdot \nabla v$ is coercive and continuous on X, we have $DF(u) \in \mathrm{Isom}(X, Y^*)$ for all $u \in X$.

The discrete Problem (2.10) is obtained by setting

$$X_h = Y_h := S_h^{1,0} \cap X,$$
$$\langle F_h(u_h), \varphi_h \rangle := \langle F(u_h), \varphi_h \rangle \quad \forall u_h, \varphi_h \in X_h.$$

Obviously, $F_h(u_h) = 0$ if and only if u_h is a solution of Problem (1.4).

We set $R_h := I_h$ and define $\widetilde{F}_h$ and $\widetilde{Y}_h$ by

$$\langle \widetilde{F}_h(u), \varphi \rangle := \int_\Omega \nabla u \cdot \nabla \varphi - \sum_{T \in \mathcal{T}_h} \int_T f_T \varphi - \sum_{E \in \mathcal{E}_{h,N}} \int_E g_E \varphi \quad \forall u, \varphi \in X$$

and

$$\widetilde{Y}_h := \operatorname{span}\{ b_T, b_E : T \in \mathcal{T}_h, E \in \mathcal{E}_{h,\Omega} \cup \mathcal{E}_{h,N} \}$$

where f_T and g_E are given by Equation (1.17).

Let u_h be the unique solution of Problem (2.10), i.e. the unique solution of $F_h(u_h) = 0$. Equation (1.14) then yields for all $\varphi \in Y$

$$\begin{aligned} \langle F(u_h), \varphi \rangle = &- \sum_{T \in \mathcal{T}_h} \int_T f \varphi + \sum_{E \in \mathcal{E}_{h,\Omega}} \int_E [n_E \cdot \nabla u_h]_E \varphi \\ &- \sum_{E \in \mathcal{E}_{h,N}} \int_E (g - n_E \cdot \nabla u_h) \varphi \end{aligned}$$

and – replacing elementwise f and g by f_T and g_E –

$$\begin{aligned} \langle \widetilde{F}_h(u_h), \varphi \rangle = &- \sum_{T \in \mathcal{T}_h} \int_T f_T \varphi + \sum_{E \in \mathcal{E}_{h,\Omega}} \int_E [n_E \cdot \nabla u_h]_E \varphi \\ &- \sum_{E \in \mathcal{E}_{h,N}} \int_E (g_E - n_E \cdot \nabla u_h) \varphi. \end{aligned} \tag{3.9}$$

Lemmas 3.1 and 1.3 imply

$$\begin{aligned} &\|(Id_Y - R_h)^*[F(u_h) - \widetilde{F}_h(u_h)]\|_{Y^*} \\ &= \sup_{\substack{\varphi \in Y \\ \|\varphi\|_Y = 1}} \left\{ \sum_{T \in \mathcal{T}_h} \int_T (f_T - f)(\varphi - I_h \varphi) + \sum_{E \in \mathcal{E}_{h,N}} \int_E (g_E - g)(\varphi - I_h \varphi) \right\} \\ &\le c \sup_{\substack{\varphi \in Y \\ \|\varphi\|_Y = 1}} \left\{ \sum_{T \in \mathcal{T}_h} h_T \|f - f_T\|_{0,2;T} \|\varphi\|_{1,2;\widetilde{\omega}_T} + \sum_{E \in \mathcal{E}_{h,N}} h_E^{1/2} \|g - g_E\|_{2;E} \|\varphi\|_{1,2;\widetilde{\omega}_E} \right\} \\ &\le c' \left\{ \sum_{T \in \mathcal{T}_h} h_T^2 \|f - f_T\|_{0,2;T}^2 + \sum_{E \in \mathcal{E}_{h,N}} h_E \|g - g_E\|_{2;E}^2 \right\}^{1/2} \end{aligned} \tag{3.10}$$

and

$$\begin{aligned}
&\|F(u_h) - \widetilde{F}_h(u_h)\|_{\widetilde{Y}_h^*} \\
&= \sup_{\substack{\varphi_h \in \widetilde{Y}_h \\ \|\varphi_h\|_Y = 1}} \left\{ \sum_{T \in \mathcal{T}_h} \int_T (f_T - f)\varphi_h + \sum_{E \in \mathcal{E}_{h,N}} \int_E (g_E - g)\varphi_h \right\} \\
&\leq c \sup_{\substack{\varphi_h \in \widetilde{Y}_h \\ \|\varphi_h\|_Y = 1}} \left\{ \sum_{T \in \mathcal{T}_h} h_T \|f - f_T\|_{0,2;T} \|\varphi_h\|_{1,2;T} \right. \\
&\qquad \left. + \sum_{E \in \mathcal{E}_{h,N}} h_E^{1/2} \|g - g_E\|_{2;E} \|\varphi_h\|_{1,2;\omega_E} \right\} \\
&\leq c' \left\{ \sum_{T \in \mathcal{T}_h} h_T^2 \|f - f_T\|_{0,2;T}^2 + \sum_{E \in \mathcal{E}_{h,N}} h_E \|g - g_E\|_{2;E}^2 \right\}^{1/2}. \qquad (3.11)
\end{aligned}$$

Replacing elementwise f and g by f_T and g_E, we conclude from Estimate (1.15) that

$$\begin{aligned}
&\|(Id_Y - R_h)^* \widetilde{F}_h(u_h)\|_{Y^*} \\
&= \sup_{\substack{\varphi \in Y \\ \|\varphi\|_Y = 1}} \left\{ \int_\Omega \nabla u_h \cdot \nabla(\varphi - I_h\varphi) - \sum_{T \in \mathcal{T}_h} \int_T f_T(\varphi - I_h\varphi) \right. \\
&\qquad \left. - \sum_{E \in \mathcal{E}_{h,N}} \int_E g_E(\varphi - I_h\varphi) \right\} \\
&\leq c \left\{ \sum_{T \in \mathcal{T}_h} h_T^2 \|f_T\|_{0,2;T}^2 + \sum_{E \in \mathcal{E}_{h,\Omega}} h_E \|[n_E \cdot \nabla u_h]_E\|_{2;E}^2 \right. \\
&\qquad \left. + \sum_{E \in \mathcal{E}_{h,N}} h_E \|g_E - n_E \cdot \nabla u_h\|_{2;E}^2 \right\}^{1/2} \\
&\leq c' \left\{ \sum_{T \in \mathcal{T}_h} \eta_{R,T}^2 \right\}^{1/2}. \qquad (3.12)
\end{aligned}$$

Equation (3.9) and Lemma 1.3 on the other hand yield

$$\begin{aligned}
&\|\widetilde{F}_h(u_h)\|_{\widetilde{Y}_h^*} \\
&= \sup_{\substack{\varphi_h \in \widetilde{Y}_h \\ \|\varphi\|_Y = 1}} \left\{ \sum_{E \in \mathcal{E}_{h,\Omega}} \int_E [n_E \cdot \nabla u_h]_E \varphi_h \right. \\
&\qquad \left. - \sum_{E \in \mathcal{E}_{h,N}} \int_E (g_E - n_E \cdot \nabla u_h)\varphi_h - \sum_{T \in \mathcal{T}_h} \int_T f_T \varphi_h \right\}
\end{aligned}$$

$$\leq c \sup_{\substack{\varphi_h \in \widetilde{Y}_h \\ \|\varphi_h\|_Y = 1}} \left\{ \sum_{E \in \mathcal{E}_{h,\Omega}} h_E^{1/2} \|[n_E \cdot \nabla u_h]_E\|_{2;E} \|\varphi_h\|_{1,2;\omega_E} \right.$$

$$\left. + \sum_{E \in \mathcal{E}_{h,N}} h_E^{1/2} \|g_E - n_E \cdot \nabla u_h\|_{2;E} \|\varphi_h\|_{1,2;\omega_E} + \sum_{T \in \mathcal{T}_h} h_T \|f_T\|_{0,2;T} \|\varphi_h\|_{1,2;T} \right\}$$

$$\leq c' \left\{ \sum_{T \in \mathcal{T}_h} \eta_{R,T}^2 \right\}^{1/2}. \tag{3.13}$$

In order to establish Inequality (2.11) it is therefore sufficient to prove that

$$\left\{ \sum_{T \in \mathcal{T}_h} \eta_{R,T}^2 \right\}^{1/2} \leq c \|\widetilde{F}_h(u_h)\|_{\widetilde{Y}_h^*}. \tag{3.14}$$

To this end we proceed as in the proof of Inequalities (1.23), (1.27), and (1.31). Consider first a triangle $T \in \mathcal{T}_h$ and let $w_T := f_T b_T$ be as in Equation (1.20). Equation (1.21), Lemma 1.3, and Equation (3.9) then imply

$$\frac{9}{20} \|f_T\|_{0,2;T}^2 = \int_T f_T w_T = -\langle \widetilde{F}_h(u_h), w_T \rangle$$

$$\leq \|w_T\|_{1,2} \cdot \sup_{\substack{\varphi \in \widetilde{Y}_{h|T} \\ \|\varphi\|_Y = 1}} \langle \widetilde{F}_h(u_h), \varphi \rangle \leq c\, h_T^{-1} \|f_T\|_{0,2;T} \sup_{\substack{\varphi \in \widetilde{Y}_{h|T} \\ \|\varphi\|_Y = 1}} \langle \widetilde{F}_h(u_h), \varphi \rangle$$

and thus

$$h_T \|f_T\|_{0,2;T} \leq c \sup_{\substack{\varphi \in \widetilde{Y}_{h|T} \\ \|\varphi\|_Y = 1}} \langle \widetilde{F}_h(u_h), \varphi \rangle. \tag{3.15}$$

Here, we denote by $\widetilde{Y}_{h|\omega}$, $\omega \in \{T, \omega_E, \omega_T\}$, the set of all functions $\varphi \in \widetilde{Y}_h$ with $\operatorname{supp}(\varphi) \subset \omega$.

Next, consider an edge $E \in \mathcal{E}_{h,\Omega}$ and let $w_E := [n_E \cdot \nabla u_h]_E b_E$ be as in Equation (1.24). Equation (1.25), Lemma 1.3, and Equation (3.9) now yield

$$\frac{2}{3} \|[n_E \cdot \nabla u_h]_E\|_{2;E}^2$$

$$= \int_E [n_E \cdot \nabla u_h]_E w_E$$

$$= \langle \widetilde{F}_h(u_h), w_E \rangle + \sum_{T \subset \omega_E} \int_T f_T w_E$$

$$\leq \|w_E\|_{1,2} \sup_{\substack{\varphi \in \widetilde{Y}_{h|\omega_E} \\ \|\varphi\|_Y = 1}} \langle \widetilde{F}_h(u_h), \varphi \rangle + \sum_{T \subset \omega_E} \|f_T\|_{0,2;T} \|w_E\|_{0,2;T}$$

$$\leq c\|[n_E \cdot \nabla u_h]_E\|_{2;E}\left\{h_E^{-1/2} \sup_{\substack{\varphi\in\widetilde{Y}_{h|\omega_E}\\ \|\varphi\|_Y=1}} \langle \widetilde{F}_h(u_h),\varphi\rangle + \sum_{T\subset\omega_E} h_E^{1/2}\|f_T\|_{0,2;T}\right\}$$

and thus – using Equation (3.15) –

$$h_E^{1/2}\|[n_E \cdot \nabla u_h]_E\|_{2;E} \leq c \sup_{\substack{\varphi\in\widetilde{Y}_{h|\omega_E}\\ \|\varphi\|_Y=1}} \langle \widetilde{F}_h(u_h),\varphi\rangle. \tag{3.16}$$

Using the function $w_E = (g_E - n_E \cdot \nabla u_h)b_E$ given by Equation (1.28), we obtain with the same arguments for all $E \in \mathcal{E}_{h,N}$

$$h_E^{1/2}\|g_E - n_E \cdot \nabla u_h\|_{2;E} \leq c \sup_{\substack{\varphi\in\widetilde{Y}_{h|\omega_E}\\ \|\varphi\|_Y=1}} \langle \widetilde{F}_h(u_h),\varphi\rangle. \tag{3.17}$$

Combining Estimates (3.15)–(3.17), we get for all $T \in \mathcal{T}_h$

$$\eta_{R,T} \leq c \sup_{\substack{\varphi\in\widetilde{Y}_{h|\omega_T}\\ \|\varphi\|_Y=1}} \langle \widetilde{F}_h(u_h),\varphi\rangle. \tag{3.18}$$

Summing over all $T \in \mathcal{T}_h$, this proves the desired Estimate (3.14).
Proposition 1.5 now follows from Propositions 2.1 and 2.5, Estimates (3.10)–(3.14) and (3.18), and the fact that

$$\int_\Omega \nabla v \cdot \nabla w \leq \|v\|_{1,2;\omega_T}\|w\|_{1,2;\omega_T}$$

holds for all $v, w \in X$ with $\mathrm{supp}(w) \subset \omega_T$.

In order to recover Proposition 1.9 we choose an arbitrary vertex $x \in \mathcal{N}_{h,\Omega} \cup \mathcal{N}_{h,N}$, set

$$\widehat{X}_h := \widehat{Y}_h := \widetilde{Y}_{h|\omega_x} = V_x,$$

where V_x is given by Equation (1.32), and define $B \in \mathcal{L}(\widehat{X}_h, \widehat{Y}_h^*)$ by

$$\langle Bu,\varphi\rangle := \int_{\omega_x} \nabla u \cdot \nabla\varphi \quad \forall u,\varphi \in \widehat{X}_h.$$

One easily checks that $B \in \mathrm{Isom}(\widehat{X}_h, \widehat{Y}_h^*)$ and that $u_x \in \widehat{X}_h$ is a solution of

$$\langle Bu_x,\psi\rangle = \langle \widetilde{F}_h(u_h),\psi\rangle \quad \forall\psi \in \widehat{Y}_h$$

if and only if $-u_x$ solves Equation (1.34). Proposition 1.9 now follows from Proposition 2.7, Equation (3.18) and the arguments leading to Equation (3.13) applied to functions in $\widehat{Y}_h$.

The previous analysis and Remarks 2.4, 2.9, and 3.6 also yield L^2-error estimates. To be more precise, we set

$$\begin{aligned}
&X_- := L^2(\Omega), \|.\|_{X_-} := \|.\|_{0,2},\\
&X_+ := X, \|.\|_{X_+} := \|.\|_X,\\
&Y_+ := W^{2,2}(\Omega) \cap Y, \|.\|_{Y_+} := \|.\|_{2,2},\\
&\widetilde{Y}_h := \text{span}\{\phi_T^2, \psi_E : T \in \mathcal{T}_h, E \in \mathcal{E}_{h,\Omega} \cup \mathcal{E}_{h,N}\},\\
&\widehat{X}_h := \widehat{Y}_h := \widetilde{Y}_{h|\omega_x}.
\end{aligned} \tag{3.19}$$

Here, ψ_E is given by Equation (3.6) and $\widetilde{Y}_h$ is equipped with the norm of Y_+. We do not need the continuation operators P and $\widetilde{P}$ of Section 3.1 since we are dealing with functions which are constant on each edge. The condition $DF(u_0)^* \in \text{Isom}(Y_+, X_-^*)$ is equivalent to the $W^{2,2}$-regularity of the Poisson equation, i.e., the assumption that Problem (1.1) with $g = 0$ admits for each right-hand side $f \in L^2(\Omega)$ a unique solution $u \in W^{2,2}(\Omega)$ which depends continuously on f. This condition is satisfied if Ω is convex. The estimates leading to inequalities (3.9) – (3.18), Lemma 3.1, and Inequalities (3.7), (3.8) then prove that

$$\begin{aligned}
&\|(Id_{Y_+} - R_h)^*[F(u_h) - \widetilde{F}_h(u_h)]\|_{Y_+^*}\\
\le& c\left\{\sum_{T\in\mathcal{T}_h} h_T^4 \|f - f_T\|_{0,2;T}^2 + \sum_{E\in\mathcal{E}_{h,N}} h_E^3 \|g - g_E\|_{2;E}^2\right\}^{1/2},\\
&\|F(u_h) - \widetilde{F}_h(u_h)\|_{\widetilde{Y}_h^*}\\
\le& c\left\{\sum_{T\in\mathcal{T}_h} h_T^4 \|f - f_T\|_{0,2;T}^2 + \sum_{E\in\mathcal{E}_{h,N}} h_E^3 \|g - g_E\|_{2;E}^2\right\}^{1/2},\\
&\|(Id_{Y_+} - R_h)^* \widetilde{F}_h(u_h)\|_{Y_+^*} \le c\left\{\sum_{T\in\mathcal{T}_h} h_T^2 \eta_{R,T}^2\right\}^{1/2},\\
&\|\widetilde{F}_h(u_h)\|_{\widetilde{Y}_h^*} \le c\left\{\sum_{T\in\mathcal{T}_h} h_T^2 \eta_{R,T}^2\right\}^{1/2},\\
&h_T \eta_{R,T} \le c \sup_{\substack{\varphi\in\widetilde{Y}_{h|\omega_T}\\ \|\varphi\|_{Y_+}=1}} < \widetilde{F}_h(u_h), \varphi > .
\end{aligned}$$

The operator B is defined as before. But, now $\widehat{X}_h$ is equipped with the $L^2(\Omega)$-norm and $\widehat{Y}_h$ is endowed with the $W^{2,2}(\Omega)$-norm. In order to check that B is still an isomorphism of $\widehat{X}_h$ onto $\widehat{Y}_h^*$ and that the corresponding norms of B and B^{-1} do not depend on h, we observe that Inequalities (3.7) and (3.8) imply for all $u \in \widehat{X}_h, \varphi \in \widehat{Y}_h$

$$\begin{aligned} < Bu, \varphi > &\leq \|\nabla u\|_{0,2;\omega_x} \|\nabla \varphi\|_{0,2;\omega_x} \\ &\leq \widetilde{c}_4^2 h_T^{-2} \|u\|_{0,2;\omega_x} \|\varphi\|_{0,2;\omega_x} \\ &\leq \widetilde{c}_4^2 \widetilde{c}_3^{-1} \|u\|_{0,2;\omega_x} \|\varphi\|_{2,2;\omega_x} \end{aligned} \tag{3.20}$$

and

$$\begin{aligned} < Bu, u > &\geq \|\nabla u\|_{0,2;\omega_x}^2 \\ &\geq \widetilde{c}_3^2 h_T^{-2} \|u\|_{0,2;\omega_x}^2 \\ &\geq \widetilde{c}_3^2 \widetilde{c}_4^{-1} \|u\|_{0,2;\omega_x} \|u\|_{2,2;\omega_x}. \end{aligned} \tag{3.21}$$

This proves $B \in \mathrm{Isom}(\widehat{X}_h, \widehat{Y}_h^*)$.
Collecting all these results we obtain the following extension of Propositions 1.5 and 1.9.

Proposition 3.8. *Assume that Ω is convex. Let u and u_h be the exact solutions of Problems (1.3) and (1.4). There are two constants $\underline{c}$ and $\overline{c}$, which only depend on the smallest angle in the triangulation, such that the estimates*

$$\begin{aligned} \|u - u_h\|_{0,2} \leq \overline{c} \Bigg\{ &\sum_{T \in \mathcal{T}_h} h_T^2 \eta_{R,T}^2 + \sum_{T \in \mathcal{T}_h} h_T^4 \|f - f_T\|_{0,2;T}^2 \\ &+ \sum_{E \in \mathcal{E}_{h,N}} h_E^3 \|g - g_E\|_{2;E}^2 \Bigg\}^{1/2} \end{aligned}$$

and

$$\begin{aligned} h_T \eta_{R,T} \leq \underline{c} \Bigg\{ &\|u - u_h\|_{0,2;\omega_T}^2 + \sum_{T' \subset \omega_T} h_{T'}^4 \|f - f_{T'}\|_{0,2;T'}^2 \\ &+ \sum_{E \in \mathcal{E}(T) \cap \mathcal{E}_{h,N}} h_E^3 \|g - g_E\|_{2;E}^2 \Bigg\}^{1/2} \end{aligned}$$

hold for all $T \in \mathcal{T}_h$. Here, f_T, g_E, and $\eta_{R,T}$ are given by Equations (1.17) and (1.18).
For $x \in \mathcal{N}_{h,\Omega} \cup \mathcal{N}_{h,N}$ set

$$\widetilde{\eta}_{D,x} := \|v_x\|_{0,2;\omega_x},$$

where v_x is the unique solution of Problem (1.34) with $V_x := \widetilde{Y}_{h|\omega_x}$ and $\widetilde{Y}_h$ given by Equation (3.19). There are constants $\widetilde{c}_{\mathcal{N},1}, \ldots, \widetilde{c}_{\mathcal{N},4}$, which only depend on the smallest angle in the triangulation, such that the estimates

$$\widetilde{\eta}_{D,x} \leq \widetilde{c}_{\mathcal{N},1} \Big\{ \sum_{x \in \mathcal{N}(T)} h_T^2 \eta_{R,T}^2 \Big\}^{1/2},$$

$$h_T \eta_{R,T} \leq \widetilde{c}_{\mathcal{N},2} \Big\{ \sum_{x \in \mathcal{N}(T) \setminus \mathcal{N}_{h,D}} \widetilde{\eta}_{D,x}^2 \Big\},$$

$$\widetilde{\eta}_{D,x} \leq \widetilde{c}_{\mathcal{N},3} \Big\{ \|u - u_h\|_{0,2;\omega_x}^2 + \sum_{x \in \mathcal{N}(T)} h_T^4 \|f - f_T\|_{0,2;T}^2 + \sum_{E \subset \partial\omega_x \cap \Gamma_N} h_E^3 \|g - g_E\|_{2;E}^2 \Big\}^{1/2},$$

$$\|u - u_h\|_{0,2} \leq \widetilde{c}_{\mathcal{N},4} \Big\{ \sum_{x \in \mathcal{N}_{h,\Omega} \cup \mathcal{N}_{h,N}} \widetilde{\eta}_{D,x}^2 + \sum_{T \in \mathcal{T}_h} h_T^4 \|f - f_T\|_{0,2;T}^2 + \sum_{E \in \mathcal{E}_{h,N}} h_E^3 \|g - g_E\|_{2;E}^2 \Big\}^{1/2}$$

hold for all $x \in \mathcal{N}_{h,\Omega} \cup \mathcal{N}_{h,N}$ and all $T \in \mathcal{T}_h$.

Remark 3.9. The results of this section immediately carry over to higher order finite element discretizations, i.e., $X_h = Y_h = S_h^{k,0} \cap X$ with $k \geq 2$. One only has to perform the following modifications:

(1) Replace f_T and g_E by the corresponding L^2-projections of f and g onto the spaces of functions which are piecewise polynomials of degree at most $k-2$ and $k-1$, respectively.

(2) In the definition (1.18) of $\eta_{R,T}$ replace f_T by $f_T + \Delta u_h$.

(3) In the case of H^1-error estimates replace $\widetilde{Y}_h$ by

$$\operatorname{span}\{ b_T v, b_E P\sigma : v \in \Pi_{k-2|T}, \sigma \in \Pi_{k-1|E}, T \in \mathcal{T}_h, E \in \mathcal{E}_{h,\Omega} \cup \mathcal{E}_{h,N} \},$$

where b_T and b_E are the bubble functions of Section 1.1 and P is the continuation operator of Section 3.1.

(4) In the case of L^2-error estimates replace $\widetilde{Y}_h$ by

$$\operatorname{span}\{ b_T^2 v, \psi_E \widetilde{P}\sigma : v \in \Pi_{k-2|T}, \sigma \in \Pi_{k-1|E}, T \in \mathcal{T}_h, E \in \mathcal{E}_{h,\Omega} \cup \mathcal{E}_{h,N} \},$$

where b_T is the triangle-bubble function of Section 1.1 and ψ_E and $\widetilde{P}$ are as in Remark 3.6.

□

3.3. Quasilinear Equations of 2nd Order.

Consider the boundary value problem

$$\begin{aligned} -\nabla \cdot \underline{a}(x,u,\nabla u) &= b(x,u,\nabla u) \quad \text{in } \Omega \subset \mathbb{R}^n \\ u &= 0 \qquad\qquad\quad \text{on } \Gamma \end{aligned} \tag{3.22}$$

where $b \in C(\Omega \times \mathbb{R} \times \mathbb{R}^n, \mathbb{R})$ and $\underline{a} \in C^1(\Omega \times \mathbb{R} \times \mathbb{R}^n, \mathbb{R}^n)$ are such that the matrix $A(x,y,z) := \left(\frac{1}{2}(\partial_{z_j} a_i(x,y,z) + \partial_{z_i} a_j(x,y,z)\right)_{1 \le i,j \le n}$ is positive definite for all $x \in \Omega$, $y \in \mathbb{R}$, $z \in \mathbb{R}^n$.

Under suitable growth-conditions on $\underline{a}$, b, and their derivatives there are real numbers $1 < r, q < \infty$ such that the weak formulation of Problem (3.22) fits into the framework of Section 2.1 with

$$\begin{aligned} X &:= \{ u \in W^{1,r}(\Omega) : u = 0 \text{ on } \Gamma \}, \\ \|\cdot\|_X &:= \|\cdot\|_{1,r}, \\ Y &:= \{ \varphi \in W^{1,q}(\Omega) : u = 0 \text{ on } \Gamma \}, \\ \|\cdot\|_Y &:= \|\cdot\|_{1,q}, \\ \langle F(u), \varphi \rangle &:= \int_\Omega \underline{a}(x,u,\nabla u)\nabla\varphi - \int_\Omega b(x,u,\nabla u)\varphi. \end{aligned}$$

Denote by $p := \frac{q}{q-1}$ the dual exponent of q. Note, that $DF(u) \in \mathrm{Isom}(X, Y^*)$ if and only if the linear boundary value problem

$$\begin{aligned} -\nabla \cdot (A(x,u,\nabla u)\nabla v) - \nabla \cdot (\partial_y \underline{a}(x,u,\nabla u)v) & \\ -\nabla_z b(x,u,\nabla u) \cdot \nabla v - \partial_y b(x,u,\nabla u)v &= f \text{ in } \Omega \\ v &= 0 \text{ on } \Gamma \end{aligned}$$

admits for each $f \in Y^*$ a unique weak solution $v \in X$ which depends continuously on f.

Some examples of problems falling into the present category are given by:

(1) *The equations of prescribed mean curvature:*

$$\begin{aligned} \underline{a}(x,u,\nabla u) &:= [1 + \|\nabla u\|^2]^{-1/2}\nabla u, \\ b(x,u,\nabla u) &:= f(x) \in L^2(\Omega), \\ r &:= q := 2. \end{aligned}$$

(2) *The α-Laplacian:*

$$\begin{aligned} \underline{a}(x,u,\nabla u) &:= \|\nabla u\|^{\alpha-2}\nabla u, \quad \alpha > 1, \\ b(x,u,\nabla u) &:= f(x) \in L^p(\Omega), \\ r &:= q := \alpha. \end{aligned}$$

(3) *The subsonic flow of an irrotational, ideal, compressible gas:*

$$\begin{aligned} \underline{a}(x,u,\nabla u) &:= \left[1 - \frac{\gamma-1}{2}\|\nabla u\|^2\right]^{1/(\gamma-1)} \nabla u, \quad \gamma > 1, \\ b(x,u,\nabla u) &:= f(x) \in L^p(\Omega), \\ r &:= q := \frac{2\gamma}{\gamma-1}. \end{aligned}$$

(4) *The stationary heat equation with convection and non-linear diffusion coefficient:*

$$\begin{aligned} \underline{a}(x,u,\nabla u) &:= k(u)\nabla u, \\ b(x,u,\nabla u) &:= f - \mathbf{c}\cdot\nabla u, \\ f \in L^\infty(\Omega), \quad \mathbf{c} \in C(\Omega, I\!R^n), \quad k &\in C^2(I\!R), \\ k(s) \geq \alpha > 0, \quad |k^{(l)}(s)| \leq \gamma, \quad \forall s &\in I\!R,\ l = 0,1,2, \\ r &:= p \in (2,4). \end{aligned}$$

(5) *Bratu's equation:*

$$\begin{aligned} \underline{a}(x,u,\nabla u) &:= \nabla u, \\ b(x,u,\nabla u) &:= \lambda e^u, \quad \lambda > 0, \\ r &:= p > n. \end{aligned}$$

(6) *A non-linear eigenvalue problem:*

$$\begin{aligned} \underline{a}(x,u,\nabla u) &:= \nabla u, \\ b(x,u,\nabla u) &:= \lambda u - u^\beta, \quad \beta \geq n, \\ r &:= p \geq n. \end{aligned}$$

Example (2) fits into the framework of Proposition 2.1 if $\alpha \geq 2$. If $1 < \alpha < 2$, the corresponding function F is no longer differentiable. However, it still fits into the framework of Remark 2.3 with

$$\varrho(t) = \underline{c}\{\, \|u_0\|_X + t \,\}^{\alpha-2} t, \quad \sigma(t) = \overline{c}\, t^{\alpha-1}$$

(cf. [37], § 5.3).
In example (5) there is a critical parameter $\lambda^* > 0$ such that the problem admits two weak solutions if $0 < \lambda < \lambda^*$, exactly one weak solution if $\lambda = \lambda^*$, and no solution if $\lambda > \lambda^*$. The solution corresponding to $\lambda = \lambda^*$ is a turning point and fits into the framework of the second part of Section 2.2 (cf. [39]).
Example (6) always admits the trivial solution. If λ is a simple eigenvalue of the Laplacian there is a simple bifurcation fitting into the framework of the third part of Section 2.2 (cf. [39]).

We do not specify the discretization of Problem (3.22) in detail. We only assume that $X_h \subset X \cap W^{1,\infty}(\Omega)$ and $Y_h \subset Y \cap W^{1,\infty}(\Omega)$ are finite element spaces corresponding to $\mathcal{T}_h$ consisting of affine equivalent elements in the sense of [37] and that $S_h^{1,0} \cap Y \subset Y_h$.

In order to construct R_h, $\widetilde{F}_h$, and $\widetilde{Y}_h$, we define two integers k, l and approximations $\underline{a}_h$ of $\underline{a}$ and b_h of b as follows:

$$\underline{a}_h(x, u_h, \nabla u_h) := \begin{cases} \underline{a}(x, u_h, \nabla u_h), & \text{if } \underline{a}(x, v_h, \nabla v_h) \in S_h^{k,-1} \\ & \forall v_h \in X_h, \\ \sum_{T \in \mathcal{T}_h} \pi_{1,T} \underline{a}(x, u_h, \nabla u_h), \ k := 1, & \text{otherwise} \end{cases}$$

$$b_h(x, u_h, \nabla u_h) := \begin{cases} b(x, u_h, \nabla u_h), & \text{if } b(x, v_h, \nabla v_h) \in S_h^{l,-1} \\ & \forall v_h \in X_h, \\ \sum_{T \in \mathcal{T}_h} \pi_{0,T} b(x, u_h, \nabla u_h), \ l := 0 & \text{otherwise.} \end{cases}$$

Here, $u_h \in X_h$ is arbitrary. Now, $\widetilde{F}_h$ is defined in the same way as F with $\underline{a}$ and b replaced by $\underline{a}_h$ and b_h, respectively, $R_h := I_h$, and

$$\widetilde{Y}_h := \operatorname{span}\{\, \psi_T v, \psi_E P\sigma : v \in \Pi_{m|T},\ \sigma \in \Pi_{k|E},\ T \in \mathcal{T}_h,\ E \in \mathcal{E}_{h,\Omega} \,\},$$

where $m := \max\{\, k-1, l \,\}$.

Put for abbreviation

$$\varepsilon_T := \Big\{ h_T^p \| - \nabla \cdot (\underline{a}(\cdot, u_h, \nabla u_h) - \underline{a}_h(\cdot, u_h, \nabla u_h)) - (b(\cdot, u_h, \nabla u_h) - b_h(\cdot, u_h, \nabla u_h)) \|_{0,p;T}^p + \sum_{E \in \mathcal{E}(T) \cap \mathcal{E}_{h,\Omega}} h_E \| [n_E \cdot (\underline{a}(\cdot, u_h, \nabla u_h) - \underline{a}_h(\cdot, u_h, \nabla u_h))]_E \|_{p;E}^p \Big\}^{1/p} \quad \forall T \in \mathcal{T}_h \tag{3.23}$$

$$\eta_T := \Big\{ h_T^p \| \nabla \cdot \underline{a}_h(\cdot, u_h, \nabla u_h) + b_h(\cdot, u_h, \nabla u_h) \|_{0,p;T}^p + \sum_{E \in \mathcal{E}(T) \cap \mathcal{E}_{h,\Omega}} h_E \| [n_E \cdot \underline{a}_h(\cdot, u_h, \nabla u_h)]_E \|_{p;E}^p \Big\}^{1/p} \quad \forall T \in \mathcal{T}_h. \tag{3.24}$$

The quantity ε_T obviously measures the quality of the approximation of $\underline{a}$ and b by $\underline{a}_h$ and b_h, respectively and can be estimated explicitly. Below, we will show that $\|(\mathrm{Id}_Y - R_h)^*[F(u_h) - \widetilde{F}_h(u_h)]\|_{Y^*}$ and $\|F(u_h) - \widetilde{F}_h(u_h)\|_{\widetilde{Y}_h^*}$ are bounded from above by $c \left\{ \sum_{T \in \mathcal{T}_h} \varepsilon_T^p \right\}^{1/p}$.

Note, that

$$\varepsilon_T = h_T \| f - \pi_{0,T} f \|_{0,p;T} \quad \forall T \in \mathcal{T}_h,$$

if $X_h \subset S_h^{1,0}$ in Examples (1)–(3), that

$$\varepsilon_T \le c\, h_T^{2(1-n/p)} \| \nabla u_h \|_{0,p;T} \quad \forall T \in \mathcal{T}_h$$

if $X_h \subset S_h^{1,0}$ in Example (4), that

$$\varepsilon_T \le c\, h_T^2 \| \nabla u_h \|_{0,p;T} \exp(\|u_h\|_{0,\infty;T}) \quad \forall T \in \mathcal{T}_h$$

if X_h consists of piecewise polynomials in Example (5), and that

$$\varepsilon_T = 0 \quad \forall T \in \mathcal{T}_h$$

if X_h consists of piecewise polynomials and $\beta \in I\!N$ in Example (6).

Using integration by parts elementwise, we obtain for all $\varphi \in Y$

$$\langle F(u_h), \varphi \rangle = \sum_{T \in \mathcal{T}_h} \int_T \{ -\nabla \cdot \underline{a}(x, u_h, \nabla u_h) - b(x, u_h, \nabla u_h) \}\, \varphi + \sum_{E \in \mathcal{E}_{h,\Omega}} \int_E [n_E \cdot \underline{a}(x, u_h, \nabla u_h)]_E\, \varphi, \tag{3.25}$$

and

$$\langle \widetilde{F}_h(u_h), \varphi \rangle = \sum_{T \in \mathcal{T}_h} \int_T \{ -\nabla \cdot \underline{a}_h(x, u_h, \nabla u_h) - b_h(x, u_h, \nabla u_h) \} \varphi + \sum_{E \in \mathcal{E}_{h,\Omega}} \int_E [n_E \cdot \underline{a}_h(x, u_h, \nabla u_h)]_E \, \varphi. \tag{3.26}$$

Lemmas 3.1, 3.3, the definition of $\widetilde{Y}_h$, and Equalities (3.25), (3.26) then imply that

$$\begin{aligned} &\|(\mathrm{Id}_Y - R_h)^* [F(u_h) - \widetilde{F}_h(u_h)]\|_{Y^*} \\ &= \sup_{\substack{\varphi \in Y \\ \|\varphi\|_Y = 1}} \Bigg\{ \sum_{T \in \mathcal{T}_h} \int_T \{ -\nabla \cdot (\underline{a}(x, u_h, \nabla u_h) - \underline{a}_h(x, u_h, \nabla u_h)) \\ &\qquad - (b(x, u_h, \nabla u_h) - b_h(x, u_h, \nabla u_h)) \} \{ \varphi - I_h \varphi \} \\ &\qquad + \sum_{E \in \mathcal{E}_{h,\Omega}} \int_E [n_E \cdot (\underline{a}(x, u_h, \nabla u_h) - \underline{a}_h(x, u_h, \nabla u_h))]_E \{ \varphi - I_h \varphi \} \Bigg\} \\ &\leq c \Bigg\{ \sum_{T \in \mathcal{T}_h} \varepsilon_T^p \Bigg\}^{1/p} \end{aligned} \tag{3.27}$$

and

$$\begin{aligned} &\|F(u_h) - \widetilde{F}_h(u_h)\|_{\widetilde{Y}_h^*} \\ &= \sup_{\substack{\varphi_h \in \widetilde{Y}_h \\ \|\varphi_h\|_Y = 1}} \Bigg\{ \sum_{T \in \mathcal{T}_h} \int_T \{ -\nabla \cdot (\underline{a}(x, u_h, \nabla u_h) - \underline{a}_h(x, u_h, \nabla u_h)) \\ &\qquad - (b(x, u_h, \nabla u_h) - b_h(x, u_h, \nabla u_h)) \} \varphi_h \\ &\qquad + \sum_{E \in \mathcal{E}_{h,\Omega}} \int_E [n_E \cdot (\underline{a}(x, u_h, \nabla u_h) - \underline{a}_h(x, u_h, \nabla u_h))]_E \varphi_h \Bigg\} \\ &\leq c \Bigg\{ \sum_{T \in \mathcal{T}_h} \varepsilon_T^p \Bigg\}^{1/p}. \end{aligned} \tag{3.28}$$

Similarly, we obtain

$$\begin{aligned}
&\|(\mathrm{Id}_Y - R_h)^* \widetilde{F}_h(u_h)\|_{Y^*} \\
&= \sup_{\substack{\varphi \in Y \\ \|\varphi\|_Y = 1}} \Big\{ \sum_{T \in \mathcal{T}_h} \int_T \{ -\nabla \cdot \underline{a}_h(x, u_h, \nabla u_h) - b_h(x, u_h, \nabla u_h) \} \{ \varphi - I_h \varphi \} \\
&\qquad + \sum_{E \in \mathcal{E}_{h,\Omega}} \int_E [n_E \cdot \underline{a}_h(x, u_h, \nabla u_h)]_E \{ \varphi - I_h \varphi \} \Big\} \\
&\le c \Big\{ \sum_{T \in \mathcal{T}_h} \eta_T^p \Big\}^{1/p}
\end{aligned} \tag{3.29}$$

and

$$\begin{aligned}
&\|\widetilde{F}_h(u_h)\|_{\widetilde{Y}_h^*} \\
&= \sup_{\substack{\varphi_h \in \widetilde{Y}_h \\ \|\varphi_h\|_Y = 1}} \Big\{ \sum_{T \in \mathcal{T}_h} \int_T \{ -\nabla \cdot \underline{a}_h(x, u_h, \nabla u_h) - b_h(x, u_h, \nabla u_h) \} \varphi_h \\
&\qquad + \sum_{E \in \mathcal{E}_{h,\Omega}} \int_E [n_E \cdot \underline{a}_h(x, u_h, \nabla u_h)]_E \varphi_h \Big\} \\
&\le c \Big\{ \sum_{T \in \mathcal{T}_h} \eta_T^p \Big\}^{1/p} .
\end{aligned} \tag{3.30}$$

In order to prove Inequality (2.11), consider an arbitrary simplex $T \in \mathcal{T}_h$ and an arbitrary face $E \in \mathcal{E}_{h,\Omega}$ of T and define $\widetilde{Y}_{h|\omega}$, $\omega \in \{T, \omega_E, \omega_T\}$, as in Section 3.2. Lemma 3.3, Equation (3.26), and the definition of $\widetilde{Y}_h$ then yield

$$\begin{aligned}
&c_1 c_4^{-1} h_T \|\nabla \cdot \underline{a}_h(\cdot, u_h, \nabla u_h) + b_h(\cdot, u_h, \nabla u_h)\|_{0,p;T} \\
&\le \sup_{v \in \Pi_{m|T} \setminus \{0\}} \|\nabla(\psi_T v)\|_{0,q;T}^{-1} \int_T \{ -\nabla \cdot \underline{a}_h(x, u_h, \nabla u_h) \\
&\qquad\qquad - b_h(x, u_h, \nabla u_h) \} \psi_T v \\
&= \sup_{v \in \Pi_{m|T} \setminus \{0\}} \|\nabla(\psi_T v)\|_{0,q;T}^{-1} \langle \widetilde{F}_h(u_h), \psi_T v \rangle \\
&\le \sup_{\substack{\varphi \in \widetilde{Y}_{h|T} \\ \|\varphi\|_Y = 1}} \langle \widetilde{F}_h(u_h), \varphi \rangle
\end{aligned} \tag{3.31}$$

and, using Inequality (3.31),

$$
\begin{aligned}
& c_2 c_6^{-1} c_7^{-1} h_E^{1/p} \| [n_E \cdot \underline{a}_h(\cdot, u_h, \nabla u_h)]_E \|_{p;E} \\
\leq & \sup_{\sigma \in \Pi_{k|E} \setminus \{0\}} c_6^{-1} c_7^{-1} h_E^{1/p} \|\sigma\|_{q;E}^{-1} \int_E [n_E \cdot \underline{a}_h(x, u_h, \nabla u_h)]_E \, \psi_E P\sigma \\
= & \sup_{\sigma \in \Pi_{k|E} \setminus \{0\}} c_6^{-1} c_7^{-1} h_E^{1/p} \|\sigma\|_{q;E}^{-1} \Big\{ \langle \widetilde{F}_h(u_h), \psi_E P\sigma \rangle \\
& \qquad - \int_{\omega_E} \{ -\nabla \cdot \underline{a}_h(x, u_h, \nabla u_h) - b_h(x, u_h, \nabla u_h) \} \psi_E P\sigma \Big\} \\
\leq & \sup_{\substack{\varphi \in \widetilde{Y}_h|\omega_E \\ \|\varphi\|_Y = 1}} \langle \widetilde{F}_h(u_h), \varphi \rangle \\
& \qquad + c_6^{-1} h_E \| \nabla \cdot \underline{a}_h(\cdot, u_h, \nabla u_h) + b_h(\cdot, u_h, \nabla u_h) \|_{0,p;\omega_E} \\
\leq & \, c \sup_{\substack{\varphi \in \widetilde{Y}_{h|\omega_E} \\ \|\varphi\|_Y = 1}} \langle \widetilde{F}_h(u_h), \varphi \rangle .
\end{aligned} \tag{3.32}
$$

Inequalities (3.31) and (3.32) imply that

$$
\eta_T \leq c \sup_{\substack{\varphi \in \widetilde{Y}_{h|\omega_T} \\ \|\varphi\|_Y = 1}} \langle \widetilde{F}_h(u_h), \varphi \rangle \tag{3.33}
$$

and

$$
\Big\{ \sum_{T \in \mathcal{T}_h} \eta_T^p \Big\}^{1/p} \leq c \| \widetilde{F}_h(u_h) \|_{\widetilde{Y}_h^*} . \tag{3.34}
$$

Inequalities (3.29) and (3.34) in particular prove Inequality (2.11)

Propositions 2.1 and 2.5 and Inequalities (3.27) – (3.34) yield the following a posteriori error estimates for Problem (3.22).

Proposition 3.10. *Let $u \in X$ be a weak solution of Problem (3.22) which is regular in the sense of Proposition 2.1 and let $u_h \in X_h$ be an approximate solution of the corresponding discrete problem which is sufficiently close to u in the sense of Proposition 2.1. Then the following a posteriori error estimates hold*

$$
\begin{aligned}
\|u - u_h\|_{1,r} \leq & \, c_1 \Big\{ \sum_{T \in \mathcal{T}_h} \eta_T^p \Big\}^{1/p} + c_2 \Big\{ \sum_{T \in \mathcal{T}_h} \varepsilon_T^p \Big\}^{1/p} \\
& + c_3 \| F(u_h) - F_h(u_h) \|_{Y_h^*} + c_4 \| F_h(u_h) \|_{Y_h^*}
\end{aligned}
$$

and

$$\eta_T \leq c_5 \|u - u_h\|_{1,r;\omega_T} + c_6 \left\{ \sum_{T' \subset \omega_T} \varepsilon_{T'}^p \right\}^{1/p} \quad \forall T \in \mathcal{T}_h.$$

Here, ε_T and η_T are given by Equations (3.23) and (3.24) and $\|F(u_h) - F_h(u_h)\|_{Y_h^*}$ and $\|F_h(u_h)\|_{Y_h^*}$ are the consistency error of the discretization and the residual of the discrete problem, respectively. □

Remark 3.11. Proposition 3.10 can easily be extended to the case of Neumann boundary conditions. One only has to replace in Equations (3.23) and (3.24) Γ by the part of the boundary on which Dirichlet boundary conditions are imposed. The first estimate of Proposition 3.10 also holds if η_T is defined using the original functions $\underline{a}$ and b instead of the projected ones $\underline{a}_h$ and b_h. The ε_T-term then of course disappears. If the functions $\underline{a}$ and b are sufficiently smooth one may also use higher order approximations $\underline{a}_h$ and b_h instead of the present low-order ones. □

As mentioned before, Proposition 3.10 can be applied to Example (2) only in the case $\alpha \geq 2$. Observing that for $1 < \alpha < 2$ the strong monotonicity of F implies the unique solvability of the corresponding weak problem, we obtain from Remark 2.3 and Inequalities (3.27) - (3.30), and (3.34) the following result which complements the results of [25].

Proposition 3.12. *Let $1 < \alpha < 2$ and denote by $u \in W^{1,\alpha}(\Omega)$, $u = 0$ on Γ, the unique solution of*

$$\int_\Omega \|\nabla u\|^{\alpha-2} \nabla u \nabla v = \int_\Omega f v \quad \forall v \in W^{1,\alpha}(\Omega),\ v = 0 \text{ on } \Gamma.$$

Let $u_h \in X_h$ be an approximate solution of a discretization of the above problem. Then the following a posteriori error estimates hold

$$\|u - u_h\|_{1,\alpha} \leq c_1 \left\{ \sum_{T \in \mathcal{T}_h} \eta_T^\alpha \right\}^{1/\alpha} + c_2 \left\{ \sum_{T \in \mathcal{T}_h} \varepsilon_T^\alpha \right\}^{1/\alpha} + c_3 \|F(u_h) - F_h(u_h)\|_{Y_h^*} + c_4 \|F_h(u_h)\|_{Y_h^*}$$

and

$$\left\{ \sum_{T \in \mathcal{T}_h} \eta_T^\alpha \right\}^{1/\alpha} \leq c_5 \|u - u_h\|_{1,\alpha}^{1/(\alpha-1)} + c_6 \left\{ \sum_{T \in \mathcal{T}_h} \varepsilon_T^\alpha \right\}^{1/\alpha(\alpha-1)}.$$

Here, ε_T, η_T, $\|F(u_h) - F_h(u_h)\|_{Y_h^*}$, and $\|F_h(u_h)\|_{Y_h^*}$ are as in Proposition 3.10. Moreover,

$$\varepsilon_T = h_T \|f - \pi_{0,T} f\|_{0,\alpha;T} \quad \forall T \in \mathcal{T}_h,$$

if u_h is piecewise linear. □

As mentioned before, Example (6) exhibits a simple bifurcation from the trivial branch at the simple eigenvalues of the Laplacian. Combining the results of Section 2.2 with those of this section, we obtain the following a posteriori error estimate.

Proposition 3.13. *Denote by $\lambda^* \in I\!R$ and $u^* \in W^{1,p}(\Omega)$, $u^* = 0$ on Γ, $p > n$, a simple eigenvalue of the Laplace equation with homogeneous Dirichlet boundary conditions and a corresponding eigenfunction with $\int_\Omega u^* = 1$. Let $\lambda_h \in I\!R$ and $u_h \in X_h$ be a solution of*

$$\int_\Omega \nabla u_h \nabla v_h - \lambda_h \int_\Omega u_h v_h + \int_\Omega u_h^\beta v_h = 0 \quad \forall v_h \in X_h$$

where $X_h \subset \{ v \in W^{1,\infty}(\Omega) : v = 0 \text{ on } \Gamma \}$ is a finite element space corresponding to $\mathcal{T}_h$ consisting of piecewise polynomials and where $\beta \in I\!N$, $\beta \geq n$. If λ_h and u_h are sufficiently close to λ^* and u^*, the following a posteriori error estimates hold

$$|\lambda_h - \lambda^*| + \|u_h - u^*\|_{1,p} \leq c_1 \left\{ \left| \int_\Omega u_h - 1 \right| + \left\{ \sum_{T \in \mathcal{T}_h} \eta_T^p \right\}^{1/p} \right\}$$

and

$$\left| \int_\Omega u_h - 1 \right| + \left\{ \sum_{T \in \mathcal{T}_h} \eta_T^p \right\}^{1/p} \leq c_2 \{ |\lambda_h - \lambda^*| + \|u_h - u^*\|_{1,p} \},$$

where

$$\eta_T := \left\{ h_T^p \|\Delta u_h + \lambda_h u_h\|_{0,p;T}^p + \sum_{E \in \mathcal{E}(T) \cap \mathcal{E}_{h,\Omega}} h_E \| [n_E \cdot \nabla u_h]_E \|_{p;E}^p \right\}^{1/p}.$$

PROOF: Observing that the consistency error of the above discretization vanishes, Proposition 3.13 follows from Inequalities (3.27) – (3.30) and the results of the third part of Section 2.2 with $l \in \mathcal{L}(V, I\!R)$ given by

$$l(v) := \int_\Omega v \quad \forall v \in W^{1,p}(\Omega),\ v = 0 \text{ on } \Gamma.$$

□

When comparing Propositions 3.10 and 3.13, we remark that the latter only yields global lower bounds on the error. This is due to the global nature of the functional l defined above.

We conclude this section with a simple example of an a posteriori error estimator which is based on the solution of auxiliary local problems and which generalizes

the estimator $\eta_{D,x}$ of Section 1.3. For simplicity we assume that $p = q = r = 2$. We choose an arbitrary vertex x_0 in the partition $\mathcal{T}_h$ and keep it fixed in what follows. Put $\omega_0 := \omega_{x_0}$, let

$$\widehat{X}_h := \widehat{Y}_h := \widetilde{Y}_{h|\omega_0}$$

and define the operator $B \in \mathcal{L}(\widehat{X}_h, \widehat{Y}_h^*)$ by

$$\langle Bu, \varphi \rangle := \int_{\omega_0} \nabla\varphi^t A_0 \nabla u \quad \forall u \in \widehat{X}_h,\ \varphi \in \widehat{Y}_h,$$

where

$$A_0 := A(x_0, u_h(x_0), \pi_{0,\omega_0}(\nabla u_h)).$$

Note, that the operator B is obtained by first linearizing around u_h the differential operator associated with Problem (3.22), then freezing the coefficients of the resulting linear operator at x_0, and finally retaining only the principal part of the linear constant-coefficient operator. Since ∇u_h may be discontinuous, its value at x_0 is approximated by the L^2-projection $\pi_{0,\omega_0}(\nabla u_h)$. Other constructions are of course also possible.

Since the matrix $A(x, y, z)$ is symmetric and positive definite for all $x \in \Omega$, $y \in I\!R$, $z \in I\!R^n$ and since the functions in $\widehat{X}_h = \widehat{Y}_h$ vanish on $\partial\omega_0$, we immediately obtain from Korn's inequality that $B \in \mathrm{Isom}(\widehat{X}_h, \widehat{Y}_h^*)$. Let $u_0 \in \widehat{X}_h$ be the unique solution of

$$\langle Bu_0, \varphi \rangle = \langle \widetilde{F}_h(u_h), \varphi \rangle \quad \forall \varphi \in \widehat{Y}_h \tag{3.35}$$

and set

$$\eta_{x_0} := \|u_0\|_{1,2;\omega_0}. \tag{3.36}$$

Note, that Problem (3.35) is equivalent to

$$\int_{\omega_0} \nabla\varphi^t A_0 \nabla u_0 = \int_{\omega_0} \underline{a}_h(x, u_h, \nabla u_h)\nabla\varphi - \int_{\omega_0} b_h(x, u_h, \nabla u_h)\varphi \quad \forall \varphi \in \widehat{Y}_h.$$

This shows that η_{x_0} is a generalization of the estimator $\eta_{D,x}$ introduced in Section 1.3.

Lemma 3.3 and Equations (3.26) and (3.33) immediately imply that

$$\underline{c}\|\widetilde{F}_h(u_h)\|_{\widehat{Y}_h^*} \le \left\{ \sum_{T \in \mathcal{T}_0} \eta_T^2 \right\}^{1/2} \le \overline{c}\|\widetilde{F}_h(u_h)\|_{\widehat{Y}_h^*}.$$

Together with Proposition 2.7 this yields the following result.

Proposition 3.14. *Let x_0 be an arbitrary vertex in the triangulation $\mathcal{T}_h$. Then there are two constants c_1, c_2, which only depend on the polynomial degree of the space X_h and on the ratio h_T/ϱ_T, such that the following inequalities hold*

$$c_1\Big\{\sum_{T\subset\omega_0}\eta_T^2\Big\}^{1/2} \le \eta_{x_0} \le c_2\Big\{\sum_{T\subset\omega_0}\eta_T^2\Big\}^{1/2}.$$

Here, η_T and η_{x_0} are given by Equations (3.24) and (3.36), respectively. □

Remark 3.15. In order to obtain bounds on the error measured in the $L^r(\Omega)$-norm, we apply Remarks 2.4, 2.9, and 3.6. More precisely, we set

$$\begin{aligned} X_- &:= L^r(\Omega), \|.\|_{X_-} := \|.\|_{0,r},\\ X_+ &:= W^{1,s}(\Omega)\cap X, \|.\|_{X_+} := \|.\|_{1,s}, \quad r\le s\le\infty,\\ Y_+ &:= W^{2,q}(\Omega)\cap Y, \|.\|_{Y_+} := \|.\|_{2,q}. \end{aligned}$$

The choice of s depends on the particular example. The condition $DF(u)^* \in \mathrm{Isom}(Y_+, X_-^*)$ is equivalent to the assumption that the adjoint linearized problem

$$\begin{aligned} &-\nabla\cdot(A(x,u,\nabla u)\nabla v) + \partial_y a(x,u,\nabla u)\cdot\nabla v\\ &+\nabla\cdot(\nabla_z b(x,u,\nabla u)v) - \partial_y b(x,u,\nabla u)v = f \quad \text{in} \quad \Omega\\ &\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad v = 0 \quad \text{on} \quad \Gamma \end{aligned}$$

admits for each $f\in L^{r'}(\Omega)$ a unique weak solution $v\in Y_+$ which depends continuously on f. Here, $r' = \frac{r}{r-1}$ denotes the dual exponent of r. This assumption is satisfied if the coefficients $\underline{a}$ and b are sufficiently smooth and Ω is convex. The spaces $\widetilde{Y}_h, \widehat{Y}_h$, and $\widehat{X}_h$ are defined as before; but now, the functions ψ_T, ψ_E, and the continuation operator P are choosen according to Remark 3.6. The arguments leading to Estimates (3.27) – (3.30) and (3.33), (3.34) carry over immediately. Since Y_+ and $\widetilde{Y}_h$ are equipped with the $W^{2,q}(\Omega)$ norm, we may now apply Lemma 3.1 with $l=2$ and Estimates (3.7) and (3.8) with $k=1$. We therefore always gain an additional power of h. The arguments used to establish Estimates (3.20) and (3.21) show that $B\in\mathrm{Isom}(\widehat{X}_h, \widehat{Y}_h^*)$ also with the present choice of norms. Collecting all

these results we obtain the following error estimates

$$\begin{aligned}\|u-u_h\|_{0,r} \leq & c_1\left\{\sum_{T\in\mathcal{T}_h} h_T^p\eta_T^p\right\}^{1/p} + c_2\left\{\sum_{T\in\mathcal{T}_h} h_T^p\epsilon_T^p\right\}^{1/p} \\ & + c_3\|R_h^*[F(u_h)-F_h(u_h)]\|_{Y_+^*} + c_4\|R_h^*F_h(u_h)\|_{Y_+^*},\end{aligned}$$

$$h_T\eta_T \leq c_5\|u-u_h\|_{0,r;\omega_T} + c_6\left\{\sum_{T'\subset\omega_T} h_{T'}^p\epsilon_{T'}^p\right\}^{1/p},$$

$$c_7\left\{\sum_{T\subset\omega_0} h_T^2\eta_T^2\right\}^{1/2} \leq \widetilde{\eta}_{x_0} \leq c_8\left\{\sum_{T\subset\omega_0} h_T^2\eta_T^2\right\}^{1/2}.$$

Here, ϵ_T and η_T are as in Equations (3.23) and (3.24) and $\widetilde{\eta}_{x_0}$ is given by

$$\widetilde{\eta}_{x_0} := \|u_0\|_{0,2;\omega_0} \tag{3.37}$$

with $u_0 \in \widehat{X}_h$ denoting the solution of the local auxiliary problem (3.35). Usually, the third and fourth term on the right-hand side of the first estimate will be smaller by a power of h than the corresponding terms in the first estimate of Proposition 3.10. □

Remark 3.16. The present analysis follows the lines of [86]. Examples (2) and (4) were originally analyzed in [25] and [62], respectively. There, however, only upper bounds on the error are established. □

3.4. Eigenvalue Problems for 2nd Order Operators.

As an example for the treatment of eigenvalue problems we consider in this section the problem

$$\begin{aligned}-\nabla\cdot(A(x)\nabla u) + d(x)u &= \lambda u \quad \text{in } \Omega\subset I\!R^n \\ u &= 0 \quad \text{on } \Gamma.\end{aligned} \tag{3.38}$$

Here, $d \in C(\Omega, I\!R_+)$ and $A \in C^1(\Omega, I\!R^{n\times n})$ are such that A is symmetric and uniformly positive definite on Ω. Of course we are only interested in solutions u which do not identically vanish on Ω.

When considering λ as a parameter, Problem (3.38) can be treated as a bifurcation problem similar to Example (6) of the previous section. Here, we adopt a different strategy and define

$$\begin{aligned}X := Y &:= I\!R \times \{\, u \in W^{1,2}(\Omega) : u = 0 \text{ on } \Gamma \,\}, \\ \|\cdot\|_X := \|\cdot\|_Y &:= \{\, |\cdot|^2 + \|\cdot\|_{1,2}^2 \,\}^{1/2}, \\ \langle F([\lambda,u]),[\mu,v]\rangle &:= \int_\Omega \{\, \nabla v^t A\nabla u + duv - \lambda uv \,\} + \mu\left\{\int_\Omega u^2 - 1\right\}.\end{aligned}$$

Then, $[\lambda, u] \in X$, $\|u\|_{0,2} = 1$, is a weak solution of Problem (3.38) if and only if it is a solution of Problem (2.1). Moreover, one easily checks that $[\lambda, u]$ is a regular solution in the sense of Proposition 2.1 if and only if λ is a simple eigenvalue of the differential operator associated with Problem (3.38).

As in the previous chapter, we do not specify the discretization of Problem (3.38) in detail. We only assume that

$$X_h = I\!R \times V_h \subset X, \quad Y_h = I\!R \times W_h \subset Y,$$
$$\langle F_h([\lambda_h, u_h]), [\mu_h, v_h]\rangle = \langle F([\lambda_h, u_h]), [\mu_h, v_h]\rangle \quad \forall [\lambda_h, u_h] \in X_h, [\mu_h, v_h] \in Y_h,$$

where V_h, W_h are finite element spaces corresponding to $\mathcal{T}_h$ which consist of affine equivalent elements in the sense of [37] and which satisfy $\{ v_h \in S_h^{1,0} : v_h = 0 \text{ on } \Gamma \} \subset W_h$. Obviously, the consistency error of the above discretization vanishes. Moreover, $[\lambda_h, u_h] \in X_h$ is a solution of Problem (2.10) if and only if

$$\int_\Omega \{ \nabla v_h^t A \nabla u_h + d u_h v_h \} = \lambda_h \int_\Omega u_h v_h \quad \forall v_h \in W_h$$
$$\int_\Omega u_h^2 = 1. \tag{3.39}$$

Hence, Problem (2.10) is equivalent to a standard finite dimensional eigenvalue problem. In what follows, we will always assume that $[\lambda_h, u_h] \in X_h$ is a solution of Problem (3.39).

Let m be the maximal polynomial degree of the functions in W_h. Proceeding as in the previous section, we set

$$A_h := \sum_{T \in \mathcal{T}_h} \pi_{1,T} A,$$
$$d_h := \sum_{T \in \mathcal{T}_h} \pi_{0,T} d,$$
$$R_h := [0, I_h],$$
$$\widetilde{Y}_h := I\!R \times \operatorname{span}\{ \psi_T v, \psi_E P\sigma : v \in \Pi_{m|T}, \sigma \in \Pi_{m|E}, T \in \mathcal{T}_h, E \in \mathcal{E}_{h,\Omega} \},$$
$$\varepsilon_T := \Big\{ h_T^2 \| -\nabla \cdot ((A - A_h)\nabla u_h) + (d - d_h) u_h \|_{0,2;T}^2$$
$$+ \sum_{E \in \mathcal{E}(T) \cap \mathcal{E}_{h,\Omega}} h_E \| [n_E \cdot ((A - A_h)\nabla u_h)]_E \|_{2;E}^2 \Big\}^{1/2}, \tag{3.40}$$
$$\eta_T := \Big\{ h_T^2 \| -\nabla \cdot (A_h \nabla u_h) + d_h u_h - \lambda_h u_h \|_{0,2;T}^2$$

$$+ \sum_{E\in\mathcal{E}(T)\cap\mathcal{E}_{h,\Omega}} h_E \|[n_E\cdot(A_h\nabla u_h)]_E\|^2_{2;E} \Big\}^{1/2}, \tag{3.41}$$

and define $\widetilde{F}_h$ in the same way as F with A and d replaced by A_h and d_h, respectively. Note, that

$$\varepsilon_T \le c\, h_T^2 \{ \|A\|_{2,\infty;T}\|u_h\|_{1,2;T} + \|d\|_{1,\infty;T}\|u_h\|_{0,2;T} \}. \tag{3.42}$$

Taking into account Equation (3.39) and using the same arguments as in the previous section we conclude that

$$\begin{aligned}
\langle F([\lambda_h,u_h]),[\mu,v]\rangle &= \sum_{T\in\mathcal{T}_h}\int_T \{-\nabla\cdot(A\nabla u_h) + du_h - \lambda_h u_h\}\, v \\
&\quad + \sum_{E\in\mathcal{E}_{h,\Omega}}\int_E [n_E\cdot(A\nabla u_h)]_E\, v \quad \forall [\mu,v]\in Y, \\
\langle \widetilde{F}_h([\lambda_h,u_h]),[\mu,v]\rangle &= \sum_{T\in\mathcal{T}_h}\int_T \{-\nabla\cdot(A_h\nabla u_h) + d_h u_h - \lambda_h u_h\}\, v \\
&\quad + \sum_{E\in\mathcal{E}_{h,\Omega}}\int_E [n_E\cdot(A_h\nabla u_h)]_E\, v \quad \forall [\mu,v]\in Y,
\end{aligned}$$

$$\|(\mathrm{Id}_Y - R_h)^*[F([\lambda_h,u_h]) - \widetilde{F}_h([\lambda_h,u_h])]\|_{Y^*} \le c\Big\{\sum_{T\in\mathcal{T}_h}\varepsilon_T^2\Big\}^{1/2}, \tag{3.43}$$

$$\|F([\lambda_h,u_h]) - \widetilde{F}_h([\lambda_h,u_h])\|_{\widetilde{Y}_h^*} \le c\Big\{\sum_{T\in\mathcal{T}_h}\varepsilon_T^2\Big\}^{1/2}, \tag{3.44}$$

$$\|(\mathrm{Id}_Y - R_h)^*\widetilde{F}_h([\lambda_h,u_h])\|_{Y^*} \le c\Big\{\sum_{T\in\mathcal{T}_h}\eta_T^2\Big\}^{1/2}, \tag{3.45}$$

$$\|\widetilde{F}_h([\lambda_h,u_h])\|_{\widetilde{Y}_h^*} \le c\Big\{\sum_{T\in\mathcal{T}_h}\eta_T^2\Big\}^{1/2}, \tag{3.46}$$

$$\eta_T \le \sup_{\substack{[0,v]\in\widetilde{Y}_h \\ \operatorname{supp} v\subset\omega_T}} \|[0,v]\|_Y^{-1}\langle \widetilde{F}_h([\lambda_h,u_h]),[0,v]\rangle. \tag{3.47}$$

Inequalities (3.45) and (3.47) in particular prove Inequality (2.11).

Propositions 2.1 and 2.5 and Inequalities (3.43)–(3.47) yield the following a posteriori error estimate for Problem (3.38).

Proposition 3.17. *Let λ be a simple eigenvalue of the differential operator associated with Problem (3.38) and let u be a corresponding eigenfunction with $\|u\|_{0,2} = 1$. Let $[\lambda_h, u_h] \in X_h$ be a solution of Problem (3.39) which is sufficiently close to $[\lambda, u]$ in the sense of Proposition 2.1. Then the following a posteriori error estimates hold*

$$|\lambda - \lambda_h| + \|u - u_h\|_{1,2} \leq c_1 \left\{ \sum_{T \in \mathcal{T}_h} \eta_T^2 \right\}^{1/2} + c_2 \left\{ \sum_{T \in \mathcal{T}_h} \varepsilon_T^2 \right\}^{1/2}$$

and

$$\left\{ \sum_{T \in \mathcal{T}_h} \eta_T^2 \right\}^{1/2} \leq c_3 \{ |\lambda - \lambda_h| + \|u - u_h\|_{1,2} \} + c_4 \left\{ \sum_{T \in \mathcal{T}_h} \varepsilon_T^2 \right\}^{1/2},$$

where the constants $c_1, \ldots, c_4$ only depend on the polynomial degree of the spaces V_h and W_h and on the ratio h_T/ϱ_T and where ε_T and η_T are given by Equations (3.40) and (3.41), respectively. □

Remark 3.18. The condition, that $[\lambda_h, u_h]$ has to be sufficiently close to $[\lambda, u]$, essentially means that $|\lambda - \lambda_h|$ has to be smaller than the distance of λ to its neighbouring eigenvalues. In contrast to Proposition 3.10 we obtain in Proposition 3.17 only a global lower bound on the error. This is due to the global nature of the constraint $\int_\Omega u^2 = 1$ inherent in the definition of F. Proposition 3.17 can easily be extended to the case of Neumann boundary conditions. One only has to replace in Equations (3.40) and (3.41) Γ by the part of the boundary on which Dirichlet boundary conditions are imposed. □

3.5. Stationary, Incompressible Navier-Stokes Equations.

As a first example for the treatment of elliptic systems we consider the stationary, incompressible Navier-Stokes equations

$$\begin{aligned} -\nu \Delta \mathbf{u} + (\mathbf{u} \cdot \nabla)\mathbf{u} + \nabla p &= \mathbf{f} \quad \text{in} \quad \Omega \subset I\!R^n \\ \nabla \cdot \mathbf{u} &= 0 \quad \text{in} \quad \Omega \\ \mathbf{u} &= 0 \quad \text{on} \quad \Gamma \end{aligned} \tag{3.48}$$

where $\nu > 0$ is the constant viscosity of the fluid.
In order to cast Problem (3.48) into the framework of Section 2.1, set

$$M := \{ \mathbf{u} \in W^{1,2}(\Omega)^n : \mathbf{u} = 0 \text{ on } \Gamma \}, \qquad Q := \left\{ p \in L^2(\Omega) : \int_\Omega p = 0 \right\}$$

and define

$$
\begin{aligned}
X := Y &:= M \times Q, \\
\|\cdot\|_X := \|\cdot\|_Y &:= \{\, \|\cdot\|_{1,2}^2 + \|\cdot\|_{0,2}^2 \,\}^{1/2}, \\
\langle F([\mathbf{u},p]),[\mathbf{v},q]\rangle &:= \nu \int_\Omega \nabla\mathbf{u}\nabla\mathbf{v} + \int_\Omega (\mathbf{u}\cdot\nabla)\mathbf{u}\,\mathbf{v} - \int_\Omega p\nabla\cdot\mathbf{v} + \int_\Omega q\nabla\cdot\mathbf{u} - \int_\Omega \mathbf{f}\,\mathbf{v}.
\end{aligned}
$$

Let $M_h \subset M$ and $Q_h \subset Q$ be two finite element spaces corresponding to $\mathcal{T}_h$ consisting of affine equivalent elements in the sense of [37]. We assume that there are two integers $k, l \geq 1$ such that

$$
[S_h^{1,0}]^n \cap M \subset M_h \subset [S_h^{k,0}]^n
$$

and

$$
S_h^{1,0} \cap Q \subset Q_h \subset S_h^{l,0} \quad \text{or} \quad S_h^{0,-1} \cap Q \subset Q_h \subset S_h^{l,-1}.
$$

Define

$$
\begin{aligned}
X_h := Y_h &:= M_h \times Q_h, \\
\langle F_h([\mathbf{u}_h,p_h]),[\mathbf{v}_h,q_h]\rangle &:= \langle F([\mathbf{u}_h,p_h]),[\mathbf{v}_h,q_h]\rangle \\
&\quad + \delta \sum_{T\in\mathcal{T}_h} h_T^2 \int_T \{\, -\nu\Delta\mathbf{u}_h + (\mathbf{u}_h\cdot\nabla)\mathbf{u}_h + \nabla p_h - \mathbf{f} \,\} \\
&\qquad\qquad \{\, (\mathbf{u}_h\cdot\nabla)\mathbf{v}_h + \nabla q_h \,\} \\
&\quad + \delta \sum_{E\in\mathcal{E}_{h,\Omega}} h_E \int_E [p_h]_E [q_h]_E \\
&\quad + \alpha\delta \int_\Omega \nabla\cdot\mathbf{u}_h \nabla\cdot\mathbf{v}_h. \qquad (3.49)
\end{aligned}
$$

Here, $\alpha \geq 0$, $\delta \geq 0$ are stability parameters. If $\alpha > 0$, $\delta > 0$, the above discretization is capable of stabilizing both the influence of the convection term and of the divergence constraint without any conditions about the spaces M_h, Q_h or the Peclet-number $h_T\nu^{-1}$ (cf. [74] where also optimal a priori error estimates are established). The case $\alpha = \delta = 0$ corresponds to the standard mixed finite element discretization of Problem (3.48). The spaces M_h, Q_h then have to satisfy the Babuška-Brezzi condition

$$
\inf_{p_h\in Q_h\setminus\{0\}} \sup_{\mathbf{u}_h\in M_h\setminus\{0\}} \frac{\int_T p_h \nabla\cdot\mathbf{u}_h}{\|p_h\|_{0,2}\|\mathbf{u}_h\|_{1,2}} \geq \beta > 0 \qquad (3.50)
$$

with a constant β independent of h. Moreover, the Peclet number $h_T\nu^{-1}$ must be sufficiently small in order to balance the influence of the convection term (cf. [48] where examples of spaces M_h, Q_h satisfying Inequality (3.50) are also given).
If $\alpha = \delta = 0$, the consistency error obviously vanishes. If $\delta > 0$, we conclude from standard inverse estimates that it is bounded by

$$\begin{aligned}
&\|F([\mathbf{u}_h,p_h]) - F_h([\mathbf{u}_h,p_h])\|_{Y_h^*} \\
&= \sup_{\substack{[\mathbf{v}_h,q_h]\in Y_h \\ \|[\mathbf{v}_h,q_h]\|_Y=1}} \Big\{ \delta \sum_{T\in\mathcal{T}_h} h_T^2 \int_T \{ -\nu\Delta\mathbf{u}_h + (\mathbf{u}_h\cdot\nabla)\mathbf{u}_h + \nabla p_h - \mathbf{f}\} \\
&\qquad\qquad \{ (\mathbf{u}_h\cdot\nabla)\mathbf{v}_h + \nabla q_h \} \\
&\qquad + \delta \sum_{E\in\mathcal{E}_h,\Omega} h_E \int_E [p_h]_E[q_h]_E + \alpha\delta \int_\Omega \nabla\cdot\mathbf{u}_h \nabla\cdot\mathbf{v}_h \Big\} \\
&\le c(1+\alpha)\delta(1+\|\mathbf{u}_h\|_{1,2}) \Big\{ \sum_{T\in\mathcal{T}_h} [h_T^2\| - \nu\Delta\mathbf{u}_h + (\mathbf{u}_h\cdot\nabla)\mathbf{u}_h + \nabla p_h - \mathbf{f}\|_{0,2;T}^2 \\
&\qquad + \|\nabla\cdot\mathbf{u}_h\|_{0,2;T}^2] + \sum_{E\in\mathcal{E}_h,\Omega} h_E\|[p_h]_E\|_{2;E}^2 \Big\}^{1/2}.
\end{aligned} \tag{3.51}$$

In order to cast this discretization into the framework of Section 2.3, we define $\widetilde{F}_h$ in the same way as F with $\mathbf{f}$ replaced by $\pi_{0,T}\mathbf{f}$ and set

$$\begin{aligned}
R_h[\mathbf{u},p] &:= [I_h u_1,\ldots,I_h u_n,0], \\
\widetilde{Y}_h &:= \operatorname{span}\{ [\psi_T\mathbf{v},0],[\psi_E P\sigma,0],[0,\psi_T p] \\
&\qquad : \mathbf{v}\in[\Pi_{m|T}]^n, \sigma\in[\Pi_{m'|E}]^n, p\in\Pi_{k-1|T}, T\in\mathcal{T}_h, E\in\mathcal{E}_{h,\Omega}\},
\end{aligned}$$

where $m := \max\{2k-1, l-1, 0\}$ and $m' := \max\{k-1, l\}$.

Lemmas 3.1 and 3.3 immediately imply

$$\begin{aligned}
\|\widetilde{F}_h([\mathbf{u}_h,p_h]) - F([\mathbf{u}_h,p_h])\|_{\widetilde{Y}_h^*} &= \sup_{\substack{[\mathbf{v}_h,q_h]\in\widetilde{Y}_h \\ \|[\mathbf{v}_h,q_h]\|_Y=1}} \sum_{T\in\mathcal{T}_h} \int_T (\mathbf{f}-\pi_{0,T}\mathbf{f})\mathbf{v}_h \\
&\le c\Big\{ \sum_{T\in\mathcal{T}_h} h_T^2\|\mathbf{f}-\pi_{0,T}\mathbf{f}\|_{0,2;T}^2 \Big\}^{1/2}
\end{aligned} \tag{3.52}$$

and

$$\begin{aligned}
&\|(\mathrm{Id}_Y - R_h)^*[\widetilde{F}_h([\mathbf{u}_h, p_h]) - F([\mathbf{u}_h, p_h])]\|_{Y^*} \\
&= \sup_{\substack{[\mathbf{v},q]\in Y \\ \|[\mathbf{v},q]\|_Y=1}} \sum_{T\in\mathcal{T}_h} \sum_{i=1}^{n} \int_T (f_i - \pi_{0,T} f_i)(v_i - I_h v_i) \\
&\le c\Big\{ \sum_{T\in\mathcal{T}_h} h_T^2 \|\mathbf{f} - \pi_{0,T}\mathbf{f}\|_{0,2;T}^2 \Big\}^{1/2}. \qquad (3.53)
\end{aligned}$$

For abbreviation, we define for all $T \in \mathcal{T}_h$

$$\begin{aligned}
\eta_T := \Big\{ & h_T^2 \| - \nu\Delta\mathbf{u}_h + (\mathbf{u}_h\cdot\nabla)\mathbf{u}_h + \nabla p_h - \pi_{0,T}\mathbf{f}\|_{0,2;T}^2 \\
&+ \sum_{E\in\mathcal{E}(T)\cap\mathcal{E}_{h,\Omega}} h_E \|[\nu\mathbf{n}_E\cdot\nabla\mathbf{u}_h - p_h\mathbf{n}_E]_E\|_{2;E}^2 \\
&+ \|\nabla\cdot\mathbf{u}_h\|_{0,2;T}^2 \Big\}^{1/2}. \qquad (3.54)
\end{aligned}$$

Observing that the identity

$$\begin{aligned}
\langle \widetilde{F}_h([\mathbf{u}_h, p_h]), [\mathbf{v}, q]\rangle = \sum_{T\in\mathcal{T}_h} \Big\{ & \int_T \{ -\nu\Delta\mathbf{u}_h + (\mathbf{u}_h\cdot\nabla)\mathbf{u}_h + \nabla p_h - \pi_{0,T}\mathbf{f}\}\mathbf{v} \\
&+ \int_T q\nabla\cdot\mathbf{u}_h \Big\} \\
&+ \sum_{E\in\mathcal{E}_{h,\Omega}} \int_E [\nu\mathbf{n}_E\cdot\nabla\mathbf{u}_h - p_h\mathbf{n}_E]_E\,\mathbf{v} \qquad (3.55)
\end{aligned}$$

holds for all $[\mathbf{v}, q] \in Y$, we conclude from Lemmas 3.1 and 3.3 that

$$\begin{aligned}
&\|\widetilde{F}_h([\mathbf{u}_h, p_h])\|_{\widetilde{Y}_h^*} \\
&= \sup_{\substack{[\mathbf{v}_h,q_h]\in\widetilde{Y}_h \\ \|[\mathbf{v}_h,q_h]\|_Y=1}} \Big\{ \sum_{T\in\mathcal{T}_h} \Big\{ \int_T \{ -\nu\Delta\mathbf{u}_h + (\mathbf{u}_h\cdot\nabla)\mathbf{u}_h + \nabla p_h - \pi_{0,T}\mathbf{f}\}\mathbf{v}_h \\
&\qquad + \int_T q_h\nabla\cdot\mathbf{u}_h \Big\} + \sum_{E\in\mathcal{E}_{h,\Omega}} \int_E [\nu\mathbf{n}_E\cdot\nabla\mathbf{u}_h - p_h\mathbf{n}_E]_E\,\mathbf{v}_h \Big\} \\
&\le c\Big\{ \sum_{T\in\mathcal{T}_h} \eta_T^2 \Big\}^{1/2} \qquad (3.56)
\end{aligned}$$

and

$$
\begin{aligned}
&\|(\mathrm{Id}_Y - R_h)^* \widetilde{F}_h([\mathbf{u}_h, p_h])\|_{Y^*} \\
&= \sup_{\substack{[\mathbf{v},q]\in Y \\ \|[\mathbf{v},q]\|_Y=1}} \Bigg\{ \sum_{T\in\mathcal{T}_h} \Bigg\{ \sum_{i=1}^{n} \int_T \{ -\nu\Delta u_{h,i} + (\mathbf{u}_h\cdot\nabla)u_{h,i} + \partial_i p_h - \pi_{0,T} f_i \} \{ v_i - I_h v_i \} \\
&\qquad\qquad + \int_T q\nabla\cdot\mathbf{u}_h \Bigg\} \\
&\qquad\qquad + \sum_{E\in\mathcal{E}_{h,\Omega}} \sum_{i=1}^{n} \int_E [\nu\mathbf{n}_E\cdot\nabla u_{h,i} - p_h n_{Ei}]_E (v_i - I_h v_i) \Bigg\} \\
&\le c \Bigg\{ \sum_{T\in\mathcal{T}_h} \eta_T^2 \Bigg\}^{1/2} .
\end{aligned} \tag{3.57}
$$

In order to prove Inequality (2.11), consider an arbitrary simplex $T \in \mathcal{T}_h$ and an arbitrary face $E \in \mathcal{E}_{h,\Omega}$ of T and define $\widetilde{Y}_{h|\omega}$, $\omega \in \{T, \omega_E, \omega_T\}$, as in Section 3.2. The definition of $\widetilde{Y}_h$, Equation (3.55), and Lemma 3.3 then yield the estimates

$$
\begin{aligned}
&c_1 \|\nabla\cdot\mathbf{u}_h\|_{0,2;T} \\
&\le \sup_{r\in\Pi_{k-1|T}\setminus\{0\}} \|r\|_{0,2;T}^{-1} \int_T \nabla\cdot u_h \psi_T r \\
&= \sup_{r\in\Pi_{k-1|T}\setminus\{0\}} \langle \widetilde{F}_h([\mathbf{u}_h, p_h]), [0, \psi_T r] \rangle \, \|r\|_{0,2;T}^{-1} \\
&\le \sup_{\substack{[\mathbf{v},q]\in\widetilde{Y}_{h|T} \\ \|[\mathbf{v},q]\|_Y=1}} \langle \widetilde{F}_h([\mathbf{u}_h, p_h]), [\mathbf{v}, q] \rangle,
\end{aligned} \tag{3.58}
$$

$$
\begin{aligned}
&c_1 c_4^{-1} h_T \| -\nu\Delta\mathbf{u}_h + (\mathbf{u}_h\cdot\nabla)\mathbf{u}_h + \nabla p_h - \pi_{0,T}\mathbf{f}\|_{0,2;T} \\
&\le \sup_{\mathbf{w}\in[\Pi_{m|T}]^n\setminus\{0\}} \|\nabla(\psi_T\mathbf{w})\|_{0,2;T}^{-1} \int_T \{ -\nu\Delta\mathbf{u}_h + (\mathbf{u}_h\cdot\nabla)\mathbf{u}_h + \nabla p_h - \pi_{0,T}\mathbf{f} \} \psi_T \mathbf{w} \\
&= \sup_{\mathbf{w}\in[\Pi_{m|T}]^n\setminus\{0\}} \|\nabla(\psi_T\mathbf{w})\|_{0,2;T}^{-1} \langle \widetilde{F}_h([\mathbf{u}_h, p_h]), [\psi_T\mathbf{w}, 0] \rangle \\
&\le \sup_{\substack{[\mathbf{v},q]\in\widetilde{Y}_{h|T} \\ \|[\mathbf{v},q]\|_Y=1}} \langle \widetilde{F}_h([\mathbf{u}_h, p_h], [\mathbf{v}, q] \rangle,
\end{aligned} \tag{3.59}
$$

and, using Inequality (3.59),

$$
\begin{aligned}
&c_2 c_6^{-1} c_7^{-1} h_E^{1/2} \|[\nu\mathbf{n}_E\cdot\nabla\mathbf{u}_h - p_h\mathbf{n}_E]_E\|_{2;E} \\
&\le \sup_{\sigma\in[\Pi_{m'|E}]^n\setminus\{0\}} c_2 c_6^{-1} c_7^{-1} h_E^{1/2} \|\sigma\|_{2;E}^{-1} \int_E [\nu\mathbf{n}_E\cdot\nabla\mathbf{u}_h - p_h\mathbf{n}_E] \psi_E P\sigma
\end{aligned}
$$

$$
\begin{aligned}
&= \sup_{\sigma\in[\Pi_{m'|E}]^n\setminus\{0\}} c_2c_6^{-1}c_7^{-1}h_E^{1/2}\|\sigma\|_{2;E}^{-1}\Big\{\langle\widetilde{F}_h([\mathbf{u}_h,p_h]),[\psi_E P\sigma,0]\rangle \\
&\qquad - \int_{\omega_E}\{-\nu\Delta\mathbf{u}_h+(\mathbf{u}_h\cdot\nabla)\mathbf{u}_h+\nabla p_h-\pi_{0,T}\mathbf{f}\}\psi_E P\sigma\Big\} \\
&\le \sup_{\substack{[\mathbf{v},q]\in\widetilde{Y}_{h|\omega_E}\\ \|[\mathbf{v},q]\|_Y=1}} \langle\widetilde{F}_h([\mathbf{u}_h,p_h]),[\mathbf{v},q]\rangle \\
&\qquad + c_6^{-1}h_E\|-\nu\Delta\mathbf{u}_h+(\mathbf{u}_h\cdot\nabla)\mathbf{u}_h+\nabla p_h-\pi_{0,T}\mathbf{f}\|_{0,2;\omega_E} \\
&\le c \sup_{\substack{[\mathbf{v},q]\in\widetilde{Y}_{h|\omega_E}\\ \|[\mathbf{v},q]\|_Y=1}} \langle\widetilde{F}_h([\mathbf{u}_h,p_h]),[\mathbf{v},q]\rangle. \qquad (3.60)
\end{aligned}
$$

Inequalities (3.58)–(3.60) imply

$$
\eta_T \le c \sup_{\substack{[\mathbf{v},q]\in\widetilde{Y}_{h|\omega_T}\\ \|[\mathbf{v},q]\|_Y=1}} \langle\widetilde{F}_h([\mathbf{u}_h,p_h]),[\mathbf{v},q]\rangle \qquad (3.61)
$$

and

$$
\Big\{\sum_{T\in\mathcal{T}_h}\eta_T^2\Big\}^{1/2} \le c\|\widetilde{F}_h([\mathbf{u}_h,p_h])\|_{\widetilde{Y}_h^*}. \qquad (3.62)
$$

Inequalities (3.56), (3.57), and (3.62) prove Inequality (2.11) and show that, up to multiplicative constants, $\|\widetilde{F}_h([\mathbf{u}_h,p_h])\|_{\widetilde{Y}_h^*}$ is bounded from above and from below by $\{\sum_T\eta_T^2\}^{1/2}$. Propositions 2.1 and 2.5 and Inequalities (3.51), (3.52), (3.53), (3.56), and (3.61) now yield the following a posteriori error estimate.

Proposition 3.19. *Let $[\mathbf{u},p]$ be a weak solution of Problem (3.48) which is regular in the sense of Proposition 2.1 and let $[\mathbf{u}_h,p_h]\in X_h$ be a solution of*

$$
\langle F_h([\mathbf{u}_h,p_h]),[\mathbf{v}_h,q_h]\rangle = 0, \quad \forall[\mathbf{v}_h,q_h]\in Y_h,
$$

where F_h is given in Equation (3.49), which is sufficiently close to $[\mathbf{u},p]$ in the sense of Proposition 2.1. Then the following a posteriori error estimates hold

$$
\begin{aligned}
\{\|\mathbf{u}-\mathbf{u}_h\|_{1,2}^2+\|p-p_h\|_{0,2}^2\}^{1/2} &\le c_1[1+(1+\alpha)\delta(1+\|\mathbf{u}_h\|_{1,2})]\Big\{\sum_{T\in\mathcal{T}_h}\eta_T^2\Big\}^{1/2} \\
&\quad + c_2\Big\{\sum_{T\in\mathcal{T}_h}h_T^2\|\mathbf{f}-\pi_{0,T}\mathbf{f}\|_{0,2;T}^2\Big\}^{1/2}, \\
\eta_T &\le c_3\{\|\mathbf{u}-\mathbf{u}_h\|_{1,2;\omega_T}^2+\|p-p_h\|_{0,2;\omega_T}^2\}^{1/2} \\
&\quad + c_4\Big\{\sum_{T'\subset\omega_T}h_{T'}^2\|\mathbf{f}-\pi_{0,T'}\mathbf{f}\|_{0,2;T'}^2\Big\}^{1/2},
\end{aligned}
$$

where η_T is given by Equation (3.54) and the constants $c_1, \ldots, c_4$ only depend on the polynomial degrees of the spaces M_h, Q_h and on the ratio h_T/ϱ_T. □

Remark 3.20. Proposition 3.19 can be extended to the case of the slip boundary condition

$$\mathbf{u} \cdot \mathbf{n} = \mathbf{T}(\nu\mathbf{u}, p) - [\mathbf{n} \cdot \mathbf{T}(\nu\mathbf{u}, p) \cdot \mathbf{n}]\mathbf{n} = 0$$

where

$$\mathbf{T}(\mathbf{u}, p) := \left(\frac{1}{2}(\partial_i u_j + \partial_j u_i) - p\delta_{ij}\right)_{1 \le i,j \le n}$$

denotes the stress tensor. One then has to replace in Equation (3.54) $\nu\nabla\mathbf{u} - p\mathbf{I}$ by $\mathbf{T}(\nu\mathbf{u}, p)$ and Γ by the part of the boundary on which the no-slip condition $\mathbf{u} = 0$ is imposed. Here, $\mathbf{I} := (\delta_{ij})_{1 \le i,j \le n}$ denotes the unit tensor. Of course, the discretization then also has to take account of the different boundary condition (cf., e. g., [77, 83]). □

Remark 3.21. The previous results can also be extended to non-Newtonian fluids. Combining the arguments used to establish Propositions 3.10, 3.12, and 3.19, one can prove that the error estimator of [25] also yields local lower bounds similar to the second estimate of Proposition 2.5. □

Next, we introduce an a posteriori error estimator for Problem (3.48) which is based on the solution of discrete local Stokes problems and which fits into the framework of Proposition 2.7. This estimator is an extension to the Navier-Stokes equations of $\eta_{D,x}$ defined in Equation (1.33).
We choose an arbitrary vertex x_0 in the partition $\mathcal{T}_h$ and keep it fixed in what follows. Let $\omega_0 = \omega_{x_0}$, $\mathcal{T}_0 := \{ T \in \mathcal{T}_h : x_0 \in \mathcal{N}(T) \}$, and $\mathcal{E}_0 := \{ E \in \mathcal{E}_h : x_0 \in \mathcal{N}(E) \}$. Choose ψ_T and ψ_E as in Remark 3.2 and put

$$M_0 := \operatorname{span}\{ \psi_T \mathbf{v}, \psi_E P\sigma : \mathbf{v} \in [\Pi_{m''|T}]^n, \sigma \in [\Pi_{m'|E}]^n, T \in \mathcal{T}_0, E \in \mathcal{E}_0 \},$$
$$Q_0 := \operatorname{span}\{ \psi_T p : p \in \Pi_{k-1|T}, T \in \mathcal{T}_0 \},$$

where $m := \max\{2k-1, l-1, 0\}$, $m' := \max\{k-1, l\}$, and $m'' := \max\{m, k+n-1\}$ and define

$$\widehat{X}_h := \widehat{Y}_h := M_0 \times Q_0,$$
$$\langle B([\mathbf{v}, q]), [\mathbf{w}, r] \rangle := \nu \int_{\omega_0} \nabla\mathbf{v}\nabla\mathbf{w} - \int_{\omega_0} q\nabla \cdot \mathbf{w} + \int_{\omega_0} r\nabla \cdot \mathbf{v} \quad \forall [\mathbf{v}, q], [\mathbf{w}, r] \in \widehat{X}_h.$$

The definition of m'' implies that $\psi_T \nabla q \in M_0$ for all $q \in Q_0$. Together with Lemma 3.3 this shows that the spaces M_0, Q_0 satisfy an analogon of Equation

(3.50). Hence, we have $B \in \mathrm{Isom}(\widehat{X}_h, \widehat{Y}_h^*)$. Let $[\mathbf{u}_0, p_0] \in \widehat{X}_h$ be the unique solution of

$$\langle B([\mathbf{u}_0, p_0]), [\mathbf{w}, r]\rangle = \langle \widetilde{F}_h([\mathbf{u}_h, p_h]), [\mathbf{w}, r]\rangle \quad \forall [\mathbf{w}, r] \in \widehat{Y}_h \tag{3.63}$$

and define

$$\eta_{x_0} := \{\, \nu \|\mathbf{u}_0\|^2_{1,2;\omega_0} + \|p_0\|^2_{0,2;\omega_0} \,\}^{1/2}. \tag{3.64}$$

Note, that Problem (3.63) is equivalent to

$$\begin{aligned} \nu \int_{\omega_0} \nabla \mathbf{u}_0 \nabla \mathbf{w} - \int_{\omega_0} p_0 \nabla \cdot \mathbf{w} &= \nu \int_{\omega_0} \nabla \mathbf{u}_h \nabla \mathbf{w} + \int_{\omega_0} (\mathbf{u}_h \cdot \nabla)\mathbf{u}_h \mathbf{w} \\ &\quad - \int_{\omega_0} p_h \nabla \cdot \mathbf{w} - \int_{\omega_0} \pi_{0,T} \mathbf{f} \mathbf{w} \quad \forall \mathbf{w} \in M_0 \\ \int_{\omega_0} r \nabla \cdot \mathbf{u}_0 &= \int_{\omega_0} r \nabla \cdot \mathbf{u}_h \quad \forall r \in Q_0. \end{aligned}$$

This shows that η_{x_0} is a generalization of the estimator $\eta_{D,x}$ of Section 1.3. Obviously, we have $\widetilde{Y}_{h|\omega_0} \subset \widehat{Y}_h$. Lemma 3.3 and Equation (3.55) on the other hand immediately imply

$$\begin{aligned} &\|\widetilde{F}_h([\mathbf{u}_h, p_h])\|_{\widehat{Y}_h^*} \\ &= \sup_{\substack{[\mathbf{v},q]\in\widehat{Y}_h \\ \|[\mathbf{v},q]\|_Y=1}} \left\{ \sum_{T\in\mathcal{T}_0} \left\{ \int_T \{ -\nu\Delta\mathbf{u}_h + (\mathbf{u}_h\cdot\nabla)\mathbf{u}_h + \nabla p_h - \pi_{0,T}\mathbf{f} \}\mathbf{v} + \int_T q\nabla\cdot\mathbf{u}_h \right\} \right. \\ &\qquad \left. + \sum_{E\in\mathcal{E}_0} \int_E [\nu \mathbf{n}_E \cdot \nabla \mathbf{u}_h - p_h \mathbf{n}_E]_E \mathbf{v} \right\} \\ &\le c \left\{ \sum_{T\in\mathcal{T}_0} \eta_T^2 \right\}^{1/2}. \end{aligned}$$

Together with Inequality (3.61) this proves

$$\|\widetilde{F}_h([\mathbf{u}_h, p_h])\|_{\widehat{Y}_h^*} \le c \sup_{\substack{[\mathbf{v},q]\in\widetilde{Y}_{h|\omega_0} \\ \|[\mathbf{v},q]\|_Y=1}} \langle \widetilde{F}_h([\mathbf{u}_h, p_h]), [\mathbf{v}, q]\rangle.$$

These results and Proposition 2.7 yield the following a posteriori error estimate which is a generalization of Proposition 1.9.

Proposition 3.22. *Let x_0 be an arbitrary vertex in the partition $\mathcal{T}_h$. Then there are two constants c_1, c_2, which only depend on the polynomial degree of the spaces M_h, Q_h and on the ratio h_T/ϱ_T, such that the following inequalities hold*

$$c_1\left\{\sum_{T\in\mathcal{T}_0}\eta_T^2\right\}^{1/2} \leq \eta_{x_0} \leq c_2\left\{\sum_{T\in\mathcal{T}_0}\eta_T^2\right\}^{1/2}.$$

Here, η_T and η_{x_0} are given by Equations (3.54) and (3.64), respectively. □

Remark 3.23. Remarks 2.4, 2.9, and 3.6 yield estimates on the error measured in the L^2-norm for the velocity and in the $W^{-1,2}$-norm for the pressure. More precisely, we set

$$\begin{aligned} X_- &:= L^2(\Omega)^n \times [W^{1,2}(\Omega)\cap Q]^*, \\ X_+ &:= [W^{1,3}(\Omega)^n \cap M] \times Q, \\ Y_+ &:= [W^{2,2}(\Omega)^n \cap M] \times [W^{1,2}(\Omega)\cap Q]. \end{aligned}$$

The choice of X_+ and the Sobolev embedding theorem ensure the Lipschitz continuity of $DF([\mathbf{u},p])$. The condition $DF([\mathbf{u},p])^* \in \mathrm{Isom}(X_-, Y_+^*)$ is equivalent to the assumption that the linearized adjoint Navier-Stokes equations

$$\begin{aligned} -\nu\Delta\mathbf{v} - (\mathbf{u}\cdot\nabla)\mathbf{v} + (\mathbf{v}\cdot\nabla)\mathbf{u} - \nabla q &= \mathbf{w} \quad \text{in} \quad \Omega \\ -\nabla\cdot\mathbf{v} &= r \quad \text{in} \quad \Omega \\ \mathbf{v} &= 0 \quad \text{on} \quad \Gamma \end{aligned}$$

admit for each right-hand side $[\mathbf{w},r] \in L^2(\Omega)^n \times [W^{1,2}(\Omega)\cap Q]$ a unique weak solution $[\mathbf{v},q] \in Y_+$ which depends continuously on $[\mathbf{w},r]$. This assumption is certainly fulfilled if Ω is convex and $\nu^{-2}\|\mathbf{f}\|_{0,2} < 1$. The spaces $\widehat{Y}_h, \widehat{Y}_h$, and $\widehat{X}_h$ are defined as before; but now, the functions ψ_T, ψ_E, and the continuation operator P are choosen according to Remark 3.6. The arguments leading to Estimates (3.52), (3.53), (3.56), (3.57), (3.61), and (3.62) carry over immediately. But now, we always gain an additional power of h. The arguments used for establishing Estimates (3.20) and (3.21) show that $B \in \mathrm{Isom}(\widehat{X}_h, \widehat{Y}_h^*)$ also with the present choice of norms. Assuming that the consistency error $\|F([\mathbf{u}_h,p_h]) - F_h([\mathbf{u}_h,p_h])\|_{Y_h^*}$

vanishes, we obtain the following error estimates

$$\begin{aligned}
\|\mathbf{u}-\mathbf{u}_h\|_{0,2}+\|p-p_h\|_{-1,2} \leq & c_1\left\{\sum_{T\in\mathcal{T}_h} h_T^2\eta_T^2\right\}^{1/2} \\
& + c_2\left\{\sum_{T\in\mathcal{T}_h} h_T^4\|\mathbf{f}-\pi_{0,T}\mathbf{f}\|_{0,2;T}^2\right\}^{1/2}, \\
h_T\eta_T \leq & c_3\left\{\|\mathbf{u}-\mathbf{u}_h\|_{0,2;\omega_T}^2+\|p-p_h\|_{-1,2;\omega_T}^2\right\}^{1/2} \\
& + c_4\left\{\sum_{T'\subset\omega_T} h_{T'}^4\|\mathbf{f}-\pi_{0,T'}\mathbf{f}\|_{0,2;T'}^2\right\}^{1/2}, \\
c_5\left\{\sum_{T\in\mathcal{T}_0} h_T^2\eta_T^2\right\}^{1/2} \leq & \widetilde{\eta}_{x_0} \leq c_6\left\{\sum_{T\in\mathcal{T}_0} h_T^2\eta_T^2\right\}^{1/2}.
\end{aligned}$$

Here, η_T is as in Equations (3.54) and $\widetilde{\eta}_{x_0}$ is given by

$$\widetilde{\eta}_{x_0} := \{\nu\|\mathbf{u}_0\|_{0,2;\omega_0}^2+h_0^2\|p_0\|_{0,2;\omega_0}^2\}^{1/2} \tag{3.65}$$

with $h_0 := \mathrm{diam}\omega_0$ and $[\mathbf{u}_0, p_0]$ denoting the solution of the auxiliary local problem (3.63). In order to obtain an expression for $\widetilde{\eta}_{x_0}$ which is easy to compute we have used the fact that $\{\nu\|\mathbf{v}_0\|_{0,2;\omega_0}^2+\|q\|_{-1,2;\omega_0}^2\}^{1/2}$ and $\{\nu\|\mathbf{v}_0\|_{0,2;\omega_0}^2+h_0^2\|q\|_{0,2;\omega_0}^2\}^{1/2}$ define equivalent norms on $\widehat{Y}_h$. □

Remark 3.24. The present analysis follows the lines of [86]. Residual a posteriori error estimators for the Stokes and Navier-Stokes equations have been considered in [2, 23, 24, 27, 78, 79, 80, 84]. Estimaters based on the solution of auxiliary local problems are considered in [23, 24, 27, 89]. □

3.6. The Equations of Linearized Elasticity.

As a second example for an elliptic system we consider the equations of linearized elasticity in $I\!R^n, n = 2, 3$:

$$\begin{aligned}
-\nabla\cdot\sigma(u) &= f \quad \text{in} \quad \Omega\subset I\!R^n \\
u &= 0 \quad \text{on} \quad \Gamma_D \\
n\cdot\sigma(u) &= g \quad \text{on} \quad \Gamma_N.
\end{aligned} \tag{3.66}$$

Here, $u : \Omega \to I\!R^n$ is the sought displacement of the elastic structure, f denotes the external load, and g describes prescribed boundary tractions. The stress tensor $\sigma(u)$ is related to the strain tensor

$$\epsilon(u) := (\frac{1}{2}(\partial_i u_j+\partial_j u_i))_{1\leq i,j\leq n}$$

by the constitutive law

$$\sigma(u) = 2\mu\epsilon(u) + \lambda tr\epsilon(u)I. \tag{3.67}$$

Here, $I = (\delta_{ij})_{1\le i,j\le n}$ denotes the unit tensor,

$$tr\epsilon(u) := \sum_{i=1}^{n} \epsilon_{ii}(u)$$

is the trace of $\epsilon(u)$, and $\lambda > 0$ and $\mu > 0$ are the Lamé constants. As in the previous sections we assume for simplicity that Ω is a polyhedron. In order to guarantee the unique solvability of Problem (3.66), the Dirichlet part Γ_D of the boundary must have a positive $(n-1)$-dimensional Lebesgue measure.

Remark 3.25. For simplicity we assume throughout this section that the displacement u vanishes on Γ_D. Non-vanishing displacements on Γ_D can be treated in a completely similar fashion. □

Problem (3.66) can be treated in the same way as the model problem (1.1) in Section 3.2. We set

$$\begin{aligned} X := Y :=& \{u \in W^{1,2}(\Omega)^n : u = 0 \quad \text{on} \quad \Gamma_D\}, \\ \|.\|_X :=& \|.\|_Y := \|.\|_{1,2}, \\ \langle F(u), \varphi\rangle :=& \int_\Omega \sigma(u) : \epsilon(v) - \int_\Omega f \cdot v - \int_{\Gamma_N} g \cdot v. \end{aligned}$$

Then $u \in X$ is a solution of Problem (2.1), i.e. $F(u) = 0$, if and only if it is a weak solution of Problem (3.66). Since the bilinear form $u, v \to \int_\Omega \sigma(u) : \epsilon(v)$ is continuous and coercive on X (cf., e.g., [37; Ch.1, §1.2]), we have $DF(u) \in \mathrm{Isom}(X, Y^*)$ for all $u \in X$.

We do not specify the discretization of Problem (3.66) in detail. We only assume that $X_h \subset X \cap W^{1,\infty}(\Omega)^n$ and $Y_h \subset Y \cap W^{1,\infty}(\Omega)^n$ are finite element spaces corresponding to $\mathcal{T}_h$ consisting of affine equivalent elements in the sense of [37] and that $[S_h^{1,0}]^n \cap Y \subset Y_h$. F_h is defined by

$$\langle F_h(u_h), \varphi_h\rangle = \langle F(u_h), \varphi_h\rangle \quad \forall u_h \in X_h, \varphi_h \in Y_h.$$

In particular this implies that the consistency error $\|F(u_h) - F_h(u_h)\|_{Y_h^*}$ vanishes.

Denote by $k \ge 1$ the maximal polynomial degree of the functions in X_h. Set $m := \max\{0, k-2\}$. Then, $\widetilde{F}_h$ is defined in the same way as F with f and g

replaced by $\sum_{T\in\mathcal{T}_h} \pi_{m,T} f$ and $\sum_{E\in\mathcal{E}_{h,N}} \pi_{k-1,E} g$, respectively. The space $\widetilde{Y}_h$ is given by

$$\widetilde{Y}_h := \operatorname{span}\{b_T v, b_E P\tau : v \in \Pi^n_{m|T}, \\ \tau \in \Pi^n_{k-1|E}, T \in \mathcal{T}_h, E \in \mathcal{E}_{h,\Omega} \cup \mathcal{E}_{h,N}\},$$

where b_T and b_E are the bubble functions of Section 1.1 und P is the continuation operator of Section 3.1. The restriction operator R_h is defined by applying the Clément interpolation operator I_h componentwise.

In order to derive a representation of the residual which is similar to Equations (1.14), (3.25), and (3.26), we observe that the Green's formula

$$\int_\omega \sigma(u) : \epsilon(v) = -\int_\omega \nabla\cdot\sigma(u)\cdot v + \int_{\partial\omega} n\cdot\sigma(u)\cdot v \tag{3.68}$$

holds for all $u \in W^{2,2}(\omega)^n, v \in W^{1,2}(\omega)^n$ and all $\omega \subset I\!R^n$ having a Lipschitz boundary $\partial\omega$ (cf. [37; Equ. (1.2.38)]). Applying (3.68) elementwise we immediately obtain for all $\varphi \in Y$

$$\begin{aligned}\langle F(u_h),\varphi\rangle = &\sum_{T\in\mathcal{T}_h}\int_T \{-\nabla\cdot\sigma(u_h) - f\}\cdot\varphi \\ &+ \sum_{E\in\mathcal{E}_{h,\Omega}}\int_E [n_E\cdot\sigma(u_h)]_E\cdot\varphi \\ &+ \sum_{E\in\mathcal{E}_{h,N}}\int_E \{n\cdot\sigma(u_h) - g\}\cdot\varphi\end{aligned}$$

and

$$\begin{aligned}\langle \widetilde{F}_h(u_h),\varphi\rangle = &\sum_{T\in\mathcal{T}_h}\int_T \{-\nabla\cdot\sigma(u_h) - \pi_{m,T} f\}\cdot\varphi \\ &+ \sum_{E\in\mathcal{E}_{h,\Omega}}\int_E [n_E\cdot\sigma(u_h)]_E\cdot\varphi \\ &+ \sum_{E\in\mathcal{E}_{h,N}}\int_E \{n\cdot\sigma(u_h) - \pi_{k-1,E} g\}\cdot\varphi.\end{aligned}$$

We therefore define the residual error estimator $\eta_T, T \in T_h$, by

$$\begin{aligned}\eta_T := \Big\{ &h_T^2\|\nabla\cdot\sigma(u_h) + f\|^2_{0,2;T} + \sum_{E\in\mathcal{E}(T)\cap\mathcal{E}_{h,\Omega}} h_E\|[n_E\cdot\sigma(u_h)]_E\|^2_E \\ &+ \sum_{E\in\mathcal{E}(T)\cap\mathcal{E}_{h,N}} h_E\|n\cdot\sigma(u_h) - \pi_{k-1,E} g\|^2_{2;E}\Big\}^{1/2}.\end{aligned} \tag{3.69}$$

The quantities $\|(\mathrm{Id}_Y - R_h)^*[F(u_h) - \widetilde{F}_h(u_h)]\|_{Y^*}$, $\|F(u_h) - \widetilde{F}_h(u_h)\|_{\widetilde{Y}_h^*}$, $\|(\mathrm{Id}_Y - R_h)^*\widetilde{F}_h(u_h)\|_{Y^*}$, and $\|\widetilde{F}_h(u_h)\|_{\widetilde{Y}_h^*}$ can now be estimated in the same way as in Sections 3.2 and 3.3.

In order to obtain an error estimator which is based on the solution of auxiliary local problems and which generalizes the estimator $\eta_{D,x}$ Section 1.3, we proceed as in Sections 3.2 and 3.3. Choose a vertex x_0 in $\mathcal{N}_h$, set $\omega_0 := \omega_{x_0}$ and

$$\widehat{X}_h := \widehat{Y}_h := \widetilde{Y}_{h|\omega_0}$$

and define the operator $B \in \mathcal{L}(\widehat{X}_h, \widehat{Y}_h^*)$ by

$$\langle Bu, \varphi\rangle := \int_{\omega_0} \sigma(u) : \epsilon(v) \quad \forall u \in \widehat{X}_h, \varphi \in \widehat{Y}_h.$$

Korn's inequality (cf. [37; Eq. (1.2.35)]) implies that $B \in \mathrm{Isom}(\widehat{X}_h, \widehat{Y}_h^*)$. Denote by $u_0 \in \widehat{X}_h$ the unique solution of

$$\langle Bu_0, \varphi\rangle = \langle \widetilde{F}_h(u_h), \varphi\rangle \quad \forall \varphi \in \widehat{Y}_h \tag{3.70}$$

and set

$$\eta_{x_0} := \|\sigma(u_0)\|_{0,2;\omega_0}. \tag{3.71}$$

Note, that Problem (3.70) is a discrete local displacement problem; the load and the boundary tractions on $\partial\omega_0 \cap \Gamma_N$ are implicitely given by the right-hand side.

In order to obtain L^2-error estimates, we proceed as in the second part of Section 3.2 and Remark 3.15 and set

$$\begin{aligned}
X_- :=& L^2(\Omega)^n, \|.\|_{X_-} := \|.\|_{0,2},\\
X_+ :=& X, \|.\|_{X_+} := \|.\|_X,\\
Y_+ :=& W^{2,2}(\Omega)^n \cap Y, \|.\|_{Y_+} := \|.\|_{2,2},\\
\widetilde{Y}_h :=& \mathrm{span}\Big\{ b_T^2 v, \psi_E \widetilde{P}\tau : v \in \Pi^n_{m|T},\\
&\qquad \tau \in \Pi^n_{k-1|E}, T \in \mathcal{T}_h, E \in \mathcal{E}_{h,\Omega} \cup \mathcal{E}_{h,N} \Big\},\\
\widehat{X}_h :=& \widehat{Y}_h := \widetilde{Y}_{h|\omega_0}.
\end{aligned}$$

Here, ψ_E and $\widetilde{P}$ are as in Remark 3.6, and $\widetilde{Y}_h$ is equipped with the norm of Y_+. The condition $DF(u)^* \in \mathrm{Isom}(Y_+, X_-^*)$ is equivalent to the $W^{2,2}$-regularity

of Problem (3.66) with $g = 0$ and arbitrary load $f \in L^2(\Omega)^n$. This condition is satisfied if Ω is convex. As in Section 3.2, we gain an additional power of h when estimating the quantities $\|(\mathrm{Id}_{Y_+} - R_h)^*[F(u_h) - \widetilde{F}_h(u_h)]\|_{Y_+^*}$, $\|F(u_h) - \widetilde{F}_h(u_h)\|_{\widetilde{Y}_h^*}$, $\|(\mathrm{Id}_{Y_+} - R_h)^*\widetilde{F}_h(u_h)\|_{Y_+^*}$, and $\|\widetilde{F}_h(u_h)\|_{\widetilde{Y}_h^*}$. The scaling arguments used in the proof of Inequalitites (3.20) and (3.21) show that $B \in \mathrm{Isom}(\widehat{X}_h, \widehat{Y}_h^*)$ also with the present choice of norms. Set

$$\widetilde{\eta}_{x_0} := \|u_0\|_{0,2;\omega_0} \tag{3.72}$$

where u_0 is the solution of Problem (3.70).

Collecting the previous results, we obtain:

Proposition 3.26. *Let $u \in X$ be the unique weak solution of Problem (3.66) and let $u_h \in X_h$ be an approximate solution of the corresponding discrete problem. Then the following a posteriori error estimates hold*

$$\begin{aligned}\|u - u_h\|_{1,2} \le c_1 &\Big\{ \sum_{T\in\mathcal{T}_h} \eta_T^2 \Big\}^{1/2} \\ &+ c_2 \Big\{ \sum_{T\in\mathcal{T}_h} h_T^2 \|f - \pi_{m,T} f\|_{0,2;T}^2 \\ &\qquad + \sum_{E\in\mathcal{E}_{h,N}} h_E \|g - \pi_{k-1,E} g\|_{2;E}^2 \Big\}^{1/2} \\ &+ c_3 \|F_h(u_h)\|_{Y_h^*},\end{aligned}$$

$$\begin{aligned}\eta_T \le c_4 \|u - u_h\|_{1,2;\omega_T} + c_5 &\Big\{ \sum_{T'\subset\omega_T} h_{T'}^2 \|f - \pi_{m,T'} f\|_{0,2;T'}^2 \\ &+ \sum_{E\in\mathcal{E}(T)\cap\mathcal{E}_{h,N}} h_E \|g - \pi_{k-1,E} g\|_{2;E}^2 \Big\}^{1/2},\end{aligned}$$

$$c_6 \Big\{ \sum_{T\subset\omega_0} \eta_T^2 \Big\}^{1/2} \le \eta_{x_0} \le c_7 \Big\{ \sum_{T\subset\omega_0} \eta_T^2 \Big\}^{1/2}.$$

Here, η_T and η_{x_0} are given by Equations (3.69) and (3.71).

If in addition Ω is convex, the following a posteriori error are also valid

$$\begin{aligned}
\|u-u_h\|_{0,2} \leq & c_8\left\{\sum_{T\in\mathcal{T}_h} h_T^2\eta_T^2\right\}^{1/2} \\
& + c_9\left\{\sum_{T\in\mathcal{T}_h} h_T^4\|f-\pi_{m,T}f\|_{0,2;T}^2\right. \\
& \qquad \left. + \sum_{E\in\mathcal{E}_{h,N}} h_E^3\|g-\pi_{k-1,E}g\|_{2;E}^2\right\}^{1/2} \\
& + c_{10}\|R_h^* F_h(u_h)\|_{Y_+^*}, \\
h_T\eta_T \leq & c_{11}\|u-u_h\|_{0,2;\omega_T} \\
& + c_{12}\left\{\sum_{T'\subset\omega_T} h_{T'}^4\|f-\pi_{m,T'}f\|_{0,2;T'}^2\right. \\
& \qquad \left. + \sum_{E\in\mathcal{E}(T)\cap\mathcal{E}_{h,N}} h_E^3\|g-\pi_{k-1,E}g\|_{2;E}^2\right\}^{1/2}, \\
c_{13}\left\{\sum_{T\subset\omega_0} h_T^2\eta_T^2\right\}^{1/2} & \leq \widetilde{\eta}_{x_0} \leq c_{14}\left\{\sum_{T\subset\omega_0} h_T^2\eta_T^2\right\}^{1/2}.
\end{aligned}$$

Here, $\widetilde{\eta}_{x_0}$ is given by Equation (3.72). The constants $c_1,\ldots,c_{14}$ only depend on Ω, the Lamé constants λ,μ, the polynomial degree of the finite element space X_h, and the shape parameter of Condition (2) in Section 3.1. □

Remark 3.27. Proceeding as in Section 3.3 one may extend Proposition 3.26 to problems of nonlinear elasticity where the load and boundary tractions may depend on the displacement and the stress tensor and where the Lamé coefficients λ,μ may be functions of the displacement. □

Remark 3.28. In order to avoid locking phenomena one often works with mixed formulations of Problem (3.66) which involve the stress tensor as an additional unknown and which are related to dual or hybrid primal/dual variational problems. For such problems a posteriori error estimates can be derived by combining the techniques of this and the preceeding section. Some technical difficulties, however, occur. They are due to the fact, that one has to work with the space

$$H(\mathrm{div};\Omega) := \{\sigma\in L^2(\Omega)^{n\times n} : \nabla\cdot\sigma\in L^2(\Omega)^n\}$$

and that this space has approximation and scaling properties which are less favorable than those of the standard Sobolev spaces (cf., e.g., [32]). □

Remark 3.29. In [87] various a posteriori error estimators for Problem (3.66) are described and compared. Error estimators for mixed formulations of Problem (3.66) are investigated in, e.g., [32, 50]. □

3.7. The Biharmonic Equation.

As an example for the treatment of higher order elliptic equations we consider the two dimensional biharmonic equation

$$\begin{aligned} \Delta^2 u &= f && \text{in } \Omega \subset I\!R^2 \\ u = \frac{\partial u}{\partial n} &= 0 && \text{on } \Gamma \end{aligned} \tag{3.73}$$

which models the vertical displacement of the mid-surface of a clamped plate. Problem (3.73) fits into the abstract framework of Section 2.1 with

$$\begin{aligned} X = Y &:= \{\, u \in W^{2,2}(\Omega) : u = \frac{\partial u}{\partial n} = 0 \text{ on } \Gamma \,\}, \\ \|\cdot\|_X = \|\cdot\|_Y &:= \|\cdot\|_{2,2}, \\ \langle F(u), \varphi \rangle &:= \int_\Omega \Delta u \Delta \varphi - \int_\Omega f\,\varphi. \end{aligned} \tag{3.74}$$

Since the bilinear form $u, v \to \int_\Omega \Delta u \Delta v$ is continuous and coercive on X (cf. [37]), we have $DF(u) \in \mathrm{Isom}(X, Y^*)$ for all $u \in X$.

For the discretization of Problem (3.73) we assume that $X_h \subset X$ and $Y_h \subset Y$ are finite element spaces corresponding to $\mathcal{T}_h$ and consisting of piecewise polynomials. These conditions in particular imply that the functions in X_h and Y_h are of class C^1. Examples of C^1-conforming finite element spaces are given in, e.g., [37; Chapter 6]. Denote by $k \geq 1$ the maximal polynomial degree of the functions in X_h. We denote by $J_h : Y \to Y_h$ the Clément interpolation operator which satisfies the following analogues of Lemmas 1.4 and 3.1 [cf. 38]

$$\|\varphi - J_h\varphi\|_{0,2;T} \leq c_1 h_T^2 \|\varphi\|_{2,2;\widetilde{\omega}_T} \quad \forall \varphi \in W^{2,2}(\widetilde{\omega}_T),\ T \in \mathcal{T}_h \tag{3.75}$$

$$\begin{aligned} &\|\varphi - J_h\varphi\|_{2;E} + h_E \|n_E \cdot \nabla(\varphi - J_h\varphi)\|_{2;E} \\ &\qquad \leq c_2 h_E^{3/2} \|\varphi\|_{2,2;\widetilde{\omega}_E} \quad \forall \varphi \in W^{2,2}(\widetilde{\omega}_E),\ E \in \mathcal{E}_{h,\Omega}. \end{aligned} \tag{3.76}$$

We set $R_h := J_h$ and define $\widetilde{F}_h$ in the same way as F with f replaced by

$$f_h := \sum_{T \in \mathcal{T}_h} \pi_{l,T} f.$$

Here, $l \geq 0$ is an arbitrary integer which will be kept fixed in what follows.

Using integration by parts elementwise, we obtain for all $\varphi \in Y$ and $u_h \in X_h$

$$\begin{aligned}\langle F(u_h), \varphi \rangle = & \sum_{T \in \mathcal{T}_h} \int_T (\Delta^2 u_h - f)\varphi \\ & + \sum_{E \in \mathcal{E}_{h,\Omega}} \left\{ \int_E [\Delta u_h]_E \, n_E \cdot \nabla \varphi - \int_E [n_E \cdot \nabla \Delta u_h]_E \, \varphi \right\}\end{aligned} \tag{3.77}$$

and

$$\begin{aligned}\langle \widetilde{F}_h(u_h), \varphi \rangle = & \sum_{T \in \mathcal{T}_h} \int_T (\Delta^2 u_h - f_h)\varphi \\ & + \sum_{E \in \mathcal{E}_{h,\Omega}} \left\{ \int_E [\Delta u_h]_E \, n_E \cdot \nabla \varphi - \int_E [n_E \cdot \nabla \Delta u_h]_E \, \varphi \right\}.\end{aligned} \tag{3.78}$$

Equations (3.77) and (3.78) together with Inequality (3.75) immediately imply

$$\begin{aligned}& \|(Id_Y - R_h)^*[F(u_h) - \widetilde{F}_h(u_h)]\|_{Y^*} \\ = & \sup_{\substack{\varphi \in Y \\ \|\varphi\|_Y = 1}} \sum_{T \in \mathcal{T}_h} \int_T (f_h - f)(\varphi - J_h \varphi) \\ \leq & \, c_1 \left\{ \sum_{T \in \mathcal{T}_h} h_T^4 \|f - f_h\|_{0,2;T}^2 \right\}^{1/2} \\ = & \, c_1 \left\{ \sum_{T \in \mathcal{T}_h} \varepsilon_T^2 \right\}^{1/2},\end{aligned} \tag{3.79}$$

where we have set for abbreviation

$$\varepsilon_T := h_T^2 \|f - f_h\|_{0,2;T} \quad \forall T \in \mathcal{T}_h. \tag{3.80}$$

Equations (3.78) and Estimates (3.75), (3.76) on the other hand yield

$$
\begin{aligned}
&\|(Id_Y - R_h)^* \widetilde{F}_h(u_h)\|_{Y^*} \\
&= \sup_{\substack{\varphi \in Y \\ \|\varphi\|_Y = 1}} \left\{ \sum_{T \in \mathcal{T}_h} \int_T (\Delta^2 u_h - f_h)(\varphi - J_h \varphi) \right. \\
&\qquad \left. + \sum_{E \in \mathcal{E}_{h,\Omega}} \left\{ \int_E [\Delta u_h]_E \, n_E \cdot \nabla(\varphi - J_h \varphi) - \int_E [n_E \cdot \nabla \Delta u_h]_E (\varphi - J_h \varphi) \right\} \right\} \\
&\le c \left\{ \sum_{T \in \mathcal{T}_h} h_T^4 \|\Delta^2 u_h - f_h\|_{0,2;T}^2 \right. \\
&\qquad \left. + \sum_{E \in \mathcal{E}_{h,\Omega}} [h_E \|[\Delta u_h]_E\|_{2;E}^2 + h_E^3 \|[n_E \cdot \nabla \Delta u_h]_E\|_{2;E}^2] \right\}^{1/2} \\
&\le c' \left\{ \sum_{T \in \mathcal{T}_h} \eta_T^2 \right\}^{1/2},
\end{aligned}
\tag{3.81}
$$

where we have set for abbreviation

$$
\begin{aligned}
\eta_T := \Big\{ & h_T^4 \|\Delta^2 u_h - f_h\|_{0,2;T}^2 \\
& + \sum_{E \in \mathcal{E}(T) \cap \mathcal{E}_{h,\Omega}} [h_E \|[\Delta u_h]_E\|_{2;E}^2 \\
& \qquad + h_E^3 \|[n_E \cdot \nabla \Delta u_h]_E\|_{2;E}^2] \Big\}^{1/2} \quad \forall T \in \mathcal{T}_h.
\end{aligned}
\tag{3.82}
$$

Relations (3.78), (3.81), and (3.82) and the results of Sections 3.1 and 3.3 show that the space $\widetilde{Y}_h$ must balance the following quantities:

on each triangle: $\Delta^2 u - f_h$,

on each interior edge: $[\Delta u_h]_E$ and $[n_E \cdot \nabla \Delta u_h]_E$.

To achieve this, we adopt the strategy of Sections 3.1 and 3.3 and make the following Ansatz

$$
\begin{aligned}
\widetilde{Y}_h := \operatorname{span}\{ & \varphi_T v, \varphi_{E,1} P_1 \sigma, \varphi_{E,2} P_2 \tau : \\
& v \in \Pi_{m_0|T}, \sigma \in \Pi_{m_1|E}, \tau \in \Pi_{m_2|E}, T \in \mathcal{T}_h, E \in \mathcal{E}_{h,\Omega} \}
\end{aligned}
\tag{3.83}
$$

where $m_0 := \max\{k-4, l\}$, $m_1 := \max\{k-2, 0\}$ and $m_2 := \max\{k-3, 0\}$. The functions φ_T, $\varphi_{E,1}$, and $\varphi_{E,2}$ and the prolongation operators P_1 and P_2 will be constructed in what follows. In doing this we have to keep in mind that, due to the condition $\widetilde{Y}_h \subset Y \subset W^{2,2}(\Omega)$, the functions in $\widetilde{Y}_h$ must be continuously differentiable.

Consider first an arbitrary $T \in \mathcal{T}_h$ and define

$$\varphi_T := b_T^2,$$

where b_T is the triangle-bubble function introduced in Section 1.1. Since φ_T together with its first order derivatives vanishes on ∂T, we have $\varphi_T v \in Y$ for all $v \in \Pi_{m_0|T}$. The arguments used in the proof of Lemma 3.3 imply for all $v \in \Pi_{m_0|T}$

$$\begin{aligned} c_1 \|v\|_{0,2;T}^2 &\le \int_T \varphi_T v^2, \\ \|\varphi_T v\|_{0,2;T} &\le \|v\|_{0,2;T}, \\ c_2 h_T^{-2} \|\varphi_T v\|_{0,2;T} \le \|\varphi_T v\|_{2,2;T} &\le c_3 h_T^{-2} \|\varphi_T v\|_{0,2;T}, \end{aligned}$$

where the constants $c_1, \ldots, c_3$ depend on m_0 and h_T/ϱ_T. Put

$$w_T := \varphi_T(\Delta^2 u_h - f_h).$$

The above estimates and Equation (3.78) then imply

$$\begin{aligned} c_1 \|\Delta^2 u_h - f_h\|_{0,2;T}^2 &\le \int_T (\Delta^2 u_h - f_h) w_T \\ &= \langle \widetilde{F}_h(u_h), w_T \rangle \\ &\le c_3 h_T^{-2} \|\Delta^2 u_h - f_h\|_{0,2;T} \sup_{\substack{\varphi \in \widetilde{Y}_{h|T} \\ \|\varphi\|_Y = 1}} \langle \widetilde{F}_h(u_h), \varphi \rangle \end{aligned}$$

and thus

$$h_T^2 \|\Delta^2 u_h - f_h\|_{0,2;T} \le c \sup_{\substack{\varphi \in \widetilde{Y}_{h|T} \\ \|\varphi\|_Y = 1}} \langle \widetilde{F}_h(u_h), \varphi \rangle. \tag{3.84}$$

Next we consider an arbitrary $E \in \mathcal{E}_{h,\Omega}$. Let $T_1, T_2 \in \mathcal{T}_h$ be such that $\omega_E = T_1 \cup T_2$ and that n_E is the exterior normal to T_1. The orientation of E is chosen such that T_1 is lying on the left when passing through E in this orientation. Set

$$\varphi_{E,1} := (b_{T_2} - b_{T_1}) b_E \quad \text{and} \quad P_1 := P,$$

where b_E is the edge-bubble function introduced in Section 1.1 and where P is the prolongation operator of Section 3.1. Consider an arbitrary $\sigma \in \Pi_{m_1|E}$. An easy calculation shows that $\varphi_{E,1}P\sigma$ is continuously differentiable across E and that

$$n_E \cdot \nabla(\varphi_{E,1}P\sigma) = \frac{27}{8}\left(\frac{h_E}{|T_1|} + \frac{h_E}{|T_2|}\right) b_E^2 P\sigma \tag{3.85}$$

holds on E. Since $\varphi_{E,1}P\sigma$ together with its first order derivatives vanishes on $\partial\omega_E$, we conclude that $\varphi_{E,1}P\sigma \in Y$. Moreover, Equation (3.85) and the arguments used in the proof of Lemma 3.3 imply

$$\begin{aligned}
c_4 h_E^{-1}\|\sigma\|_{2;E}^2 &\le \int_E \sigma n_E \cdot \nabla(\varphi_{E,1}P\sigma),\\
\|n_E \cdot \nabla(\varphi_{E,1}P\sigma)\|_{2;E} &\le c_5 h_E^{-1}\|\sigma\|_{2;E},\\
c_6 h_E^{-2}\|\varphi_{E,1}P\sigma\|_{0,2;\omega_E} &\le \|\varphi_{E,1}P\sigma\|_{2,2;\omega_E} \le c_7 h_E^{-2}\|\varphi_{E,1}P\sigma\|_{0,2;\omega_E},\\
\|\varphi_{E,1}P\sigma\|_{0,2;\omega_E} &\le c_8 h_E^{1/2}\|\sigma\|_{2;E},
\end{aligned}$$

where the constants $c_4, \dots, c_8$ depend on m_1 and h_T/ϱ_T. Put

$$w_{E,1} := \varphi_{E,1}P([\Delta u_h]_E).$$

The above estimates, Equations (3.78), (3.85), and Inequality (3.84) then yield

$$\begin{aligned}
c_4 h_E^{-1}\|[\Delta u_h]_E\|_{2;E}^2 &\le \int_E [\Delta u_h]_E\, n_E \cdot \nabla w_{E,1}\\
&= \langle \widetilde{F}_h(u_h), w_{E,1}\rangle - \sum_{i=1,2}\int_{T_i} (\Delta^2 u_h - f_h) w_{E,1}\\
&\le c\, h_E^{-2}\|w_{E,1}\|_{0,2;\omega_E} \sup_{\substack{\varphi \in \widetilde{Y}_{h|\omega_E}\\ \|\varphi\|_Y = 1}} \langle \widetilde{F}_h(u_h), \varphi\rangle\\
&\le c c_8 h_E^{-3/2}\|[\Delta u_h]_E\|_{2;E} \sup_{\substack{\varphi \in \widetilde{Y}_{h|\omega_E}\\ \|\varphi\|_Y = 1}} \langle \widetilde{F}_h(u_h), \varphi\rangle
\end{aligned}$$

and thus

$$h_E^{1/2}\|[\Delta u_h]_E\|_{2;E} \le c' \sup_{\substack{\varphi \in \widetilde{Y}_{h|\omega_E}\\ \|\varphi\|_Y = 1}} \langle \widetilde{F}_h(u_h), \varphi\rangle. \tag{3.86}$$

In order to balance the $[n_E \cdot \nabla\Delta u_h]_E$-term, we use Remark 3.6 and set $\varphi_{E,2} := \psi_E$, $P_2 := \widetilde{P}$, and

$$w_{E,2} := \psi_E \widetilde{P}([n_E \cdot \nabla\Delta u_h]_E).$$

Remark 3.6, Equation (3.78), and Inequalities (3.84) and (3.86) then give

$$
\begin{aligned}
& c_9 \|[n_E \cdot \nabla \Delta u_h]_E\|_{2;E}^2 \\
\leq & \int_E [n_E \cdot \nabla \Delta u_h]_E \, w_{E,2} \\
= & \sum_{i=1,2} \int_{T_i} (\Delta^2 u_h - f_h) w_{E,2} + \int_E [\Delta u_h]_E n_E \cdot \nabla w_{E,2} - \langle \widetilde{F}_h(u_h), w_{E,2} \rangle \\
\leq & \|[n_E \cdot \nabla \Delta u_h]_E\|_{2;E} \Bigg\{ c_{13} \Bigg[\sum_{i=1,2} h_E \|\Delta^2 u_h - f_h\|_{0,2;T_i}^2 \Bigg]^{1/2} \\
& \qquad + c_{12} h_E^{-1} \|[\Delta u_h]_E\|_{2;E} \\
& \qquad + c_{11} c_{13} h_E^{-3/2} \sup_{\substack{\varphi \in \widetilde{Y}_{h|\omega_E} \\ \|\varphi\|_Y = 1}} \langle \widetilde{F}_h(u_h), \varphi \rangle \Bigg\} \\
\leq & \, c \, h_E^{-3/2} \|[n_E \cdot \nabla \Delta u_h]_E\|_{2;E} \sup_{\substack{\varphi \in \widetilde{Y}_{h|\omega_E} \\ \|\varphi\|_Y = 1}} \langle \widetilde{F}_h(u_h), \varphi \rangle
\end{aligned}
$$

and thus

$$
h_E^{3/2} \|[n_E \cdot \nabla \Delta u_h]_E\|_{2;E} \leq c' \sup_{\substack{\varphi \in \widetilde{Y}_{h|\omega_E} \\ \|\varphi\|_Y = 1}} \langle \widetilde{F}_h(u_h), \varphi \rangle. \tag{3.87}
$$

Inequalities (3.84), (3.86), and (3.87) imply for all $T \in \mathcal{T}_h$

$$
\eta_T \leq c \sup_{\substack{\varphi \in \widetilde{Y}_{h|\omega_T} \\ \|\varphi\|_Y = 1}} \langle \widetilde{F}_h(u_h), \varphi \rangle. \tag{3.88}
$$

Using the same arguments as in the preceding sections this together with Inequality (3.81) proves Estimate (2.11). Finally, the above estimates for the functions in $\widetilde{Y}_h$ and Equations (3.77) and (3.78) imply in the usual way that

$$
\|\widetilde{F}_h(u_h) - F(u_h)\|_{\widetilde{Y}_h^*} \leq c \Bigg\{ \sum_{T \in \mathcal{T}_h} \varepsilon_T^2 \Bigg\}^{1/2} \tag{3.89}
$$

and

$$
\|\widetilde{F}_h(u_h)\|_{\widetilde{Y}_h^*} \leq c \Bigg\{ \sum_{T \in \mathcal{T}_h} \eta_T^2 \Bigg\}^{1/2}. \tag{3.90}
$$

Propositions 2.1 and 2.5 and Inequalities (3.79) – (3.90) therefore yield the following a posteriori error estimate for Problem (3.73).

Proposition 3.30. *Let $u \in X$ be the unique weak solution of Problem (3.73), i. e. of $F(u) = 0$, and let $u_h \in X_h$ be an approximate solution of the corresponding discrete problem. Then the following a posteriori error estimates hold*

$$\|u - u_h\|_{2,2} \leq c_1 \left\{ \sum_{T \in \mathcal{T}_h} \eta_T^2 \right\}^{1/2} + c_2 \left\{ \sum_{T \in \mathcal{T}_h} \varepsilon_T^2 \right\}^{1/2} + c_3 \|F(u_h) - F_h(u_h)\|_{Y_h^*} + c_4 \|F_h(u_h)\|_{Y_h^*}$$

and

$$\eta_T \leq c_5 \|u - u_h\|_{2,2;\omega_T} + c_6 \left\{ \sum_{T' \subset \omega_T} \varepsilon_{T'}^2 \right\}^{1/2} \quad \forall T \in \mathcal{T}_h.$$

Here, ε_T and η_T are given by Equations (3.80) and (3.82), $\|F(u_h) - F_h(u_h)\|_{Y_h^*}$ and $\|F_h(u_h)\|_{Y_h^*}$ denote the consistency error of the discretization and the residual of the discrete problem, and the constants $c_1, \ldots, c_6$ depend on h_T/ϱ_T and the integers k and l.

Remark 3.31. In order to avoid the use of C^1-elements, one often considers mixed finite element approximations of Problem (3.73) which are obtained by discretizing the weak formulation of an appropriate equivalent elliptic system of 2nd order. Since these mixed finite element discretizations only involve C^0-elements, they can be analyzed along the lines of Sections 3.1, 3.3, and 3.5. □

Remark 3.32. Equations (3.77) and (3.78) rely on the fact that $u_{h|T} \in W^{4,2}(T)$ for all $T \in \mathcal{T}_h$. When using composite elements, one has to perform the integration by parts on each composite seperately. This gives rise to additional jump terms across the boundaries of the composites which may be handled in the same way as the jump terms in Equations (3.77) and (3.78). □

Remark 3.33. To our knowledge, a posteriori error estimators for 4th order problems have not been considered before. □

4. Practical Implementation

4.1. Mesh-Refinement Techniques.

Disposing of an a posteriori error estimator one has to decide which elements must be refined or eventually be coarsened (cf. Step (4) in the algorithm given in the Introduction). In [12, 13, 63] heuristic arguments are given that for a linear finite element discretization of Problem (1.1) among all partitions with a given number of elements that one is optimal which equilibrates the error, i.e., the errors in all elements are the same. Many practical examples demonstrate the efficiency of this strategy.

Based on this strategy, there are two main possibilities to decide, based on an error estimator, which elements have to be refined:

(1) Let T be an element of $\mathcal{T}_h$ and let τ be obtained by subdividing T. It is quite reasonable to assume that the errors E_T and E_τ in T and τ behave like ch_T^λ and ch_τ^λ with unknown constants c and λ. Having computed error estimators η_T and η_τ for E_T and E_τ, one can then roughly determine c and λ and thus approximately predict the error in an element τ' obtained by a further subdivision of T.

(2) Suppose that one has computed for each element T of the partition $\mathcal{T}_h$ an estimator η_T of the error in T. Put $\eta := \max_{T\in\mathcal{T}_h} \eta_T$. Then an element T is subdivided if $\eta_T \geq \gamma\eta$, where $0 < \gamma < 1$ is a prescribed threshold; typically $\gamma = 0.5$.

The first possibility clearly is more sophisticated. It is used in the code PLTMG of R. Bank (cf. [19]). In practice, however, the second possibility, which is cheaper, gives satisfactory results. For linear finite element approximations of the model problem (1.1) it is shown in [42] that this strategy based on the residual error estimator $\eta_{R,T}$ of Section 1.2 is able to reduce the error in each refinement step by a given fixed factor provided the initial coarse mesh is sufficiently fine. The condition on the initial mesh is quantified exactly and can explicitly be checked for any given mesh and any given set of data f and g.

A different refinement strategy is proposed in [52]. There a cost functional is introduced and the error, approximated by an error estimator, is minimized subject to a prescribed bound for the cost functional. This strategy allows the construction of optimal meshes with fewer intermediate refinement steps. However, it is much more costly than the two strategies described before.

A close inspection of the proofs in Sections 1 and 3 shows that it is mandatory to preserve the shape regularity of the partition during the refinement process, i. e., to respect Condition (2) of Section 3.1. When using a quadrilateral mesh in two or three dimensions, this condition is satisfied when subdividing the elements by joining their midpoints of edges. However, this process introduces hanging nodes, i.e., midpoints of edges where not all of the neighbouring elements have been divided (see Figure 6). At those points the compatibility conditions for finite element partitions are violated. There are two posibilities to overcome this difficulty:

(1) Treat the unknowns corresponding to hanging nodes as spurious degrees of freedom, i.e., their values are fixed to be a suitable interpolation of the unknowns corresponding to neighbouring regular nodes.
(2) Introduce additional triangular or tetrahedral elements such that the resulting partition satisfies the regularity assumptions for finite element partitions.

In practice, the second possibility, however, seems to be used only occasionally.

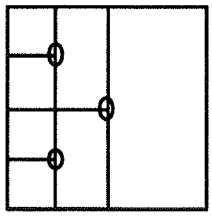


Fig. 6: hanging nodes (marked by ○)

For triangular meshes there exist essentially three different refinement strategies which preserve the shape regularity:

(1) Bisect triangles by joining the midpoint of the longest edge with the vertex opposite to this edge (*longest edge bisection*).
(2) Bisect triangles by joining the midpoint of a marked edge with the vertex opposite to this edge (*marked edge bisection*).
(3) Divide triangles into four by joining the midpoints of edges (*regular refinement*).

The longest edge bisection strategy is described in detail in [66, 67]. In order to obtain an admissible triangulation, one has to add the following two rules:

(1a) A triangle having two hanging nodes on edges, which are not the longest one, is divided by joining the midpoints of edges (*red refinement*, see Figure 7).

(1b) A triangle having exactly one hanging node on an edge, which is not the longest one, is divided by joining the hanging node to the midpoint of the longest edge and by joining the latter one to the vertex opposite to the longest edge (*blue refinement*, see Figure 7).

In [70] it is proved that, when using the longest edge bisection, the smallest angle in all refined triangulations is at least half of the smallest angle in the initial triangulation.

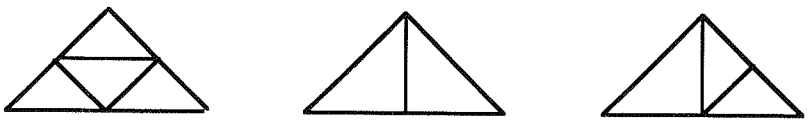


Fig. 7: red, green, and blue refinement of a triangle

The marked edge bisection is described in [55]. Sometimes it also called newest vertex bisection. It obeys to the following rules:

(2a) Each triangle has exactly one marked edge.

(2b) When bisecting a triangle its unmarked edges become the marked edges of its two sons (see Figure 8).

(2c) An edge may only be bisected, if it is the marked edge of both its adjacent triangles.

Rule (2b) ensures that there are only four different types of triangles in the triangulation [55, 71]. Thanks to rules (2a) and (2c) triangles are refined only by bisection.

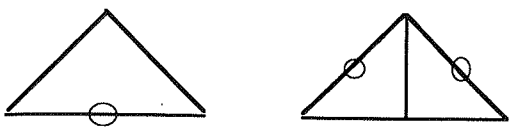


Fig. 8: bisection of marked edges (indicated by ○)

The regular refinement of triangles is described, e. g., in [18, 19, 20, 78, 80]. It has the advantage that the sons of a regularly refined triangle are similar to their father. On the other hand it introduces hanging nodes. In order to ensure the compatibility of the triangulation, the refinement process has to obey to the following rules:

(3a) A triangle having at least two hanging nodes is refined regularly.

(3b) A triangle having exactly one hanging node is bisected by joining the hanging node to the vertex opposite to it (*green refinement*).

(3c) A triangle having a hanging node on an edge, which has been previously bisected, must be refined regularly.

When refining a triangle with two hanging nodes according to rule (3a), a new hanging node is introduced. One can prove, however, that the refinement process obeying to the above rules is finite. In order to avoid the introduction of new hanging nodes and to keep the refinement zone as narrow as possible, one can modify rule (3a) as follows:

(3a′) A triangle having three hanging nodes is refined regularly; one having two hanging nodes is refined according to rule (1b) (*blue refinement*).

Rule (3c) tries to keep the number of bisected triangles as small as possible and ensures that the minimal angle in all refined triangulations is at least half of the minimal angle in the original triangulation. In order to keep the number of bisected triangles as small as possible and the refinement zone as narrow as possible, rule (3c) may be modified in two ways:

(3c′) A triangle having a hanging node on an edge, which has been previously bisected, must be refined according to rule (1b) (*blue refinement*).

(3c″) Let τ be obtained by bisecting T and let x be a hanging node of τ on the previously bisected edge. Then τ is removed from the triangulation, T is refined regularly, and the son of T, which has x as a hanging node, is bisected.

Rule (3c″) was introduced in [18] and is used in the code PLTMG [19]. It is illustrated in Figure 9. It has the disadvantage of introducing new hanging nodes and of leading to triangulations which are not nested.

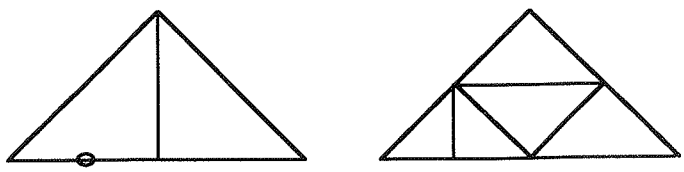


Fig. 9: rule (3c′ı), hanging node marked by ○

For some applications such as, e.g., semi-conductor device simulation, one is interested in having a discrete maximum principle, which requires weakly acute

triangulations. In order to preserve the weak acuteness of a triangulation during the refinement process, rule (3b) can be sharpened in the sense that a triangle may only be bisected along its perpendicular bisector (cf. [15]).

For tetrahedral meshes it is much more complicated to preserve the shape regularity during a refinement process, since a tetrahedron cannot be divided into similar tetrahedra.

A bisection strategy which extends the marked edge bisection to three dimensions is described in [16, 17, 55, 58]. The main idea is to bisect a tetrahedron such that its faces are divided as in the two-dimensional marked edge bisection. It is shown in [16, 17] that this strategy preserves the shape regularity.

An extension of the regular refinement strategy to three dimensions is described in [28]. As in two dimensions this strategy consists of two sets of rules which describe the regular and irregular refinement of a tetrahedron. The regular refinement corresponds to the red refinement of a triangle. The irregular refinement corresponds to the green and blue refinement of a triangle and generalizes rules (3a) - (3c).

In order to describe the regular refinement of a tetrahedron T we identify it with the ordered list $[x_0, x_1, x_2, x_3]$ of its vertices and denote by $x_{ij} := \frac{1}{2}(x_i + x_j), 0 \leq i < j \leq 3$, the midpoint of the edge connecting x_i and x_j. Then $T = [x_0, x_1, x_2, x_3]$ is cut into the following 8 tetrahedra

$$
\begin{aligned}
T_1 &:= [x_0, x_{01}, x_{02}, x_{03}], & T_5 &:= [x_{01}, x_{02}, x_{03}, x_{13}],\\
T_2 &:= [x_{01}, x_1, x_{12}, x_{13}], & T_6 &:= [x_{01}, x_{02}, x_{12}, x_{13}],\\
T_3 &:= [x_{02}, x_{12}, x_2, x_{23}], & T_7 &:= [x_{02}, x_{12}, x_{13}, x_{23}],\\
T_4 &:= [x_{03}, x_{13}, x_{23}, x_3], & T_8 &:= [x_{02}, x_{03}, x_{13}, x_{23}].
\end{aligned}
$$

$T_1, ..., T_4$ are obtained by chopping off the vertices of T. These tetrahedra are similar to T. $T_5, ..., T_8$ form the remaining octahedron. They are not similar to T. Nevertheless, one can prove that the above refinement process produces at most three different similarity classes of tetrahedra. In particular, it preserves the shape regularity.

There are four types of irregularly refined tetrahedra. They can be characterized by the refinement of the faces:

(T1) one red and three green faces,
(T2) two green faces,
(T3) one blue and two green faces,
(T4) three green faces.

A tetrahedron of type (T1) arises if one neighboring tetrahedron is refined regularly but no other edge is subdivided. A tetrahedron of type (T2) corresponds to an element with exactly one refined edge. Types (T3) and (T4) apply to two refined edges on a common face and in opposition, respectively.

The refinement algorithm is completed by the following rule corresponding to rule (3a):

If three or more edges are refined and do not belong to a common face, then the tetrahedron is refined regularly.

4.2. Data Structures.

Data structures for adaptive mesh-refinement are described, e. g., in [18, 19, 20, 64, 66, 81]. Their implementation depends among others on the refinement strategy and on the programming language used and on whether the underlying data structure of the finite element code is triangle- or node-oriented. As an example, we shortly describe the data structure used in the FORTRAN-code FEMFLOW for the mixed finite element discretization of the stationary, incompressible Navier-Stokes equations in general two-dimensional domains (cf. [81]). For a more detailed description of a similar data structure, we refer to [19].

FEMFLOW is strictly triangle-oriented. The connection between local and global enumeration of the nodes is given by an array ITNODE where

$$\text{ITNODE}(i,j) = \text{global number of node } i \text{ of triangle } j.$$

The neighbourhood-relations of the triangles are described by an array ITEDGE where

$$\text{ITEDGE}(i,j) = k \text{ if and only if edge } j \text{ of triangle } i \text{ is adjacent to triangle } k.$$

Here, edge j is the edge opposite to the vertex with the local number j. The information stored in ITEDGE is not only used for the refinement algorithm. It is also needed for the multi-grid algorithm and it is useful for algorithms such as, e. g., the transport diffusion algorithm, which require the efficient search for triangles having a given node as vertex.

The information particularly needed for the refinement process is stored in an array ITMARK where

$$\begin{aligned}
&\text{ITMARK}(1,j) = \text{number of the father of triangle } j,\\
&\text{ITMARK}(2,j) = \begin{cases} 0 & \text{if triangle } j \text{ is not refined,}\\ i & 1 \le i \le 3, \text{ if edge } i \text{ of triangle } j \text{ is bisected,}\\ 4 & \text{if triangle } j \text{ is refined regularly,}\\ 4k+i & \text{if triangle } j \text{ is refined blue by bisecting edges } i \\ & \text{and } (i+k) \text{ modulo } 3, \end{cases}\\
&\text{ITMARK}(3,j) = \text{number of a son of triangle } j.
\end{aligned}$$

The remaining sons are numbered according to Figure 10, where ITMARK(3 , j) = JS and ITMARK(2 , j) = i for the green refinement. In that case, the numbers i+1 and i+2 have to be taken modulo 3. Figure 10 also shows the local enumeration of the vertices in the refined triangles. A blue refinement will be treated as if it is the result of two successive green refinements.

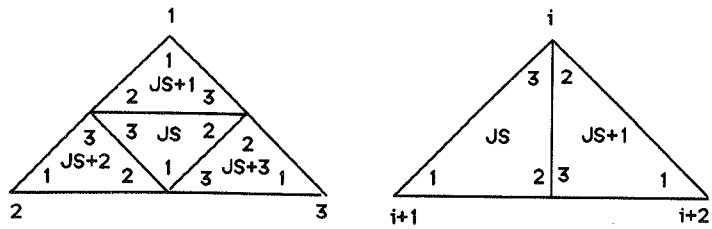


Fig. 10: Enumeration of nodes and triangles for red and green refinement

4.3. Numerical Examples.

We consider three numerical examples:

(1) The Stokes equations in a circular segment of radius 1 and angle 270°.
(2) The Stokes equations in a circular segment of radius 1 and angle 360°.
(3) The Navier-Stokes equations at Reynold's number 50 in a backward facing step with channel width/length of 1/3 units upstream and 2/19 units downstream and unit parabolic in- and outflow.

Recall that the Stokes equations are obtained by dropping the convective term $(\mathbf{u} \cdot \nabla)\mathbf{u}$ in Equation (3.48) and setting $\nu = 1$. In Examples 1 and 2, the exact

solutions are given in polar co-ordinates by

$$\mathbf{u} = \begin{pmatrix} r^{\alpha}[(1+\alpha)\sin(\varphi)\psi(\varphi) + \cos(\varphi)\partial_{\varphi}\psi(\varphi)] \\ r^{\alpha}[\sin(\varphi)\partial_{\varphi}\psi(\varphi) - (1+\alpha)\cos(\varphi)\psi(\varphi)] \end{pmatrix}$$
$$p = -r^{\alpha-1}[(1+\alpha)^2\partial_{\varphi}\psi(\varphi) + \partial_{\varphi}^3\psi(\varphi)]/(1-\alpha)$$

with

$$\psi(\varphi) = \sin((1+\alpha)\varphi)\cos(\alpha\omega)/(1+\alpha) - \cos((1+\alpha)\varphi)$$
$$- \sin((1-\alpha)\varphi)\cos(\alpha\omega)/(1-\alpha) + \cos((1-\alpha)\varphi),$$

$$\alpha = 856399/1572864, \quad \omega = 3\pi/2$$

for Example 1 and

$$\psi(\varphi) = 3\sin(0.5\varphi) - \sin(1.5\varphi),$$
$$\alpha = 0.5, \quad \omega = 2\pi$$

for Example 2. The velocity fields are shown in Figures 11 and 12.

All examples are discretized using the modified Taylor-Hood element described in [75], using continuous, piecewise linear finite elements on $\mathcal{T}_h$ for the pressure and continuous piecewise linear finite elements on $\mathcal{T}_{h/2}$ for the velocity, where $\mathcal{T}_{h/2}$ is obtained from $\mathcal{T}_h$ by a uniform refinement. Since this element satisfies the stability condition (3.50) the parameters α and δ of Equation (3.49) are set to zero. The error estimator is η_T as given by Equation (3.54) with the obvious modifications for the Stokes equations, i.e., the convection term $(\mathbf{u}\cdot\nabla)\mathbf{u}$ is dropped and the viscosity ν is set to 1. A triangle is refined if its estimated error is at least half of the estimated maximal error. We use the regular refinement strategy described in Section 4.1 according to the rules (3a'), (3b), and (3c'). Figures 13 – 25 show for all examples the initial triangulations together with two refined triangulations and zooms of the refinement near the critical regions.

Table 2 gives a comparison of uniform and adaptive refinement strategies for Examples 1 and 2. Here L, NT, NN, $\varepsilon_{\mathbf{u}}$, ε_p, q, Tt, and Ts are the number of refinement steps, the number of triangles, the number of unknowns, the relative error of the velocity in the H^1-norm, the relative error of the pressure in the L^2-norm, the ratio of the predicted value of $\{||\mathbf{u}-\mathbf{u}_h||_{1,2} + ||p-p_h||_{0,2}\}^{1/2}$ to the true one, the time in seconds needed for mesh-generation, and the time in seconds required for the solution of the discrete problems. The discrete linear problems are solved by

the combined conjugate gradient-multigrid method described in [76]. The discrete nonlinear problems are solved by a damped Newton iteration with the combined conjugate gradient-multigrid algorithm as a linear solver. All computations were done on a MacIntosh IIx. The solution algorithm could be accelerated, since it does superfluous work when only few unknowns are added during the refinement process.

Ex.	Ref.	L	NT	NN	$\varepsilon_{\mathbf{u}}$	ε_p	q	Tt	Ts
	uniform	5	3072	3338	0.0375	0.0741		10	322
1	η_T	7	2298	2534	0.0211	0.0437	0.6	14	316
	uniform	5	4096	4466	0.0828	0.2580		13	431
2	η_T	7	2832	3138	0.0428	0.0811	0.5	17	310

Table 2: comparison of uniform and adaptive refinement

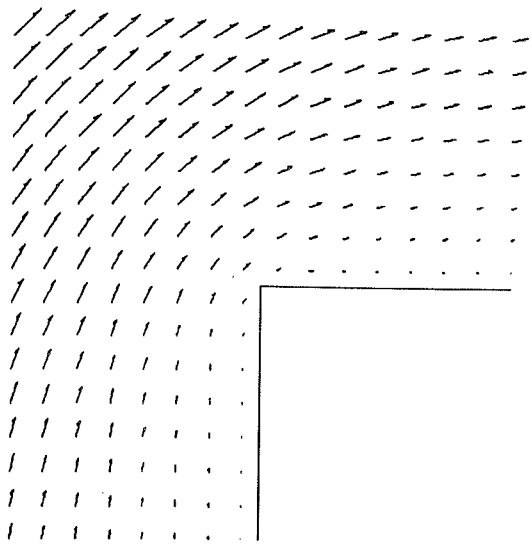


Fig. 11: Example 1, velocity field

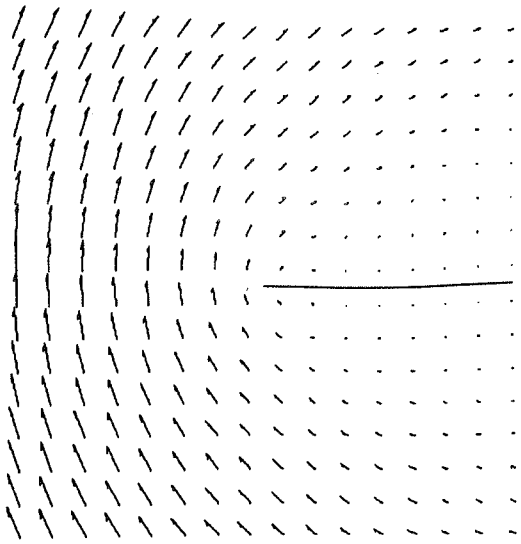


Fig. 12: Example 2, velocity field

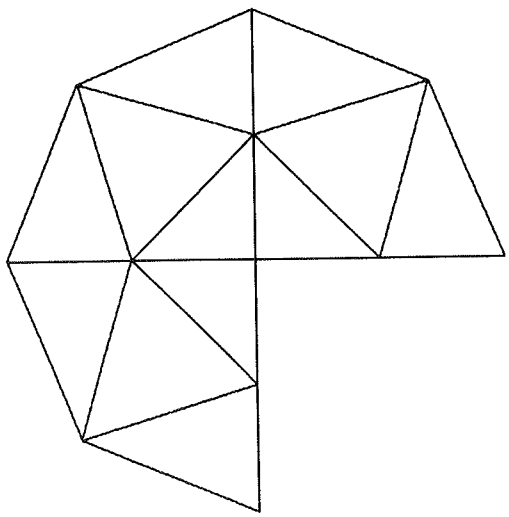


Fig. 13: Example 1, initial triangulation

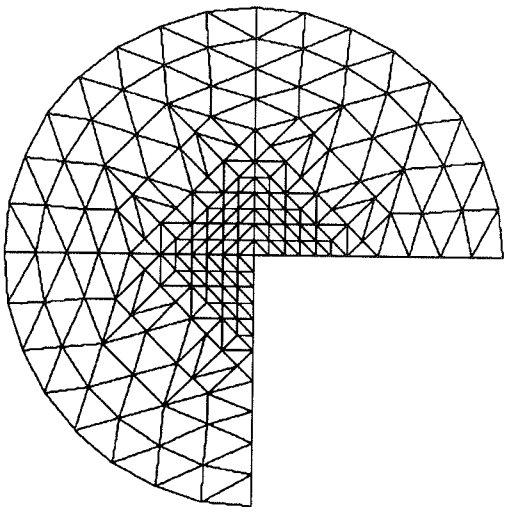


Fig. 14: Example 1, 4 refinement steps, 318 triangles, 183 nodes

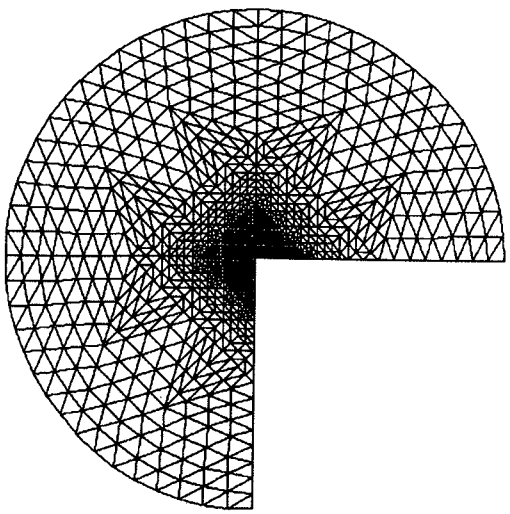


Fig. 15: Example 1, 7 refinement steps, 2298 triangles, 1209 nodes

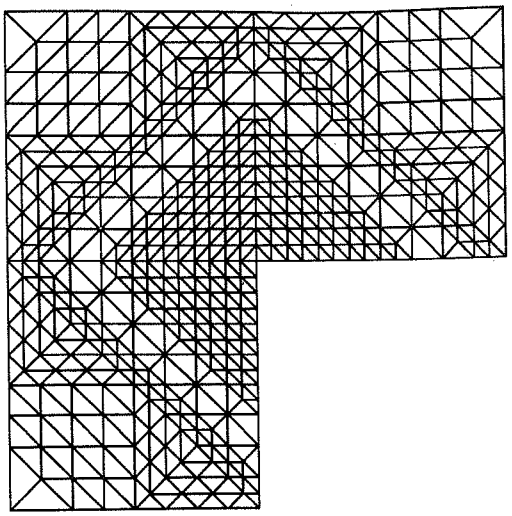


Fig. 16: zoom of $[-\frac{1}{8}, \frac{1}{8}] \times [-\frac{1}{8}, \frac{1}{8}]$ *in Fig. 15*

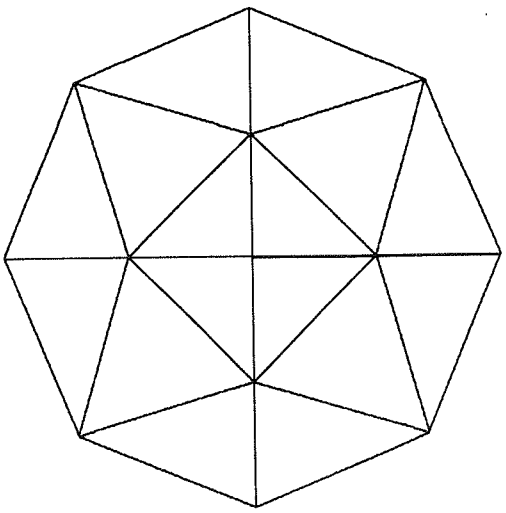


Fig. 17: Example 2, initial triangulation

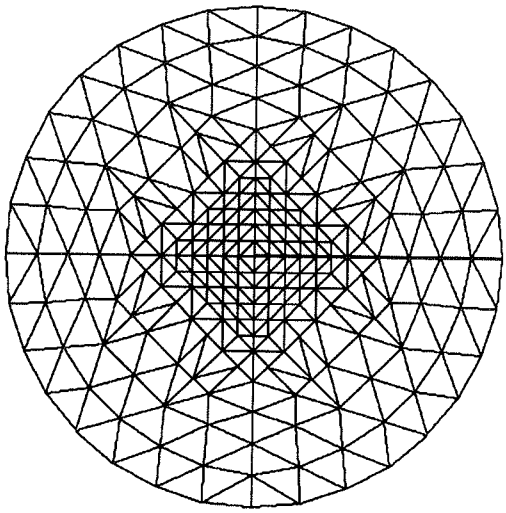


Fig. 18: Example 2, 4 refinement steps, 424 triangles, 240 nodes

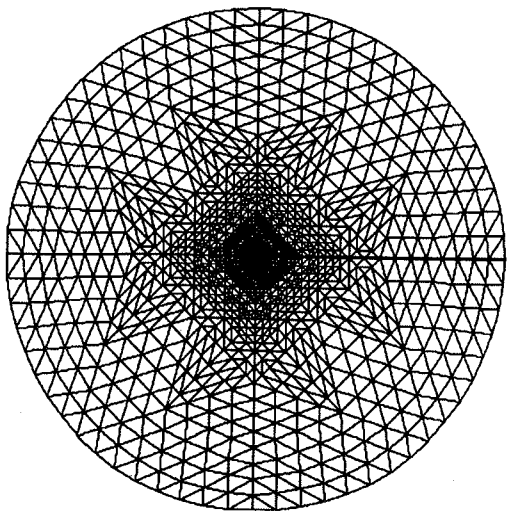


Fig. 19: Example 2, 7 refinement steps, 2832 triangles, 1482 nodes

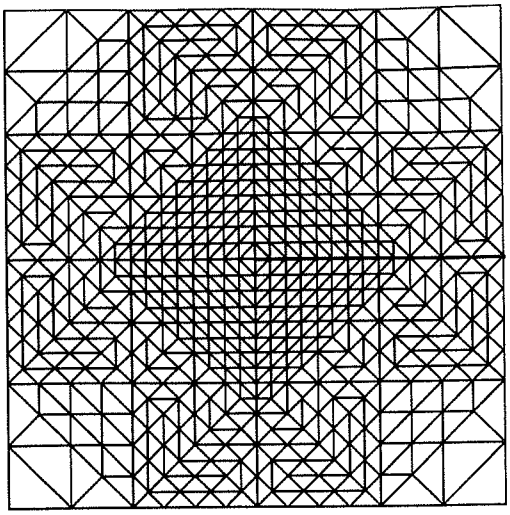


Fig. 20: zoom of $[-\frac{1}{8},\frac{1}{8}]\times[-\frac{1}{8},\frac{1}{8}]$ *in Fig. 19*

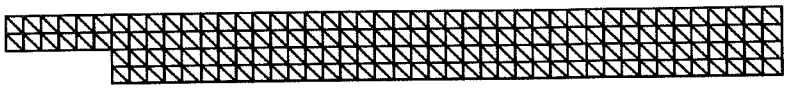


Fig. 21: Example 3, initial triangulation

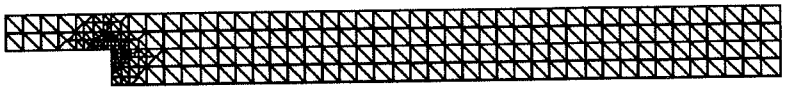


Fig. 22: Example 3, 3 refinement steps, 503 triangles, 285 nodes

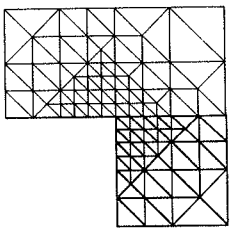


Fig. 23: zoom of $[-1,1]\times[-1,1]$ *in Fig. 22*

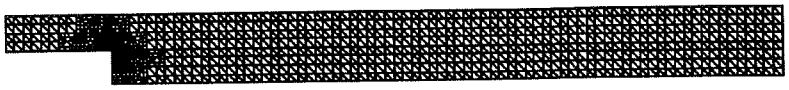


Fig. 24: Example 3, 6 refinement steps, 2399 triangles, 1319 nodes

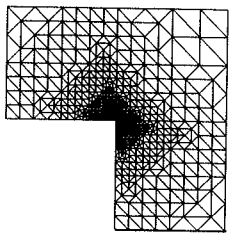


Fig. 25: zoom of $[-1,1] \times [-1,1]$ *in Fig. 24*

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